

KALASALINGAM UNIVERSITY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
ELECTROCOMM

ARTISTIC ^{Care} **WORK**

TECH HI-NOTES

LITERATURE

INNOVATION

CD-ROM
AVAILABLE
WITH THIS ISSUE

SEPTEMBER 2010



ISSUE 3

KALASALINGAM UNIVERSITY
DEPARTMENT
OF
ELECTRONICS & COMMUNICATION
ENGINEERING.

ELECTROCOMM
Issue-3

CONTENTS

❖ INSPIRATION	- 07
❖ LEADING BIRDS	-13
❖ TECH HI- NOTES	-15
❖ NON TECH ARTICLE	-96
❖ ARTISTIC WORK	-157
❖ POLL QUESTIONS	-169

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	- Mr.G.Pandiarajan	Lect/ECE
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	- S.Sumith	III yr/F

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“Coming together is a beginning , Keeping together is progress, Working together is success”

VC'S MESSAGE



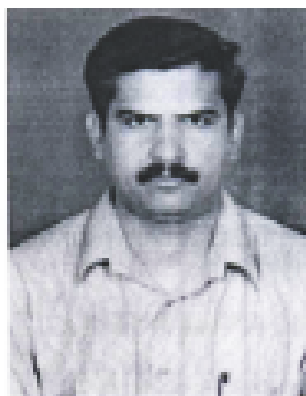
With the explosion of electronic media, communication industry has become one of the fast growing sectors in every corner of the world. The electronics and communication engineering field plays a major role in our day to day life and its technology has helped us to see, hear and communicate over vast distances.

In this scenario, I am delighted that our Department of Electronics and Communication Engineering is bringing out their department magazine "Electrocomm". I hope this will help our students to bring out their innovative ideas. Let this magazine be the centre stage for innovation, creativity, technology and credibility.

I wish the magazine give opportunities for the students to exhibit their technical knowledge and artistic skills.


(Dr. S. RADHAKRISHNAN)
VICE CHANCELLOR

HOD'S MESSAGE



As being the Head of the Department and Editor of this Magazine "ELECTROCOMM", I am pleased to wish all my students who contributed their hard work and valuable time in designing this wonderful magazine. Reading is a practice which facilitates the people to march towards the successful career. Growing up without books is growing up deprived and with a deprivation that puts one at risk for failure, said Dr. Perri Klass. As the above quote defines, we develop our students to be augmented in knowledge by providing the chance to create this magazine which brings out the 3 I's (Ideas, Interest and Innovation) of our students. Finally, I conclude my message by thanking the entire team and I wish all the students to make this effort as a successful one.


(Dr. S. Durairaj)
Prof. & Head/ECE

INSPIRATION



MR.T.ARIVOZHI

Associate Professor, ECE Department,
Kalasalingam University.

EDUCATION:

M.Sc. (Physics): The American college, Madurai.

M.Sc. (Engineering) and Ph.D.: Indian Institute of Science, Bangalore

EXPERIENCE:

1986-1992: Member Technical Staff, Centre for Development Of Telematics (C-DOT), Bangalore.

1994-1998: Project Engineer, Macquarie University, NSW, Australia

1999-2001: Principal Engineer, Radiata communications Pty. Ltd., NSW, Australia

2003-2006: Digital Lead, Nulife Semiconductors, Chennai

2006-2009: Senior Research Scientist, ICT centre, CSIRO, NSW, Australia

2010 : Kalasalingam University

DOMAIN KNOWLEDGE:

Solid State Electronic Devices

IC Evaluation

Solar Energy Harvesting

Semiconductor Sensor Devices

ACHIEVEMENTS:

Contribution to the World's First WLAN (802,11a) baseband processor

DEPARTMENT ACTIVITIES

❖ CAREER GUIDANCE PROGRAM

Our department organized a career guidance program on 27th July 2010 by **Dr.P.Selvaraj** .Prof. MBA Dept. who gave an enthusiastic and provoking speech that clearly guided the students to pave a path for their future.

❖ GUEST LECTURES

Our department organized three guest lecturers in the past 6 months

- The guest lecture on the topic “**EMBEDDED SYSTEMS**” was conducted on 27th July 2010.
The lecture was addressed by **Mr.Krishnakanth, The Managing Director of EI labs, Bangalore.**
- The guest lecture on the topic “**CONTROL SYSTEMS**” by **Dr.S.RajPandian**, Prof. south eastern Louisiana university, HAMMOND, USA.
- The guest lecture on the topic “**THE RECENT TRENDS IN WIRELESS TECHNOLOGY**” was conducted on 13th September 2010 by **Mr.V.Dasarathan** ,senior software engineer, **L&T, MYSORE.**

❖ STUDENT’S COUNSELLING

Our department organized a regular students counseling after 15th of every month and the counseling is carried out by **Shri.X.P.White, resource person**, Madurai.

❖ CAMPUS RECRUITMENT PROGRAM

Our department organized campus recruitment program on 25th august 2010. HR officers of TATA communication transformation services conducted this event.

❖ ELECTROCOMM 2nd edition

The second edition of our department magazine was released on 22nd April 2010 by **Dr.P.Sivaprakasm.HOD** civil department, Kalasalingam University in the presence of **Dr.P.Subbaraj**, principal, Theni kammavar institute of technology.

The edition was received by beloved parents of 2nd & 3rd year students.

❖ WIRELESS MAVERICKS

The **WIRELESS MAVERICKS** was inaugurated on 22.04.2010 and its website was released by Prof. Sudhakar Gummadi, project director, network security group tific core Kalasalingam University. The function was presided over by our beloved HoD Dr.S.Duraiaraj.

ELECTROCOMM MAGAZINE RELEASE FUNCTION – 2ND issue



Dr.Kim receiving the edition from Dr.V.Subbaraj



Hon'ble HoD addressing the gathering

INAUGURATION OF THE CLUB WIRELESS MAVERICKS



Dr.V.Subbaraj inaugurating the club



Dr.Sudhakar Gummadi addressing the students

Dr.KYUNG TAE KIM'S FAREWELL



Our beloved **HoD** honouring **Dr.Kim**



The Teaching Fertilities

DEPARTMENT INAUGURATION - ASELCOME 2010



Prof.P.SELVARAJ LIGHTING THE KUTHU VILAKKU



Dr.RAJU ADDRESSING THE GATHERING

THE LEADING BIRDS



K.PREM KUMAR
IV YEAR
CGPA: 9.061

C.KRISHNA BHARADWAJ
IV YEAR
CGPA : 9.061

S.SIVARANJANI
IV YEAR
CGPA: 9.1428



P.SRIRAM
III YEAR
CGPA-9.46



J.Praiseline karunya
III YEAR
CGPA-9.41



S.Rajeshwari
III YEAR
CGPA- 8.79



CHANDAN KUMAR
II YEAR
CGPA:9.075



SAJAG SATAYU PRASAD
II YEAR
CGPA:9.045



RUCHITA CHANDEL
II YEAR
CGPA:8.9

TOPPERS LIST

Final year		3.Chitrupaiya	- 8.5
Section A:		Second Year	
1.A. Anisha Fatima	- 8.84	Section A:	
2.V.Aishwarya	- 8.73	1.Adwiti	-8.67
3.S.Chandramathi	- 8.58	2.R.Alagianambi	-8.66
Section E:		3.C.Ashok	-8.15
1.S.Sivaranjani	- 9.31	Section B:	
2.V.Subha	- 9.09	1.Chandan kumar	-9.075
3.G.Thiyagaraja	- 8.89	2.Gaurav kumar sinha	-8.131
Third year		3.Dhanesh Raam .S	-8.1
Section A:		4.Kalayan kumar .V.P	-8.1
1.K.Arun kumar	- 8.67	Section C:	
2.S.Dei vanayagi	- 8.58	1.Keerthana Priya	- 8.80
3.Arjun R.Palaniappan	-8.57	2.Nikhil Joy	- 8.68
		3.Kushboo	- 8.64
Section B:		Section D:	
1.K.Jeneta	- 8.67	1.Sajag Satayu Prasad Singh	- 9.045
2.D.A.Jaya preethy	- 8.59	2.Ruchitha Chandel	- 8.9
3.K.Illam kuttalam	- 8.37	3.Ramparani.J	- 8.75
		4.Ruchi Bharathi	- 8.75
Section C:		Section E:	
1.S.Mohana priya	- 9.0	1.Shilpee kumari	-8.701
2.E.Lavanya	- 9.0	2.A.Subha priyadharshini	-8.6
3.S.Mathanki	- 8.73	3.V.N.Sundarsanam	-8.5750
		4 .V.Vedhanarayanan	-8.5750
Section D:		Section F:	
1.J.Praiseline karunya	-9.41	1.Kishlay chaubey	-8.65
2.S.Rajeshewari	- 8.79	2.K.Anand	-8.57
3.C.Rajbharath	- 8.76	3.Veluru Raghunath	-8.55
Section E:			
1.P.Sriram	- 9.46		
2. G.Rohini	- 9.05		
3 R.Shivashankar	- 8.97		
Section F:			
1.Timbur vijayarajeswar Sharma	- 9.02		
2.Ramdev Gotiya	- 8.6		

Staff achievement

International Conference on Intelligent Information Systems and Management (IISM'2010), June 10-12, 2010

R. Shyamala Sundari and S.B. Chitra Devi

Paper title:

Performance Comparison of Hybrid Methods Combining Clustering and Region Merging For Segmentation of Color Images.

K. Kanimozhi and J. Booma

Modeling and Small Signal Analysis of Single Phase PWM Rectifier

TECH HI-NOTES

INVITED ARTICLE

No World without water

What would the world be like without water? Despite being an invaluable resource, we often forget how dependent we are on water for our survival. The conflicts in West Asia in the 21st Century would be over water and not petroleum. That has not yet happened. West Asian politics remains dominated by Israeli aggression, Arab dictatorships and U.S. interference. The water conflicts have surfaced elsewhere — to South India. One of the reasons behind this is Global Warming. Most of the observed temperature increase since the middle of the 20th century was caused by increasing concentrations of greenhouse gases, which results from human activity such as the burning of fossil fuel and deforestation. It is melting glaciers in every region of the world, putting millions of people at risk from floods, droughts and lack of drinking water.

Glaciers are ancient rivers of compressed snow that creep through the landscape, shaping the planet's surface.

They are the Earth's largest freshwater reservoir, collectively covering an area the size of South America. Glaciers have been retreating worldwide since the end of the Little Ice Age (around 1850), but in recent decades glaciers have begun melting at rates that cannot be explained by historical trends. Projected climate change over the next century will further affect the rate at which glaciers melt. Average global temperatures are expected to rise 1.4-5.8°C by the end of the 21st century (2). Simulations project that a 4°C rise in temperature would eliminate nearly all of the world's glaciers (the melt-down of the Greenland ice sheets could be triggered at a temperature increase of 2 to 3°C). Even in the least damaging scenario – a 1°C rise along with an increase in rain and snow – glaciers will continue to lose volume over the coming century. (3).

The vast majority of all Himalayan glaciers have been retreating and thinning over the past 30 years, with accelerated losses in the last decade. For example, glaciers in the Bhutan Himalayas are now retreating at an average rate of 30-40 m per year (22). In

Central Asia, glaciers are wasting at exceptionally high rates. There are many causes for water pollution but two general categories exist: direct and indirect contaminant sources.

Direct sources include effluent outfalls from factories, refineries, waste treatment plants etc. That emits fluids of varying quality directly into urban water supplies. In the United States and other countries, these practices are regulated, although this doesn't mean that pollutants can't be found in these waters.

Indirect sources include contaminants that enter the water supply from soils/groundwater systems and from the atmosphere via rain water. Soils and ground waters contain the residue of human agricultural practices (fertilizers, pesticides, etc.) and improperly disposed of industrial wastes. Atmospheric contaminants are also derived from human practices (such as gaseous emissions from automobiles, factories and even bakeries).

"Man has lost the capacity to foresee and to forestall. He will end by destroying the earth." Albert Schweitzer
That is a pretty shocking quote, but when you study the impacts and choices the public and corporations in the US are making regarding the production and farming of food along with the pollution and additives deposited into our water you'll begin to get the point.

The causes of water contamination are numerous and range from agricultural runoff to improper use of household chemicals and everything in between. While the standard use in our society of over 75,000 different chemical compounds has offered added convenience and productivity in our lives, it has also come at a tremendous price.

In the early 1900s, before chlorine, pesticides, herbicides and the tens of thousands of other chemicals that we are exposed to, the average person had a 1 in 50 chance of getting cancer; today 1 in 3 can expect to get cancer in their lifetime, one out of every 2 men. Our municipal water treatment facilities are not designed or effective for removing these synthetic chemicals and typically only consist of sand bed filtration and disinfection, much like a standard swimming pool filter.

For the most part today's water treatment facilities are much the same as they were at the turn of the century. "Drinking water plants are old and out of date, and water supplies are increasingly threatened by and contaminated by chemicals and microorganisms." Natural Resources Defense Council. "The way we guarantee safe drinking water is broken and needs to be fixed." Carol Browner, U.S. EPA. A recent report by the Ralph Nader Study Group, after reviewing over 10,000 documents acquired through the Freedom Of Information Act, stated that "U.S. drinking water contains more than 2100 toxic chemicals that can cause cancer." The Federal Council on Environmental Quality reports that "Up to two thirds of all cancers may be attributed to these low level toxins." and that "once contaminated our ground water will remain so for tens of thousands of years. One of America's leading authorities on water contamination, Dr. David Ozonoff of the Boston University Of Public Health warns that, "the risk of disease associated with public drinking water has passed from the theoretical to the real." Many illnesses that in the past could not be linked to a probable cause can now are directly linked to toxins in our drinking water. Waterborne diseases

are caused by pathogenic microorganisms which are directly transmitted when contaminated fresh water is consumed. Contaminated fresh water, used in the preparation of food, can be the source of food borne through consumption of the same microorganisms. According to the World Health Organization, diarrheal disease accounts for an estimated 4.1% of the total DALY global burden of disease and is responsible for the deaths of 1.8 million people every year. It was estimated that 88% of that burden is attributable to unsafe water supply, sanitation and hygiene, and is mostly concentrated in children in developing countries.

The use of pesticides and herbicides has become so excessive that they are now commonly found in household tap water with alarming frequency. It is no doubt true that many of the rivers in the north of India are frequently affected by floods and discharge copious amounts of water during the monsoon months of June to September. However, no single dam or a series of dams can be expected to store this water and bring the flow under control. Also, in the case of existing major dams, instances are known where they have been held responsible for causing floods rather than controlling them! It will not be possible to control floods where the magnitude of the river water flow is of the order of 5 to 16 m above the danger level.

Instead of indulging in such fanciful schemes with disastrous consequences, it would be more sensible to encourage the traditional practice of conserving rainwater where it falls. This is the only measure which will mitigate the ill-effects of droughts over a major part of India. Before indulging in implementation of projects costing

enormous amounts of money it would be far safer and prudent to examine whether we have used available resources wisely and well. It is only after such an exercise that we can go seeking for water from neighboring states. In major dams that have been constructed, loss by evaporation is nearly 30%. Are we constructing these structures at enormous cost only to allow nearly half of the water to evaporate? In the open system of channels we have adopted for transporting water over long distances, loss through seepage and evaporation is enormous and the alternative of conveying through pipes has apparently not been considered. There are no regulatory structures and fields are literally flooded with water. There is no control over the crops to be grown in times of water deficiency. Farmers are allowed to grow more and more of sugarcane and go even for a third crop of rice! Can this be termed wise utilization of water? India is not badly off when compared to many other parts of the world with respect to water availability. The average amount of rainfall received over the plains of India is 117 cm as against the global average of only 70 cm. This annual precipitation amounts to as much as 370 million hectare metres of water which is adequate for our requirement. It is for us to manage these resources wisely and well. Government organizations from the Prime Minister downward are for linking of rivers, while rainwater harvesting, a much saner and practical proposition, has not even entered their minds. A task force has been formed to examine the modalities of this proposal for interlinking of rivers. Similar agency has not been thought of for promoting rainwater harvesting which is more urgent.

"Inter-basin transfer of water is quite common," says Chetan Pandit, Chief Engineer in the Water Resources Ministry in New Delhi, which is spearheading the move to launch this project. "It has been done in the US, Canada and China. Even within India, the Beas and Sutlej rivers have been linked and the Indira Gandhi Canal has brought water from the Sutlej (in Punjab) in the Bhakra canal to Rajasthan."

Making out a case that the project would increase irrigation and generate power to solve the country's major problems, Pandit recalled the time half a century ago when India went with a begging bowl to seek food aid. "We don't want to go back to those days," he said, claiming this (the river link scheme) was for him personally, like a "directive principle" Addressing a group of South Asian journalists at a workshop on "Water and Sanitation for the Poor" in Hyderabad last week, Pandit said the river linking scheme would irrigate 22 million hectares and generate 30 million units (kWh) of electricity.

According to the judgment of Supreme Court of India, the interlinking of rivers must be implemented so that the water that causes flood in Northern part of India can be effectively used in the southern part.

BY,
Ms.PREETHIYA,
Lect/ECE

SIXTH SENSE TECHNOLOGY



All of us are aware of the five basic senses – seeing, feeling, smelling, tasting and hearing. But there is also another sense called the sixth sense. It is basically a connection to something greater than what their physical senses are able to perceive. To a layman, it would be something supernatural. Some might just consider it to be a superstition or something psychological. But the invention of sixth sense technology has completely shocked the world. Although it is not widely known as of now but the time is not far when this technology will change our perception of the world.

Sixth sense is an extrasensory perception of information, not acquired through experience or knowledge. In short you do things that a normal human being can't. Sixth sense has evolved as a technology in the recent past that has been implemented over multiple applications.

We've evolved over millions of years to sense the world around us. When we encounter something, someone or some place, we use our five natural senses to perceive information about it; that

information helps us make decisions and chose the right actions to take. But arguably the most useful information that can help us make the right decision is not naturally perceivable with our five senses, namely the data, information and knowledge that mankind has accumulated about everything and which is increasingly all available online. Although the miniaturization of computing devices allows us to carry computers in our pockets, keeping us continually connected to the digital world, there is no link between our digital devices and our interactions with the physical world. Information is confined traditionally on paper or digitally on a screen. Sixth Sense bridges this gap, bringing intangible, digital information out into the tangible world, and allowing us to interact with this information via natural hand gestures. 'SixthSense' frees information from its confines by seamlessly integrating it with reality, and thus making the entire world your computer.

Pranav Mistry, 28 year old, of Indian origin is the mastermind behind the sixth sense technology. He invented ' Sixth Sense / WUW (Wear UR World) ' which is a wearable gestural , user friendly interface which links the physical world around us with digital information and uses hand gestures to interact with them. He is a PhD student at MIT and he won the 'Invention of the Year 2009 '- by Popular Science. The device sees what we see but it lets out information that we want to know while viewing the object. It can project information on any surface, be it a wall, table or any other object and uses hand / arm movements to help us interact with the projected information. The device brings us closer to reality and assists us in making right decisions by providing the relevant information, thereby, making the entire world a computer.

The Sixth Sense prototype consists of a pocket projector, mirror and a camera. The device is pendant shaped like mobile wearing devices. Both the projector and the camera are connected to the mobile computing device in the user's pocket. The projector projects visual information

enabling surfaces, walls and physical objects around us to be used as interfaces; while the camera recognizes and tracks user's hand gestures and physical objects using computer-vision based techniques. The software program processes the video stream data captured by the camera and tracks the locations of the colored markers (visual tracking fiducials) at the tip of the user's fingers using simple computer-vision techniques. It also supports multi touch and multi user interaction.

The device has a huge number of applications. Firstly, it is portable and easily to carry as you can wear it in your neck. The drawing application lets user draw on any surface by observing the movement of index finger. Mapping can also be done anywhere with the features of zooming in or zooming out. The camera also helps user to take pictures of the scene he is viewing and later he can arrange them on any surface. That's not it. Some of the more practical uses are reading a newspaper. Imagine reading a newspaper and viewing videos instead of the photos in the paper. Or live sports updates while reading the newspaper. The device can also tell you arrival, departure or delay time of your air plane on your tickets. For book lovers it is nothing less than a blessing. Open any book and you will find the Amazon ratings of the book. To add to it, pick any page and the device gives additional information on the text, comments and lot more add on features. While picking up any good at the grocery store, the user can get to know whether the product is eco friendly or not. To know the time, all one has to do is to just gesture drawing circle on the wrist and there appears a wrist watch. The device serves the purpose of a computer plus saves time spent on searching information. Currently the prototype of the device costs around \$350 to build. Still more work is being done on the device and when fully developed, it will definitely revolutionize the world

**BY,
S.NITHYA DEVI,
III-D .**

NANOMETER CMOS TECHNOLOGY:

The 11 nanometer (11 nm) node is the technology node following 16 nm node. The exact naming of this technology node comes from the International Technology Roadmap for Semiconductors (ITRS). According to the 2007 edition of this roadmap, by the year 2022, the half-pitch (i.e., half the distance between identical features in an array) for a DRAM should be 11 nm, although Intel's "Architecture and Silicon Cadence Model" places its 11 nm node closer to the year 2015. Intel's Pat Gelsinger claims that Intel sees a 'clear way' towards the 11 nm node. At the 11 nm node in 2015, Intel expects to use a half-pitch of around 21 nm. Nvidia's chief scientist, William Dally, claims that they will also reach 11 nm semiconductors by 2015, a transition he claims will be facilitated principally through new electronic design automation tools. How the use of such design tools will help Nvidia overcome the physical limitations of CMOS technology and conventional lithography is unclear. This design rule is likely to be realized by doublepatterning.

While the roadmap has been based on the continuing extension of CMOS technology, even this roadmap does not guarantee that silicon-based CMOS will extend that far. This is to be expected, since the gate length for this node may be smaller than 6 nm, and the corresponding gate dielectric thickness would scale down to a monolayer or even less. Reported estimates indicate that transistors at these dimensions are significantly affected by quantum tunnelling. As a result, non-silicon extensions of CMOS, using III-V materials or nanotubes/nanowires, as well as non-CMOS platforms, including molecular electronics, spin-based computing, and single-electron devices, have been proposed. Hence, this node marks the practical beginning of nanoelectronics.

Due to the extensive use of ultra-low-k dielectrics such as spin-on polymers or other porous materials, it is unlikely conventional lithography, etch or even chemical-mechanical polishing processes can continue to be used, because these materials contain a high density of voids or gaps. At scales of ~10 nm, quantum tunneling, especially through gaps, becomes a significant phenomenon. Controlling gaps on these scales by means of electromigration can produce interesting electrical properties themselves.

**BY,
C.RAJ BHARATH,
III – D.**

360 DEGREE DEFENSE

Attacks to the body may include overhead strikes, straight and hook punches, and upward attacks.

The Seven 360 Defense Positions

In these positions, the blocking surface is the blade of the forearm from the fingertips to the elbow, with fingers extended. Put weight behind the defense with a slight chopping motion outward to meet the attack, instead of just leaving the arms stationary in position. Attack the attack.

The seven positions for the arms are:

1. Arms raised overhead, crossing forearms, forming a triangle. The biceps are close to ears, and chin tucked in slightly to ensure the head is within the triangle. Position 1 defends against overhead attacks.
2. Arms bent at 90-degree angle, upper arms are 45 degrees from the ground, palms facing out. Position 2 defends against attacks

- on a downward diagonal attack to the head.
3. Arms bent at 90-degree angle, with biceps parallel to the ground, forearms perpendicular to the ground, palms facing slightly out. Position 3 defends against horizontal attacks from the side, aimed at the head, such as a hook punch, slap, and roundhouse kick.
 4. Elbows drawn against your sides with fingers pointed upward and back of hands just in front of shoulders. Position 4 defends against attacks to the ribs or midsection.
 5. Arms bent at 90-degree angle, biceps parallel to ground, forearm at either side of the torso perpendicular to the ground, fingertips pointing downward, palms facing back. Position 5 defends against attacks to ribs and midsection.
 6. Arms bent at 45-degree angle, hands in front of hips and slightly forward, elbows out. Bend body slightly forward from hips (not knees). Position 6 blocks against diagonally upward attacks towards the upper body.
 7. Starting with position 6, place one forearm on top of the other, elbows out to sides. Lean forward at the hips, with hips as far back as possible. Position 7 blocks attacks to the groin and solar plexus and upward attacks, such as kicks.

**BY,
RAM BALAJI .T.R.S,
RAM PRASAD .D,
III -D.**

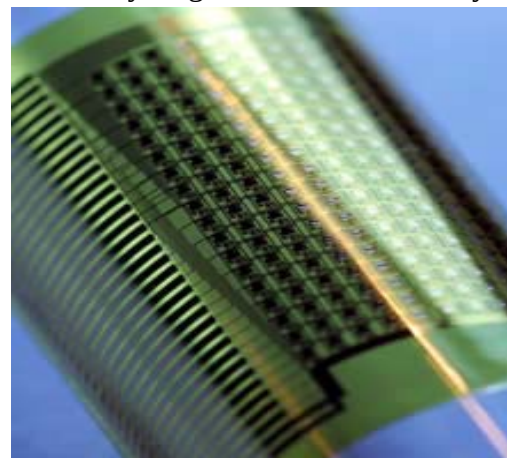
A BENDABLE HEART SENSOR

New flexible electronics can better chart the heart's electrical activity.

A new flexible and biocompatible electronic device can produce a more detailed picture of the electrical activity of a beating heart. This high-resolution electrical map could help improve the diagnosis and treatment of heart abnormalities by pinpointing areas of damage or misfiring circuitry.

Today, the best way to map the electrical activity of a person's heart is to insert a probe tipped with a few electrical sensors through a vein and into the heart. The probe is used to measure activity at different locations in the tissue to slowly build up a picture of electrical activity. An electrocardiogram, which picks up signals from outside the body, offers a less precise picture.

"It can take hours to map where these heart rhythms are coming from," says Brian Litt, a neurologist and biomedical engineer at the University of Pennsylvania, and one of the senior researchers on the project. "If you map at a very high resolution, it may be



possible that you can pick up, in local areas, precursors to arrhythmias before they occur."

The flexible device can be used to attach multiple sensors to the wall of a beating heart, measuring electrical activity at multiple sites despite the heart's movement. The electronics needed to record this activity are also built into the device, meaning more data can be collected. The new device is 25 microns thick and covers 1.5 square centimeters. It contains over 2,000 transistors sealed within a flexible coating and is covered with 288 sensor electrodes. So far the device has been tested successfully in pigs.

It is the first time that flexible electronics have been used in such high density in a medical device, says John Rogers, an engineer at the University of Illinois, Urbana-Champaign, who collaborated on the work. "These devices contain more transistors than any previously reported flexible device," he says.

The flexible device could be used in other kinds of biological sensors, says Litt, including devices for monitoring neurological conditions such as Parkinson's and epilepsy.

The key to making the device is what Rogers calls "nanomembrane transistors." These components are made out of thin ribbons of silicon, about 100 nanometers thick; on this scale the material loses its characteristic rigidity and becomes flexible. "It's much like a piece of two-by-four," Rogers says. "Wood is not very bendable, but a sheet of paper is."

Making these transistors required a completely new fabrication technique. Rogers etched out the ribbon circuits from larger blocks of silicon and then used chemical etching to remove silicon from underneath. The circuits could then be peeled away when brought into contact with a stamp-like device.

Once the circuit has been deposited on a substrate, it is encased in a photocurable water-tight epoxy material. This material was difficult to develop, since it had to have the same mechanical properties as the circuit in order to bend with it, but also needed to be resilient enough to prevent any seepage, even at points where the electrodes protrude. "It probably took us half a year to develop a recipe for that," says Rogers.

The next step, says Litt, is to build a power supply into the device so that it can be used for chronic implantation, and to find a way to transmit data from it wirelessly. The researchers are also developing a version that can also be used to ablate damaged heart tissue through localized heating.

**BY,
RAM BALAJI .T.R.S
RAM PRASAD .D
III -D**

AERODYNAMIC:

Aerodynamic" redirects here. For other uses, see Aerodynamic (disambiguation).

Aerodynamics is a branch of dynamics concerned with studying the motion of air, particularly when it interacts with a moving object. Aerodynamics is a subfield of fluid dynamics and gas dynamics, with much theory shared between them. Aerodynamics is often used synonymously with gas dynamics, with the difference being that gas dynamics applies to all gases. Understanding the motion of air (often called a flow field) around an object enables the calculation of forces and moments acting on the object. Typical properties calculated for a flow field include velocity, pressure, density and temperature as a function of position and time. By defining a control volume

around the flow field, equations for the conservation of mass, momentum, and energy can be defined and used to solve for the properties. The use of aerodynamics through mathematical analysis, empirical approximation and wind tunnel experimentation form the scientific basis for heavier-than-air flight.

Aerodynamic problems can be identified in a number of ways. The flow environment defines the first classification criterion. External aerodynamics is the study of flow around solid objects of various shapes. Evaluating the lift and drag on an airplane or the shock waves that form in front of the nose of a rocket are examples of external aerodynamics.

Internal aerodynamics is the study of flow through passages in solid objects. through a jet engine or through an air conditioning pipe.

The ratio of the problem's characteristic flow speed to the speed of sound comprises a second classification of aerodynamic problems. A problem is called subsonic if all the speeds in the problem are less than the speed of sound, transonic if speeds both below and above the speed of sound are present (normally when the characteristic speed is approximately the speed of sound), supersonic when the characteristic flow speed is greater than the speed of sound, and hypersonic when the flow speed is much greater than the speed of sound. Aerodynamicists disagree over the precise definition of hypersonic flow; minimum Mach numbers for hypersonic flow range from 3 to 12.

The influence of viscosity in the flow dictates a third classification. Some problems may encounter only very small viscous effects on the solution, in which

case viscosity can be considered to be negligible. The approximations to these problems are called inviscid flows. Flows for which viscosity cannot be neglected are called viscous flows.



A vortex is created by the passage of an aircraft wing, revealed by smoke. Vortices are one of the many phenomena associated to the study of aerodynamics. The equations of aerodynamics show that the vortex is created by the difference in pressure between the upper and lower surface of the wing. At the end of the wing, the lower surface effectively tries to 'reach over' to the low pressure side, creating rotation and the vortex.

Compressible aerodynamics:

According to the theory of aerodynamics, a flow is considered to be compressible if its change in density with respect to pressure is non-zero along a streamline. This means that - unlike incompressible flow - changes in density must be considered. In general, this is the case where the Mach number in part or all of the flow exceeds 0.3. The Mach .3 value is rather arbitrary, but it is used because gas flows with a Mach number below that value demonstrate changes in density with respect to the change in pressure of less than 5%. Furthermore, that maximum 5% density change occurs at the stagnation point of an object immersed in the gas flow and the density changes around the rest of

the object will be significantly lower. Transonic, supersonic, and hypersonic flows are all compressible.

Transonic flow:

The term Transonic refers to a range of velocities just below and above the local speed of sound (generally taken as Mach 0.8–1.2). It is defined as the range of speeds between the critical Mach number, when some parts of the airflow over an aircraft become supersonic, and a higher speed, typically near Mach 1.2, when all of the airflow is supersonic. Between these speeds some of the airflow is supersonic, and some is not.

Supersonic flow:

Supersonic aerodynamic problems are those involving flow speeds greater than the speed of sound. Calculating the lift on the Concorde during cruise can be an example of a supersonic aerodynamic problem.

Supersonic flow behaves very differently from subsonic flow. Fluids react to differences in pressure; pressure changes are how a fluid is "told" to respond to its environment. Therefore, since sound is in fact an infinitesimal pressure difference propagating through a fluid, the speed of sound in that fluid can be considered the fastest speed that "information" can travel in the flow. This difference most obviously manifests itself in the case of a fluid striking an object. In front of that object, the fluid builds up a stagnation pressure as impact with the object brings the moving fluid to rest. In fluid traveling at subsonic speed, this pressure disturbance can propagate upstream, changing the flow pattern ahead of the object and giving the impression that the fluid "knows" the object is there and is avoiding it. However, in a supersonic flow, the pressure disturbance cannot propagate upstream. Thus, when the

fluid finally does strike the object, it is forced to change its properties -- temperature, density, pressure, and Mach number -- in an extremely violent and irreversible fashion called a shock wave. The presence of shock waves, along with the compressibility effects of high-velocity (see Reynolds number) fluids, is the central difference between supersonic and subsonic aerodynamics problems.

Hypersonic flow

In aerodynamics, hypersonic speeds are speeds that are highly supersonic. In the 1970s, the term generally came to refer to speeds of Mach 5 (5 times the speed of sound) and above. The hypersonic regime is a subset of the supersonic regime. Hypersonic flow is characterized by high temperature flow behind a shock wave, viscous interaction, and chemical dissociation of gas.

Associated terminology

The incompressible and compressible flow regimes produce many associated phenomena, such as boundary layers and turbulence.

Boundary layers

The concept of a boundary layer is important in many aerodynamic problems. The viscosity and fluid friction in the air is approximated as being significant only in this thin layer. This principle makes aerodynamics much more tractable mathematically.

Turbulence

In aerodynamics, turbulence is characterized by chaotic, stochastic property changes in the flow. This includes low momentum diffusion, high momentum convection, and rapid variation of pressure and velocity in space and time. Flow that is not turbulent is called laminar flow.

Aerodynamics in other fields

Aerodynamics is important in a number of applications other than aerospace engineering. It is a significant factor in any type of vehicle design, including automobiles. It is important in the prediction of forces and moments in sailing. It is used in the design of large components such as hard drive heads. Structural engineers also use aerodynamics, and particularly aeroelasticity, to calculate wind loads in the design of large buildings and bridges. Urban aerodynamics seeks to help town planners and designers improve comfort in outdoor spaces, create urban microclimates and reduce the effects of urban pollution. The field of environmental aerodynamics studies the ways atmospheric circulation and flight mechanics affect ecosystems. The aerodynamics of internal passages is important in heating/ventilation, gas piping, and in automotive engines where detailed flow patterns strongly affect the performance of the engine

**BY,
VANISHREE.A,
II-E.**

TECHNOLOGY AND HUMAN COMMUNICATION:

Human beings are said to be social animals. Communication is the foundation on which societies can be built as it allows organisms to interact productively. The term is defined as a process that allows entities to exchange information by several methods. In simpler words, communication is the act or process of exchanging ideas, messages or information, as by speech, signals or writing. The quality of communication is marked by its effectiveness.

In today's world, it seems imperative to have effective communication skills for the success of personal relationships and to

carve a niche in the professional field as well. However, successful communication is of one the toughest battles for many today.

The two fundamental methods of communication are verbal and non-verbal. Auditory means such as speaking and singing represent verbal communication; whereas physical means such as body language, paralanguage, touch, eye contact and the use of writing signify the category of non-verbal communication. Language is the backbone of verbal communication and written text as well. Languages can be described as a system of symbols and grammars rules by which the symbols are manipulated. Non-verbal communication accompanies and supplements verbal communication. The medium of communication could be any, what matters is that the information transmitted can be received and interpreted correctly.

Fascination for the subject by scientists has transported the process of communication beyond human beings or any living creatures. Instead, today technological advancement has made communication possible between machines or parts of machines. In the 20th century, a revolution in telecommunications has greatly altered communication by providing new media for long distance communication. Analog telecommunications include traditional telephony, radio and television broadcasts and digital telecommunications symbolize the computer-mediated communication, telegraphy and computer networks.

The progression in technology has opened many new avenues of communication. We live in the communication world of cellular phone, faxes, video conferences, internet connection and many more. Gone are the days of the traditional landline telephones, today mobile phones are the trend setters. A mobile phone is a long range electronic device used for mobile voice or data communication over a network of specialized base stations known as cell sites. In addition to the standard voice function,

current mobile phones support many additional services and accessories such as SMS for text messaging, email, access to the internet, games, blue tooth, camera with video recorder, MMS for sending and receiving photographs and video, MP3 player, radio and GPS. Fax or facsimile is a telecommunication technology used to transfer copies of documents, especially using affordable devices operating over the telephone network. Fax has enabled the instantaneous transfer of documents to people at large distances. The electronic mail or email is one of the most extensively used internet application in terms of communication. It is the method of exchanging digital messages via the computer. All one requires are a personal email account and the recipient's email address, and then one can communicate with any one in the world.

Modern communication media has allowed intense long distance exchanges between larger numbers of people. The combining of technology and communication brings immediacy, integration, access and collaboration. Living in a world of instant results, many of these new technological tools have become life savers for day to day activities. Undoubtedly, technology has taken human communication to just a push of a button!

ADVANTAGES OF EMAIL COMMUNICATION:

Communication is the process of transferring information from one source to another. The term is commonly defined as 'the imparting or interchange of thoughts, opinions or information by speech, writing or signs'. Communication can be perceived as a two-way process, wherein there is an exchange and progression of thoughts, feelings and ideas towards a mutually accepted goal or direction.

Communication can be broadly categorized into verbal and non-verbal communication. In the 21st century, there is hardly any person between the age of 15 and

65, who is not familiar with email communication. Electronic mail, often abbreviated as email, is a method of exchanging digital messages, designed primarily for human use. A message at the very least consists of its content, an author address and one or more recipient addresses. Email communication falls under the group of mediated communication that includes messages that pass through a channel outside of the sender and receiver. This category of communication comprises of traditional mass media (print, radio, television and film) as well as digital devices (smart phones, MP3 players and computers). With the advancement of technology, media has become much more integrated and it is essential for communicators to comprehend its usage properly.

Email is the most widely used internet application. There are various strengths of using email communication. An email contains all details of correspondence and one does not require relying on memory for facts of interaction. If one has a habit of never deleting project related emails, then the contents can be recalled as per requirement. While communicating through email, one can edit and rephrase sentences as many times before sending it to the recipient, thus safeguarding the sender from incorrect grammar or embarrassing statements. This aspect of emailing works to the advantage of especially those who are less gifted with articulation skills. Email also provides time stamped proof of an interaction..

Email is a relatively inexpensive means of communication. It allows one to send messages to a large number of recipients simultaneously at a price that is much lesser than traditional postal rates. All one requires is access to the internet and the recipient's email address. These two things can render you the capability of communicating with anyone in the world. A very important advantage of email communication lies in the level of convenience that it offers. At any time of the day or night, one can communicate with

friends, relatives, professors and business associates. This ease increases the communication between the concerned parties. It is a known fact that these days college administrators prefer email communication because it is much cheaper and more efficient than traditional mail. A newsletter mailed to two thousand students would cost approximately 400 dollars to print and 680 dollars in postage. Labeling and stuffing envelopes would take twelve working hours. This same message could be sent by email to multiple recipients in less than an hour with no printing or postage expense.

An email address has a linked permanency factor. It is more permanent than a telephone number or a mailing address and it stays in place even while transiting. Apart from just communication between two parties, an email account offers assistance in many other areas as well. One can apply for online scholarships or grants; telephone, electricity and various other bills can be paid online; bank statements and tax refunds also require an email account..

**BY,
P.DIVYA,
II –B.**

GO GREEN! - WITH ELECTRONICS

When you buy a new cell phone, computer or printer, you may be excited to check out all of its new features. However, even after you hook it up and start using it, you're still left with an old gadget you will likely never use again. Though it may be easy to throw it away in the garbage, doing so not only creates excess waste but also introduces corrosive chemicals into landfills, which can seep into the ground. To prevent harming the environment, consider recycling your electronics instead.

Yes, electronic devices are becoming a bigger and bigger part of our lives, especially as they get smaller and smaller. We use them as tools and toys to communicate, work, enjoy media, and be expressive. Being green with electronics doesn't mean listening to truckers squawk on the old short-wave. Greening your electronics is a matter of knowing what tech to get, how to use it best, and what to do with it when its useful life is done.

One of the best and most effective ways to go green is by making eco friendly adjustments to our home. It doesn't have to be anything drastic, but every little change makes a difference. The best part is that not only are there tons of different ways to make our home more energy efficient, but an eco friendly home saves us a lot of money in the long run.

There are plenty of reasons to recycle old electronics. The Environmental Protection Agency (EPA) estimates that more than 40 million computers become obsolete and are discarded every year. However, only 15 percent of these computers are recycled, which means that 85 percent of them end up in landfills across the country. In addition to computers, millions more cell phones, computer peripherals and audio equipment end up in dumps. These electronic devices contain dangerous chemicals, including mercury, which can contaminate the earth.

There are two main ways we can recycle electronics: donating and repurposing. Donating is a fairly popular method for recycling telephones and computers, as many underprivileged people and communities around the world are in need of these technology basics. Repurposing, however, is good for older electronics, as it takes the components from the electronic devices, including metals, wires and battery parts, and turns them into new devices. There's no better time to start going green on the road. Electric and hybrid cars are being thrust into the spotlight more than ever before.

If you are trying to recycle cell phones or computers, finding a recycling centre is

fairly easy. Most major cell phone carriers also serve as cell recycling centres, and select electronics stores (especially repair shops) will take old computer hardware for repurposing. After you have found a recycling centre, it is important to prepare your device for recycling. Although rare, identity thieves have been known to reboot items discarded for recycling. So before you recycle it, make sure to wipe the device's memory completely and drain the battery to 0 percent.

Many online social communities like Green Bin bring together those devoted to making a conscious effort for the environment. Efforts can be made by joining websites or local organizations to share ideas and generate new program ideas, or form a focus group on one specific issue and take it to a general public meeting at city hall. Becoming involved first hand is one of the best ways to contribute to a positive change throughout the year and for years to come.

Ending with the famous quotes of 'Max Gluckman' -

"A science is any discipline in which the fool of this generation can go beyond the point reached by the genius of the last generation."

BY,
Ruchita Chandel,
II-D

HOW TELEPHONES WORK

Although most of us take it completely for granted, the telephone you have in your house is one of the most amazing devices ever created. If you want to talk to someone, all you have to do is pick up the phone and dial a few digits. You are instantly connected to that person, and you can have a two-way conversation.

The telephone network extends worldwide, so you can reach nearly anyone on the planet. When you compare that to the state of the world

just 100 years ago, when it might have taken several weeks to get a one-way written message to someone, you realize just how amazing the telephone is!



Telephones are actually one of the simplest devices in your house.

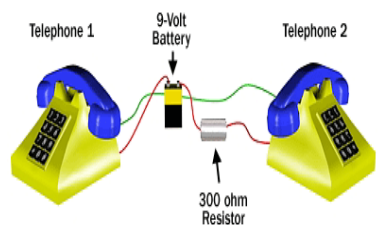
Surprisingly, a telephone is one of the *simplest* devices you have in your house. It is so simple because the telephone connection to your house has not changed in nearly a century. If you have an antique phone from the 1920s, you could connect it to the wall jack in your house and it would work fine!

In this article, we will look at the telephone device that you have in your house as well as the telephone network it connects to so you can make and receive calls. See the next page to get started.

Creating our own Telephone Network

Not only is a telephone a simple device, but the connection between you and the phone company is even simpler. In fact, you can easily create your own intercom system using two telephones, a 9-volt battery (or some other simple power supply) and a 300-ohm resistor that you can get for a dollar at Radio Shack. You can wire it up like this:

Your connection to the phone company consists of two copper wires. Usually they are red and



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green. The green wire is common, and the red wire supplies your phone with 6 to 12 volts DC at about 30 milliamps. If you think about a simple carbon granule microphone, all it is doing is modulating that current (letting more or less current through depending on how the sound waves compress and relax the granules), and the speaker at the other end "plays" that modulated signal. That's all there is to it!

The easiest way to wire up a private intercom like this is to go to a hardware or discount store and buy a 100-foot phone cord. Cut it, strip the wires and hook in the battery and resistor as shown. (Most cheap phone cords contain only two wires, but if the one you buy happens to have four, then use the center two.) When two people pick up the phones together, they can talk to each other just fine. This sort of arrangement will work at distances of up to several miles apart.

**BY,
P.DIVYA,
II –B.**

CELL PHONE TOWERS DON'T RAISE CANCER RISK:



A recent study by British researchers at Imperial College London's School of

Public Health has found that, despite widespread concern over the safety of cell phones, children born to mothers who lived near cell phone towers while pregnant do not have an elevated risk of cancer.

The team examined nearly 2,000 cases of childhood cancer in Britain between 1999 and 2001, and found that there was no correlation between how close their mother lived to cell phone towers and incidence of cancer.

Various cancers were examined, including cancers of the brain and central nervous system, leukemia, and non-Hodgkin's lymphoma. The researchers concluded, "There is no association between risk of early childhood cancers and estimates of the mother's exposure to mobile phone base stations during pregnancy." The study was published in the British medical journal

Despite the alarm and media scares, repeated studies have found no link between cell phone use and disease. In fact, as physicist S.T. Lakshmi Kumar noted in skeptical inquirer magazine the power in a standard cell phone would not be enough to cause problems: "The small amount of power being transmitted by the phone is traveling several kilometers to the tower.

Also, the cell phone has to transmit this very little power in all directions. The small power in the direction of the tower passes through several walls and other obstructions, even people, without impeding the communication"

Thus, cell phone signals are everywhere—and probably passing through you as you read this. If the signals were carcinogenic, nearly everyone would have cancer.

There's no question that cell phones can be dangerous. Many scientific, reputable

studies have proved that. In fact, cell phone use has been banned in many states for exactly this reason. But the chief threat of cell phones is the distraction they cause, not the electromagnetic fields they emit..

**By,
N.Nishanth,
III-D**

ELECTRONIC WASTE PRODUCES ALGAE FOR BIOFUEL

A different way to recycle old computers



Fig: An old Apple G4 tower, an Apple iMac CRT and a modified Dell Latitude CPX laptop are just a few of the electronics recycled for this algae-growing tank.

When you think of recycling electronics, no doubt you imagine the old PC or mobile phone being disassembled, and it's metal and plastic parts melted down to be repurposed. But for some people, it means reusing the parts to grow algae.

Students at the University of Illinois at Urbana-Champaign created Bio-Grow, a device made from various computer parts that serves as a reservoir to cultivate algae.

The algae can then be used in biodiesel production, which could potentially replace petroleum in the future.

"If someone had one of these in their homes, they would cultivate algae and extract it," said team member and undergraduate student Megan Kenney.

"Then they could take it into a gas company that was set up with an oil filtration facility and get credit off their gas."

Kenney, along with undergraduate students Timothy Harvey, Elliot DeVries and Mark Schnitzer, and graduate student Saeidreza Shiftehfar was focused on finding a way to be green, and ended up winning second place in the International Electronic Waste Competition at the University of Illinois at Urbana-Champaign with their design.

"The whole point of our project was two different concepts," Schnitzer said. "To use electronic waste and to solve a green issue in the world." This is why the team chose to use old electronics to build the device called an algae bioreactor. It encourages photosynthesis, the chemical reaction that happens in plants, which uses sunlight to convert carbon dioxide into sugar.

"By using e-waste, you are giving a second life to all these electronics," Kenney said.

The algae-growing tank was made from the side panels of an Apple G4 CPU tower, with PVC pipes and acrylic panels for structural support. The team used an Apple iMac CRT to emit the light and heat the algae needs to grow. The entire structure had to be sealed and housed within an outer cowell made of high-density foam, which provides stability as well as insulation.

A modified Dell Latitude CPX laptop was programmed to monitor and control the iMac CRT so that it would turn on a specific light spectrum at different intervals of time and adjust the temperature within the tank.

"Algae's best growth factors are within the red and blue spectrums of light at a ratio of four to one," Kenney said. "We

also knew that it needed to be 62 to 82 degrees.”

The tank also has a water pump, which aerates the algae and provides it with the maximum exposure to sunlight. A faucet allows the user to extract the algae.

The process of creating Biofuel from algae is complex and expensive and to date remains in research labs. But the team hopes this project will bring Biofuel down to the household level, which would drive up production and lower costs.

“We are imagining this product will eventually become part of a larger system,” Harvey said. “You won’t just have it in your house, you would take the algae to a biomass collection point, at which point it would be transported to a refinery.”

The refinery would then extract the lipids from the algae, which is up to 50 percent of the algae’s components to create biodiesel. The byproducts can be used for feedstock, fertilizer and high-end pharmaceuticals because algae are so rich in protein and nutrients.

The teams’ calculations showed that if just 6.5 percent of Americans had one of these in their homes, it could generate the amount of algae needed to replace petroleum with biodiesel.

And algae growers would make money, too. Kenney said the algae could be sold for a dollar a gallon, and because algae grow so fast, it could be harvested every three days.

If this device becomes commonplace like the team is hoping, perhaps the vases that adorn end tables in homes across the nation will be replaced by an algae-growing tank which could produce gas for the cars of the future.

By,
N.Nishanth,
III-D.

RADIOACTIVE :



Radioactive batteries could not only power electronics for months or even years, but they can also help to put radioactive substances to good use. Minuscule amounts of radioactive elements could safely power handheld electrical devices for months or even years, according to new research from the University of Missouri. Such electronics could help to eliminate mildly radioactive substances while providing cheap and reliable power.

"This is not a nuclear reaction. Nothing will explode; there won't be a mushroom cloud on your property," said Jae Kwon, a professor at the University of Missouri developing radioactive batteries.

"We are using the radiation coming from the material and turning that into electricity," he added. The material in question is sulfur 35, an energy packed isotope of an element most people associate with the smell, and color, of rotten eggs. In nature, sulfur 35 forms when cosmic rays break apart argon in the upper atmosphere. Within a battery, sulfur 35 spits out mildly radioactive electrons as it decays. To harvest those electrons and convert them into electricity, Kwon surrounds the sulfur with selenium, a liquid semiconductor. The selenium traps the electrons trying to escape, channeling them into a source of electricity. Less than one milligram of sulfur 35 is contained in a package

smaller than a penny. Using such a small amount helps to ensure that the levels of radiation are small, but also limits the amount of power that can be produced.

"We are talking about nano watts of power here," said Kwon. "That power level can be increased if we use a material with a higher energy density."

Uranium or other materials that emit either more radiation or more energetic radiation should boost the power supply. Kwon is currently studying other potential power sources, but won't specify which ones. Although the half-life of sulfur 35 is only 87 days, as opposed to the more than 700 million-year half-life of some uranium isotopes, scientists wouldn't want even mildly radioactive material like sulfur 35 to leak into the environment where it could cause harmful mutations in animals or cancer in humans. However, Kwon is quick to downplay such concerns. Kwon notes that current AA and AAA batteries need a special disposal system. A similar system could be developed for radioactive batteries as well. "All the technology exists to create a radioactive battery," said Mehregany. "It just needs to be put in the right package."

**By,
G.D.Nivaas,
III-D.**

Is 4G Better Than 3G?



3G phones were first introduced in 2001, but it wasn't until six years later, with the introduction of Apple's iPhone in 2007, that 3G exploded.

But while many of us are just growing comfortable with our 3G smart phones which let us access the Internet, all the major mobile operators in the United States are moving to build out 4G networks.

Why? And is 4G better than 3G? Let's go back and look at where we came from:

1G, the first generation of mobile phones transmitted sound using an analog signal, a continuous signal able to reflect the variations in loudness of a person's voice.

The signals lacked range, however, required huge batteries, were susceptible to interference and were unable to accommodate security encryption codes, meaning anyone could listen to your phone call with the right equipment, or worse, clone your telephone number and run up your phone bill. The second generation, or 2G, of mobile phone was introduced in 1991, representing a significant leap forward in technology because they used digital signals. These are discontinuous signals that transmit voice information as electrical pulses representing ones and zeros. Digital signals can be encoded with security and because they took up less bandwidth, they required less power to operate, which meant smaller batteries. The costs of phones started to decline. 2G also heralded the beginning of SMS (texting) and email over a mobile phone. The phone companies loved the new digital signaling because they were able to jam more conversations into the same amount of radio bandwidth. As people began to adopt the use of the 2G cell phones, the demand began to exceed the

capacity of the phone companies to provide enough tower coverage both in the urban, suburban and especially rural settings. The next generation in mobile telephony, defined by the International Telecommunication Union was 3G, heralded in a family of standards for mobile telecommunications called IMT-2000. The standards improved how wide-area wireless voice, video and data were transmitted over the radio spectrum, making it more efficient. These phones were an improvement over 2G phones because they allowed the mobile operators to transmit more calls. The consumer was able to simultaneously use voice and data services and, for the first time, was provided enough bandwidth to use a built-in browser to surf the Internet. 3G provided substantially more upload and download capacity for surfing the Internet, viewing videos on YouTube and sending enhanced text messages that included images. The newest and fourth generation of mobile phones, 4G, is expected to provide users with an all-Internet Protocol solution, which will exponentially increase bandwidth. This will allow simultaneous use of voice, Internet access gaming and streamlined multimedia services such as TV and video streaming services. 4G has two standards for transmitting traffic: Long Term Evolution, or LTE, and Worldwide Interoperability for Microwave Access, more commonly known as Mobile WiMAX. Billions of dollars are being spent on both solutions, and the struggle of one to win out over the other may become analogous to the race between video tape recording standards VHS and Beta during the late 1970s and early 1980s. Only the winning technological approach will assure the survival of one company over another. Both LTE and

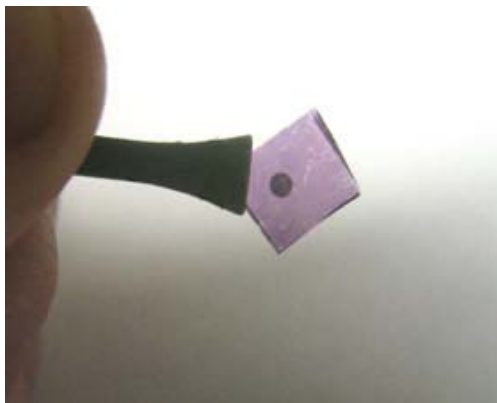
Mobile WiMAX promise unprecedented download speeds that blow current 3G cellular networks out of the water. So to answer the question posed earlier: Yes, 4G is better than 3G by a landslide.

But who will win out is still up in the air. Sprint Nextel and Comcast have placed their bets on Mobile WiMAX. In March, Sprint Nextel released the first 4G phone, Sprint Nextel has, or will have, 4G coverage in 30 cities by the end of 2010. For now, they're the only mobile operator using the WiMAX standard, which could prove to be a boost or a deathblow for this struggling company.

Comcast has deployed its own 4G networks for laptop users built on top of the same WiMax network Sprint is using, but there are no present plans for it to offer mobile phone devices. Verizon is deploying a 4G network based on the LTE standard in 30 cities but a 4G Verizon phone will not be available until 2011. AT&T plans to roll out a nationwide 4G network by 2011 using the LTE standard with some test cities up and running by the end of 2010. T-Mobile has recently upgraded its network to "3.5G," but has plans to deploy a 4G network by 2011 also using the LTE standard. So the race is on, and there is bound to be substantial consumer confusion as 3G and 4G solutions begin to compete with each other. As with all new technologies, the early adoptees will be people who enjoy being on the cutting edge of new technology. There are rumors that Apple will announce a 4G iPhone in May, and this may be what it takes to get consumers to jump to 4G.

**BY,
G.D.Nivaas,
III-D.**

World's First Light-Powered Circuit Created



THE GIST:

- Scientists have created a self-powered circuit.
- The circuit runs on solar power.
- The new technology could have a range of applications from touch screen technology to modeling the human brain.

For the first time, scientists have created a circuit that can power itself, as long as it's left in a beam of sunshine. Created by scientists from the University of Pennsylvania, the world's first photovoltaic circuit could eventually power a new line of consumer devices or even model the human brain. "This has the potential to create a new generation of optical and electronic devices," said Dawn Bonnell, a scientist from the University of Pennsylvania who co-authored a recent *ACS Nano* paper describing the research. "The touch screen of your computer could act as both the electrical charger and the computer chip."

Right now Bonnell and her colleagues can only coax minuscule amounts of electricity from their photovoltaic circuits, far too little to power consumer

electrical devices. Those amounts could quickly skyrocket.

"We would have one amp with one volt in a sample the diameter of a human hair and an inch long," said Bonnell. "If the efficiency scaled up without any additional limits."

There are plenty of other ways Bonnell can squeeze more electricity from light. Right now only about 10 percent of the photovoltaic circuits on a glass slide work. Increasing that number will boost the power output.

The photovoltaic circuit developed by Bonnell and her colleagues is a scientific breakthrough, not a technological one. These new circuits will most likely never replace their silicon counterparts.

Photovoltaic circuits could be ideal for other applications, however, such as powering tiny robotic devices or running computer calculations at the speed of light.

Far into the future, these circuits could even be used to set up as the artificial neural networks that could model the brain.

At their most basic, computers represent data as on or off, a "0" or a "1." Using light instead of electrons, these photovoltaic circuits could store data from, say, one, two, three or four. Each number would correspond to a certain wavelength or color of light -- red, green, blue and yellow, for example. To model the human nervous system, each color of light could correspond to a different neurotransmitter, say red for dopamine and blue for serotonin.

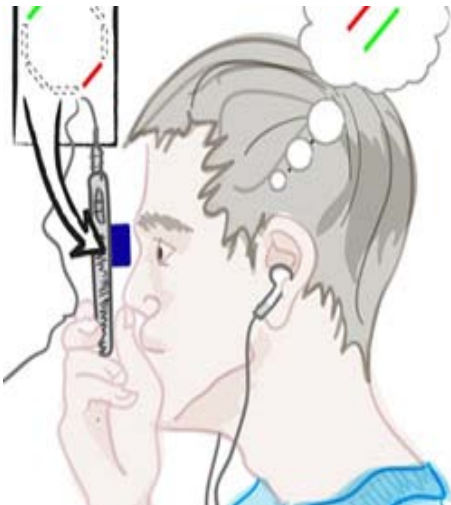
"This could open the door for many kinds of new devices," agrees Lukas Novotny, a scientist at the Institute of Optics at the University of Rochester. Novotny notes that right now the circuit turns light into energy. He wonders if the circuit could turn energy into light,

creating an entirely new class of lighting materials.

The potential applications of the technology are huge, but will take years to develop into any kind of practical equipment.

**By,
G.D.Nivaas,
III-D.**

CELL PHONE APPLICATION OFFERS EYE PRESCRIPTIONS



A quick photograph with your camera phone can now diagnose astigmatism and give you your contact prescription.

Two minutes, two dollars and a cell phone will soon be all it takes to get an eyeglass prescription.

Scientists from the Massachusetts Institute of Technology have developed a new app for smart phones that, when combined with a thin, cheap lens, could give inexpensive and accurate eyeglass prescriptions to smart phone owners. The research could help millions of people around the world see better for less money.

"Most ophthalmologic devices range from simple eye charts to expensive auto refractors," said Ramesh Raskar, a scientist at MIT who will present the app at the upcoming SIGGRAPH 2010 Conference.

"We are somewhere in between. Here you just use your cell phone and a cheap optical plate to align a series of patterns."

The combined app and screen works like an old, analog camera with a manual focus lens. Unlike today's digital cameras, which focus automatically, the new app and screen requires the user to bring patterns into focus.

After loading the application, a user attaches a short, conical viewfinder (about \$2 now, although Raskar thinks that will soon drop to 50 cents) to the screen of their high-resolution cell phones and peers in. A series of patterns appear. The user aligns the patterns by pushing buttons on the phone. The more buttons is pushed, the worse a person's eyesight.

The app repeats this process four times, one for each axis of the eye. During the process the app also measures other abnormalities, such as astigmatism. Once all the patterns have been lined up, the app spits out a person's eyeglass prescription.

Start to finish it takes about two minutes to deliver a full prescription.

Right now the new method is just as accurate as existing ophthalmologic machines, and far, far cheaper than many. Within a few months the Near-Eye Tool for Refractive Assessment (or NETRA) system will be more precise than most lenses, said Raskar.

The app does have its limitations. Only cell phones with a high-resolution touch screen, such as the Nexus One or the iPhone, can run the app. The app is also

only useful for people who can respond to the directions on the screen; most children could not use the app.

The app should not replace trips to optometrist, said Raskar. Optometrists incorporate a patient's medical history and other factors into a final eyeglass prescription. Beyond the accuracy of a patient's glasses though, optometrists also check for eye diseases like macular degeneration and cataracts.

The MIT app can't test for diseases that would rob a person of their sight, at least not right now, said Raskar. Adding disease detection to the app is one goal of the group.

The first step was developing the app. The next step is to deploy it, first in the United States and then across the world. MIT is working with various NGOs in different countries to help distribute both the app and the screen, which together will be known as Perfect Sight. Within three to six months Raskar hopes to have everything ready to roll out worldwide.

"What's really cool about this project is that it can be deployed worldwide," said David Brady, an optics expert and professor at the Duke University. "People might not have good access to high-quality ophthalmologic equipment, but throughout the world there is good access to cell phones."

"It's an amazingly simple strategy that gives you a pretty good estimate" of a person's eye prescription, he said.

By,
S.Parthiban,
III-D.

T-RAY TECH SPOTS BOMBS, DRUGS FROM A MILE AWAY



Scientists from Rensselaer Polytechnic Institute (RPI) have found an ingenious way to replicate Superman's X-ray vision.

Currently the new combination of lasers, plasma and terahertz waves penetrate objects within a distance of 67 feet. If used outside the lab, however, they could see through objects over a mile away.

The research could be applied to spot terrorists long before they can do any damage or determine if a parcel contains a bomb or drugs.

"Terahertz waves can only be used at close distances; ambient water in the atmosphere attenuates them after a few inches," said Jingle Liu, a professor at RPI and a co-author of the recent *Nature Photonics* paper that describes the research.

"We found a way to get around the water moisture problem," said Lui, and use terahertz waves for long distance scanning.

Military and homeland security officials are funding much of the T-ray research, including the RPI study, because T-rays can sense explosives and drugs as well as penetrate clothing, paper or other thin materials. T-rays also do not produce harmful, ionized radiation like X-rays.

Since T-rays are very difficult and expensive to produce, they occupied a little used part of the electromagnetic spectrum between infrared and microwave radiation until the last few years.

Since 1995, however, scientists have found ever cheaper and easier ways to produce T-rays, but they haven't quite worked out all the kinks yet.

Water molecules absorb T-rays and weaken them after only a few inches, making any kind of long-distance scanning, however desirable, extremely difficult. The RPI scientists found an ingenious way around the water vapor problem.

Rather than using terahertz waves to transmit data, the researchers instead employed them to detect information. The team then used other wavelengths of light, which are not weakened by water vapor, to bring that information back.

Here's how it works: Liu and his colleagues at RPI shoot two lasers at a target. The lasers create a localized field of terahertz waves, which penetrate clothing or other thin materials. The lasers also create a mass of ionized gas, or plasma, which reacts to the terahertz waves before they are weakened by water. The gas glows, passing on the information gleaned by the terahertz radiation.

When that information returns, it's compared with a library of other optical fingerprints. Once a match is determined, the scientists could determine whether a package contains drugs, explosives, or something more mundane and less dangerous.

Using the light from the plasma, instead of the terahertz waves themselves, to carry information was "very creative," a technique that hasn't been used before,

said Michael Silevitch, a professor at Northeastern University.

While the research was done in the lab, "it's tantalizing to hope that this would work in the field. As a proof of concept idea, it's very exciting," he added.

The next step is to test the technique in the field, said Liu. The effect currently works up to 67 feet away, the limit of the laboratory.

**By,
J.PonSenthilRajan,
III-D**

DIAMONDS ARE A SUPERCOMPUTER'S BEST FRIEND:



Diamond sheets filled with holes could be the key to the next generation of supercomputers.

Scientists in California have used commercially available technology to pattern large sheets of diamonds with tiny, nitrogen-filled holes. The nitrogen-vacancy diamonds, as the sheets are called by scientists, could store millions of times more information than current silicon-based systems and process that information dozens of times faster.

Exactly how diamond-based computing would be used has yet to be determined, but applications could range from designing more efficient silicon-based computers to drug development and cryptography.

Nitrogen has been in diamonds for as long as there have been diamonds; it's why some diamonds have a yellow hue. For years scientists have used these natural, nitrogen-infused diamonds to study various aspects of quantum mechanics.

"We've used well-known techniques to create atomic-size defects in otherwise perfect diamonds," said David Awschalom, a scientist at the University of California, Santa Barbara and co-author of a new article in the journal *ACS Nano Letters*.

A supercomputer based on quantum mechanics requires more precision than nature can provide, so scientists have searched for a way to artificially implant arrays of precisely patterned nitrogen holes inside sheets of diamond.

Scientists from the University of California, Santa Barbara, along with colleagues from the Lawrence Berkeley National Laboratory, created such an array by using an ion beam to first knock out two carbon atoms, and then replace them with one nitrogen atom. 4,000 glowing nitrogen atoms. In about one minute, the scientists had patterned several inches of flat diamond.

The scientists didn't use any overly complicated techniques to accomplish this. "You can buy it online, send it to another company for the patterning, and then explore it yourself," said Awschalom, whose students did exactly that to demonstrate the ease of the technology.

The key to a diamond-based quantum mechanical computer is an extra electron

in the hole. In a traditional computer, information is encoded as either a "0" or a "1."

In a diamond-based quantum computer, information could be stored in the spin of that electron. This means information could be stored as not only a "0" or "1," but also the direction the electron is spinning.

An exact number is hard to come by, but scientists say this would dramatically increase the computing power compared with existing silicon computers.

Diamonds likely wouldn't replace the silicon used in today's consumer computers, said Ray Beausoleil, a fellow in Information and Quantum Systems at HP. "A quantum computer won't help you add two numbers faster," said Beausoleil.

However, that doesn't mean consumers won't benefit from a diamond-based quantum computer. What it will do is help model certain extremely complex problems, said Beausoleil and David DiVincenzo, a scientist at IBM who is also familiar with the *Nano Letters* article.

"This points to the fruitful end of a very long search of all the things that you could put in diamond to make it electronically active," said DiVincenzo.

Diamonds aren't a sure bet for a quantum computer, said DiVincenzo, but they're certainly in the running because of this research.

**By,
S.PradeepKumar,
III-D**

BLACK SILICON MAKES SOLAR CELLS CHEAPER

A one-step process creates a highly antireflective layer for photovoltaics.

A simple chemical treatment could replace expensive antireflective solar cell coatings, bringing down the cost of crystalline silicon panels. The treatment, a one-step dip in a chemical bath, creates a highly antireflective layer of black silicon on the surface of silicon wafers, and it would cost just pennies per watt, say researchers at the National Renewable Energy Laboratory (NREL). They've used it to create black silicon solar cells that match the efficiency of conventional silicon cells on the market.

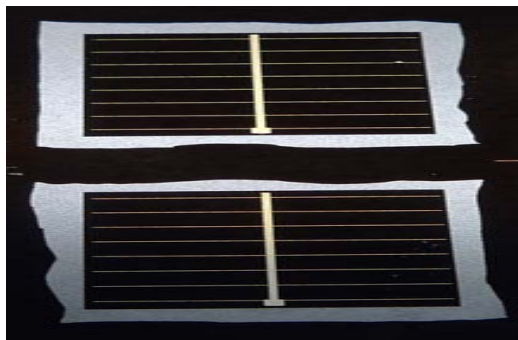
The crystalline silicon wafers used to make today's solar cells are treated to create a textured surface, then coated with an antireflective layer, usually silicon nitride, using high-vacuum processes. This additional layer increases the value of a solar cell by improving its efficiency--it suppress reflection so that more photons actually enter the silicon wafer instead of bouncing off its surface, increasing the flow of electricity off the cell. But the extra layer also adds to the expense. "We believe it can be cheaper," says Howard Branz, principal scientist in silicon materials and devices at NREL. Even with a coating, the best-quality silicon solar cells typically reflect 3 percent of the light that hits them. Branz's lab is developing inexpensive ways to create black silicon, which reflects almost no light.

Prototype solar cells made at NREL have the best efficiency ever reported for black silicon cells. Monocrystalline silicon cells with the black surface, and no additional antireflective coating,

convert 16.8 percent of the light that hits them into electricity, about the same efficiency offered by a typical crystalline silicon solar cell coated with antireflective material. The previous record for black silicon cells was 13.9 percent.

To replace the vacuum-deposition processes used to treat the surface of a silicon wafer, Branz's lab developed a chemical process that can be performed at ambient temperature and pressure using equipment already on site at solar-panel factories. A wafer is submerged in a bath containing a water solution of hydrogen peroxide, hydrofluoric acid, and chloroauric acid, which is made up of hydrogen, chlorine, and gold. The small amount of gold in the acid bath acts as a catalyst for chemical reactions. It's not clear exactly what the chemical reactions are, but they lead to the formation of gold nanoparticles that drill nanoholes at varying depths into the wafer. Branz says the gold can be reused again and again.

This etching process takes three minutes at room temperature, and less than a



minute at 40 °C. The result is a black, highly absorbent silicon wafer whose surface is riddled with tiny tunnels of varying depths. The pores create a surface with no sharp edges to reflect the light, and the variation in their depth is key, because the length of the tunnels determines which wavelength of light it

will interact with. Variation in tunnel length allows the surface to trap a broad spectrum of light.

Several groups are developing black silicon for solar cells and other optical devices. Others have used a multistep process to create the silicon-tunneling reactions, starting by placing metal particles over the surface of a wafer and then adding the acids. "We're doing it in a single, liquid step that requires no vacuum processes at all," says Branz. Beverly, MA-based startup SiOnyx uses laser pulses to generate tiny cones on silicon surfaces. Their formula for black silicon creates a different material than NREL's, but with similar properties. Company cofounder and principle scientist James Carey says the company hopes to enter the market within a year. Carey would not disclose what photovoltaic efficiencies SiOnyx has achieved using its process.

The baths used in today's factories to clean solar cells between etching steps could be used to create the black silicon coating, says Branz, which means this process could be adopted by manufacturers at very low cost. The agency has applied for patents on the process.

BY,
RAM BALAJI .T.R.S ,
RAM PRASAD .D,
III -D.

BLACK HOLES

Most people think of a black hole as a voracious whirlpool in space, sucking down everything around it. But that's not really true! A black hole is a place where gravity has gotten so strong that the escape velocity is faster than light.

But what does that mean? Gravity is what keeps us on the Earth, but it can be overcome. If you toss a rock up in the air, it will only go up a little ways before the Earth's gravity slows it and pulls it back down. If you throw it a little harder, it goes faster and higher before coming back down. If you could throw the rock hard enough, it would have enough velocity that the Earth's gravity could not slow it down enough to stop it. The rock would have enough velocity to escape the Earth. For the Earth, that velocity is about 11 kilometers per second (7 miles/second). But an object's escape velocity depends on its gravity: more gravity means a higher escape velocity, because the gravity will "hold onto" things more strongly. The Sun has far more gravity than the Earth, so its escape velocity is much higher—more than 600 km/s (380 miles/s). That's 3000 times faster than a jet plane! If you take an object and squeeze it down in size or take an object and pile mass onto it, its gravity will go up. At some point, if you keep doing that, you'll have an object with so much gravity that the escape velocity is faster than light. Since that's the ultimate speed limit of the Universe, anything too close would get trapped forever. No light can escape, and it's like a bottomless pit: a black hole.

The most common way for a black hole to form is probably in a supernova, an exploding star. When a star with about 25 times the mass of the Sun ends its life, it explodes. The outer part of the star screams outward at high speed, but the inner part of the star, its core, collapses down. If there is enough mass, the gravity of the collapsing core will compress it so much that it can become a black hole. When it's all over, the black hole will have a few times the

mass of the Sun. This is called a “stellar-mass black hole”, what many astronomers think of as a “regular” black hole. But there are also monsters, called super massive black holes. These lurk in the centers of galaxies, and are huge: they can be millions or even billions of times the mass of the Sun! They probably formed at the same time as their parent galaxies, but exactly how is not known for sure. Perhaps each one started as a single huge star which exploded to create a black hole, and then accumulated more material (including other black holes). Astronomers think there is a super massive black hole in the center of nearly every large galaxy, including our own Milky Way. Stellar-mass black holes also form when two orbiting neutron stars – ultra-dense stellar cores left over from one kind of supernova – merge to produce a short gamma ray burst, a tremendous blast of energy detectable across the entire observable Universe. Gamma-ray bursts are in a sense the birth cries of black holes.

If one falls into a black hole, one is doomed. Sure, once one falls in one can never get back out, but it turns out one will probably be dead before one gets there. The gravity one feels from an object gets stronger the closer one gets. As one approaches a stellar-mass black hole feet-first, the force of gravity on one’s feet can be thousands of times stronger than the force on one’s head! This has the effect of stretching one, pulling one apart like taffy. Tongue-in-cheek, scientists call this “spaghettification.” By the time one reaches the black hole, one will be a thin stream of matter many miles long. It probably won’t hurt though: even falling from thousands of kilometers away, the

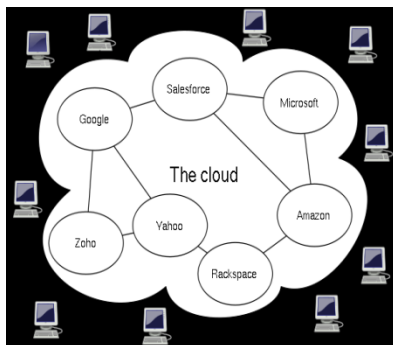
entire gory episode will be over in a few milliseconds. One may not even make it that far. Some black holes greedily gobble down matter, stealing it from an orbiting companion star or, in the case of supermassive black holes, from surrounding gas clouds. As the matter falls in, it piles up into a disk just outside the hole. Orbiting at huge speeds, the matter in this accretion disk gets extremely hot—even reaching millions of degrees. It will spew out radiation, in particular high-energy X-rays. Long before the black hole could rip one apart he’d be fired by the light. Once one passes the point where the escape velocity is faster than light, you can’t get out. This region is called the event horizon. That’s because no information from inside can escape, so any event inside is forever beyond our horizon. If the black hole is rotating, chaos awaits one inside. It’s a maelstrom as infalling matter turns back on the incoming stream, crashing into one like water churning at the bottom of a waterfall. At the very core of the black hole the seething matter finally collapses all the way down to a point. When that happens, our math fails us. It’s as if the matter has disappeared from the Universe, but its mass is still there. At the singularity, space and time come to an end.

The black hole itself may be invisible, but the ghostly fingers of its gravity leave behind fingerprints. Some stars form in pairs, called binary systems, where the stars orbit each other. Even if one of them becomes a black hole, they may remain in orbit around each other. By carefully observing such a system, astronomers can measure the orbit of the normal star and determine the mass of the black hole. Only a few

binary systems have black holes, though, so you have to know which binaries to observe. Fortunately, astronomers have discovered a signpost that points the way to black holes: X-rays. As mentioned above in What happens when you fall into a black hole?, if a black hole is “eating” matter from a companion star, that matter gets very hot and emits X-rays. This is like a signature identifying the source as a black hole. That’s why astronomers want to build spacecraft equipped with special detectors that can “see” in X-rays. In fact, black holes are so good at emitting X-rays that many thousands can be spotted this way. EXIST is one such spacecraft, designed to be able to detect tens of thousands of black holes, some of which may be billions of light years away. EXIST will create the most sensitive full-sky map locating black holes, including those which may be otherwise hidden from our view by obscuring gas and dust.

**BY,
ARINDAM DEBBARMA,
III-F**

CLOUD COMPUTING:

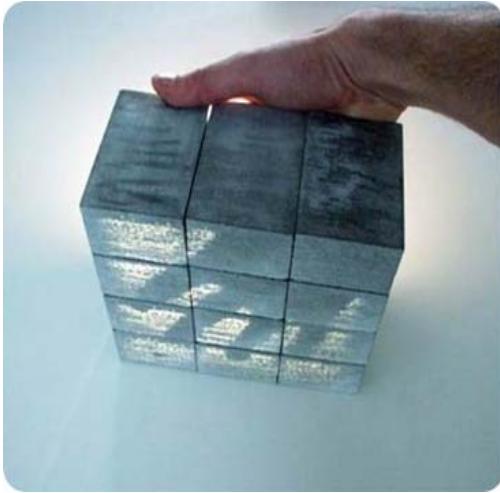


Cloud computing is Internet-based computing, whereby shared resources, software, and information are provided to computers and other devices on demand, like the electricity grid.

Cloud computing is a paradigm shift following the shift from mainframe to client-server in the early 1980s. Details are abstracted from the users, who no longer have need for expertise in, or control over, the technology infrastructure "in the cloud" that supports them. Cloud computing describes a new supplement, consumption, and delivery model for IT services based on the Internet, and it typically involves over-the-Internet provision of dynamically scalable and often virtualized resources. It is a byproduct and consequence of the ease-of-access to remote computing sites provided by the Internet. This frequently takes the form of web-based tools or applications that users can access and use through a web browser as if it were a program installed locally on their own computer. NIST provides a somewhat more objective and specific definition here. The term "cloud" is used as a metaphor for the Internet, based on the cloud drawing used in the past to represent the telephone network, and later to depict the Internet in computer network diagrams as an abstraction of the underlying infrastructure it represents. Typical cloud computing providers deliver common business applications online that are accessed from another Web service or software like a Web browser, while the software and data are stored on servers

**BY,
T.MATHUMITHA,
III-C.**

CONCRETE THAT CLEANS THE AIR:



Emissions that are the result of burning regular fossil fuels (like the fuel in your car) are made up a massive amount of different compounds. Even if we are able to reduce the emission levels of one of these compounds it is a good thing and will help bring down the pollution created by driving. That is where the research done by researchers at Eindhoven's University of Technology in the Netherlands will become a formidable method to reduce the global ecological damage caused by cars that are on the road. They have developed a concrete that strips nearly fifty percent of the nitrogen oxides from exhaust emissions.

The concrete was developed in a laboratory and after extensive testing a one thousand square meter stretch of road was laid to test the concrete in a real test of its potential. The concrete works by stripping nitrogen oxides from the air and converting them into nitrate (a nontoxic compound) as the sunlight and concrete worked together to break it down. The nitrate is then carried away by rain and never seen

again. So it not only turns a once toxic chemical into something benign – but using the forces of nature makes the problem virtually disappear.

It is assured by those who can supply the concrete that the cost incurred is not really much more than that of regular asphalt or concrete. In this regard it can be considered economically sound as well as ecologically. It can also be mixed with regular concrete and asphalt without jeopardizing the structural integrity of the latter materials. It may just be an easy way for roads to do something extra besides protect a vehicle from unnecessary wear – it will also help reduce the environmental effect of having many cars spewing forth noxious fumes from their tail pipes.

**BY,
K.Abinaya,
III –A.**

CONTROL ELECTRICAL APPLIANCES USING PC:

Here is a circuit for using the printer port of a PC, for control application using software and some interface hardware. The interface circuit along with the given software can be used with the printer port of any PC for controlling up to eight equipments.

The interface circuit shown in the figure is drawn for only one device, being controlled by D0 bit at pin 2 of the 25-pin parallel port. Identical circuits for the remaining data bits D1 through D7 (available at pins 3 through 9) have to be similarly wired. The use of opto-coupler ensures complete isolation of the PC from the relay driver circuitry.

Lots of ways to control the

hardware can be implemented using software. In C/C++ one can use the out port b (portno, value) function where port number is the parallel port address (usually 378hex for LPT1) and 'value' is the data that is to be sent to the port.

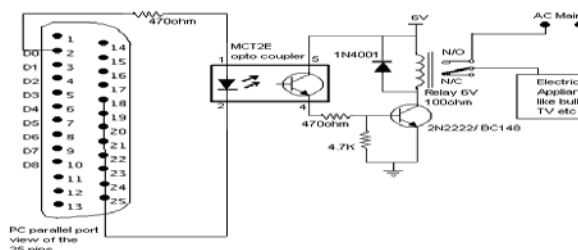
For a value 0, all the outputs (D0-D7) are off.

For 1, D0 is ON,

For 2 D1 is ON,

For 4, D2 is ON and so on.

Example: If the value is 29 in decimal which is 00011101 in binary then D0, D2, D3, D4 are ON and the rest are OFF.



Thus using our pc we can control electrical appliances around us which makes our lives even more sophisticated.

By,
Sylvia.M,
Suganya.P,
IV-E.

5G MOBILES



The cellular phone industry in North America is about to go into a period of revolution, which some people are referring to as the "5G mobile" era. Consumers await with much anticipation for the futuristic technology to impact their mobile life.

5G mobile phones will be considerably different from the norm of today. Consumers are just getting used to upgrades surrounding the Google Android phone and the Apple iPhone but they should be ready for significant improvements on the horizon. Technology is likely to change considerably even as we get used to 3G connectivity and the emergence of the smart phone. The 5G revolution is sure to take the industry by storm.

Innovation surrounds the 5G communications system. A hybrid method of transmitting is involved, featuring voice over Internet protocol, also known as the VoIP. This new technology allows a great enhancement in call quality and also allows data to be transferred at very high speeds. One day, before too long, our once humble cell phone, initially just a talking device, will be promoted to the rank of virtual personal assistant. We will not be able to do without it and will be able to tap into an amazing array of features and flexibility contained within the latest in

cutting edge hardware. Is it time to say goodbye to the clunky desktop computer, or even the laptop or note pads? Many argue that it is, as the 5G mobile phone will surely be a very powerful miniature computer by itself. Due to the huge increase in bandwidth available through voice over Internet protocol, transfer rates are going to be Avery quick and usability much improved.

Flexibility will be at the forefront of the new 5G mobile phone generation. You will not have to worry about different calling rates and whether your service provider has to "talk" to another service provider just to get you through. All this nonsense will become a thing of the past through forward thinking new service providers.

No longer will the world be a small place when the 5G mobile revolution really gets cranked up. Think about the major emerging economies of India and China and you can imagine how many people will be communicating with cell phones within the next decade or so.

How long will it be before we are in a position where technology has allowed us to instantly communicate with everyone else and to live our lives in the most highly efficient manner? Info-Tech survey released to selected CEOs this week shows that wireless customers want free calls, free phones, the ability to switch between carriers for free, with international roaming and handsets that don't need recharging every day. Carriers expressed shock and awe at the results and promised to sell off their swank marble skyscrapers, many situated in the heart of Tokyo's financial district, to facilitate the move. "We are asking shareholders to help us deal with these issues," said an obviously unsettled

CEO at a hastily convened press conference. "While we have managed to get 5G deployment ready for the end of this year, that doesn't seem to be what our customers really want," the flustered president told WWJ in an exclusive interview today.

5G-TECHNOLOGY

5G Mobile Phones Coming Soon



5G mobile technology is the name given to the post 4G generation of mobile devices such as cell phones. It will become available from at least one provider in several parts of the US on thanksgiving week 2013. There is not yet an agreed industry standard for what constitutes 5G mobile, so for now it is merely a marketing term. But most likely will improve the previous version such as relating people with same affinity and likeness. The f-one-u technology will bring the texting and chatting experience to its biggest popularity with the help of FACEBOOK.

The use of G, standing for generation, in mobile technology covers the major advances of the past 20-30 years. 1G technology involved the first widely available mobile phones. 2G technology, which began in the early 1990s,

switched to a digital format and introduced text messaging. 3G technology improved the efficiency of how data is carried, making it possible to carry enhanced information services such as websites in their original format. The latest iPhone is the best known example of 3G technology.

4G mobile was established an agreed a set of standards, so its features are currently in full use as people and customer's requirements indicated so. As well as drastically increased data transfer speeds, 4G mobile enhanced security measures. Another goal is to reduce blips in transmission when a device moves between areas covered by different networks. 5G mobile networks should also use a network based on the IP address system used for the internet plus a nanodata locator which gets rid of those blips and lots of data.

Within the United States, there are three major systems using 4G mobile technology. One is known as WiMax and is backed by Clearwire, a firm whose majority owner is Sprint Nextel. It began testing services in Baltimore in 2008 and was set to expand this into major new markets in 2009. Sprint intended to have 80 cities covered by the end of 2010. But only was met on president's day 2011. The major testing was shown on the commercial before the opening of the London olympics back in 2012.

The rival system, Long Term Evolution or LTE, is backed mainly by Verizon. It began its availability at the end of 2010 but was not available for widespread use until 2012. LTE's backers overcame this disadvantage by offering faster speeds and producing cheaper equipment. The third system was offered by Virgin

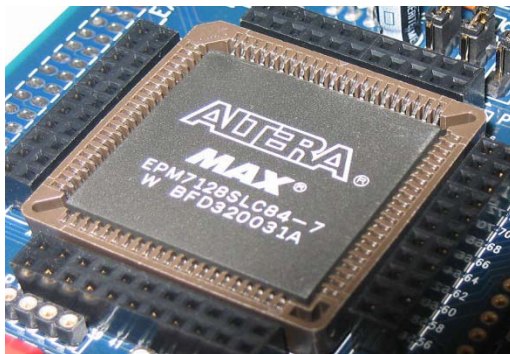
mobile The technology used in our airforce will make available the 6G-HOL technology by 2015. HOL stands for holographic technology that will make reality the tridimensional visuals shown in AVATAR 2. Unlike previous generations of mobile technology, 5G mobile will be widely used for internet access on computers as well as carrying cell phone communications and a direct interactivity with the Homeland Security in case of a major terrorist attack. Customers in areas which have strong 5G coverage will be able to use it for a home broadband connection which doesn't require any cabling to their household. It can also be used for accessing the internet on the move without having to be in a wireless hotspot such as those offered by some coffee shops, airports and libraries. But the major advantage will be the interconnectivity that people needs today.

5G Technology Features: 5G technology has powerful features as listed below 5G technology offer high resolution for crazy cell phone user and bi-directional large bandwidth shaping. The advanced billing interfaces of 5G technology makes it more attractive and effective. 5G technology also providing subscriber supervision tools for fast action. The high quality services of 5G technology based on Policy to avoid error. 5G technology is providing large broadcasting of data in Gigabit which supporting almost 65,000 connections. 5G technology offer transporter class gateway with unparalleled consistency. The traffic statistics by 5G technology makes it more accurate. Through remote management offered by 5G technology a user can get better and fast solution.

The remote diagnostics also a great feature of 5G technology. The 5G technology is providing up to 25 Mbps connectivity speed. The 5G technology also support virtual private network. The new 5G technology will take all delivery service out of business prospect. The uploading and downloading speed of 5G technology touching the peak. The 5G technology network offering enhanced and available connectivity just about the world

BY,
R.kanimozhijeeva,
III-B

DIGITAL SYSTEM DESIGN USING CPLD



Complex Programmable Logic Device is a reconfigurable hardware platform which is used to design moderately complicated digital circuits. The digital logic can be both combinational and sequential. CPLDs contains around 1000 - 10000 logic gates. Inside each CPLD is a set of macrocells. These macrocells are typically comprised of some amount of combinatorial logic (AND and OR gates, for example) and a flip-flop. In other words, a small Boolean logic equation can be built within each macrocell. The logic inside the macrocells and the

macrocells itself can be interconnected by programming it using Hardware Description Languages such as VHDL. The internal structure of a CPLD is shown in figure 1.

Figure 1: Internal Structure of CPLD

Consider a 4-bit Binary to Grey Code converter. The logic circuit consists of three XOR gates. The circuit is shown in figure 2.

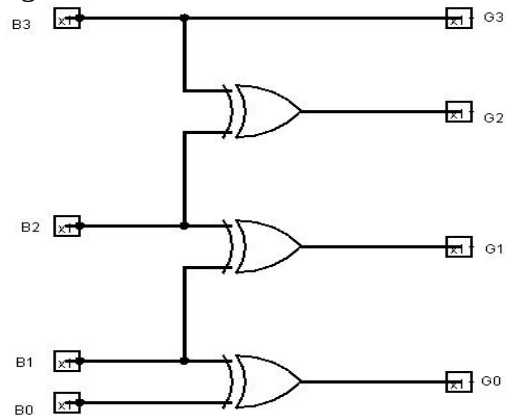


Figure 2: Binary to Grey

The VHDL script for the 4-bit Binary to Grey Code converter is given below

```
Library IEEE;
Use IEEE.STD_LOGIC_1164.ALL;
```

entity bin_to_grey is

Port (B: in STD_LOGIC_VECTOR
(3 downto 0);

G : out STD_LOGIC_VECTOR
(3 downto 0));

end bin_to_grey;

architecture Behavioral of bin_to_grey is
begin

G (0) <= B (0) XOR B (1);

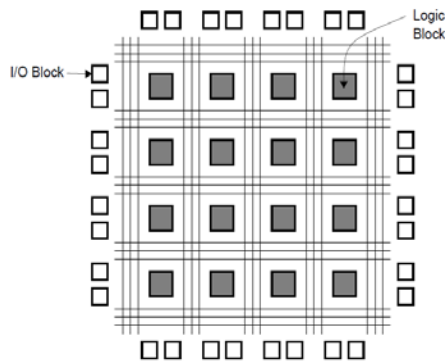
G (1) <= B (1) XOR B (2);

G (2) <= B (2) XOR B (3);

G (3) <= B (3);

end

Behavioral;



Design Flow:

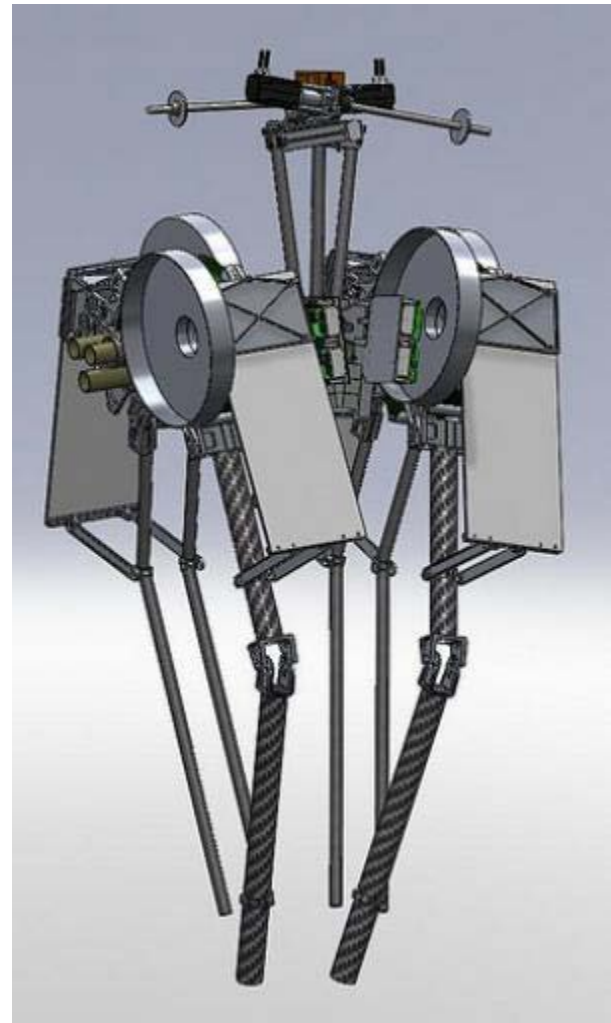
The VHDL code is self explanatory. Two data buses G & B of 4-bit length are declared and assigned according to the functionality of the Binary to Grey Code converter. '<=' is the assignment operator in VHDL. In actual hardware the XOR gates will be reduced to basic gates such as AND & OR. The design is simulated and tested on a suitable simulator. Then the port pins are mapped to actual pins on the CPLD. The program is then burned inside the CPLD using JTAG programmer and executed. Finally, verification is done on the actual hardware.

Tools & Hardware Used:

- Xilinx XC9572 CPLD
- Xilinx ISE Webpack
- ISIM simulator
- USB JTAG Programmer

BY,
Alen Antony,
III –A.

ADVANCES MADE IN WALKING, RUNNING ROBOTS



A biped robot design being built by researchers at Oregon State University holds the potential to reduce energy use while improving robotic locomotion. This work reflects some important fundamental advances in the understanding of robotic motion. (Credit: Photo courtesy of Oregon State University) ScienceDaily (May 28, 2010) — Researchers at Oregon State University have made an important fundamental advance in robotics, in work that should lead toward robots that

not only can walk and run effectively, but use little energy in the process.

By achieving an optimal approach with robotic mechanisms, studies are moving closer to robots that could take on dangerous missions in the military, create prosthetic limbs for humans that work much better, or even help some people who use wheelchairs to gain "walking" abilities. The newest findings from the Dynamic Robotics Laboratory at OSU are being presented at two conferences, including the IEEE Conference on Robotics and Automation. The OSU research program has received a new \$750,000 grant for its work from the Human Frontiers Science Program, administered through the National Science Foundation. Work to actually build a robot with some of the newest features should be completed by this summer. "Researchers have been working toward robot locomotion for a long time based mostly on experience and intuition," said Jonathan Hurst, an assistant professor of robotics and mechanical design at OSU. "What we've done is taken a step back to analyze the fundamental dynamics of the mechanical system, what behavior is really possible for a given robotic system. A rock can't fly, no matter what software you write for it." This is an important advance and gives us a new foundation to tell what actually will and won't work before we even try to build it," he said. Right now, most applications of robotics are done with machines working in a precise or controlled situation -- picking something up off an assembly line, spot welding an automobile in an exact way thousands of times a day. But in terms of locomotion, Hurst said, humans and other animals are a tough act to follow. Using limited energy, they can move easily over uneven terrain, and respond with a

fascinating balance of muscles and tendons. They have different ways to deal with forces, such as holding something in place rigidly, or also responding to outside influences -- like the delicate act of holding a cup of coffee level during a bumpy car ride. In their recent studies, the OSU researchers essentially proved that these two abilities are mutually exclusive. Humans deal with this problem by flexing opposing pairs of muscles, to change the dynamic properties of their arm. For a robot, the more it is able to do one of these tasks, the less able it is to do the other. Existing robots that can walk and run, Hurst said, tend to be as rigid as possible while still achieving a basic gait. But this approach uses a huge amount of energy, significantly reducing their value and possible applications in the real world. The OSU researchers are working toward something that has similar or better performance, but uses far less energy -- the best of both worlds, and closer to the abilities of animals. "If robotic locomotion is ever to achieve some of what we want, it will have to use less energy," Hurst said. "There are machines that can walk with no active controls at all, using barely any energy, but they fall if they run into the smallest bump. We need to use as much of that passive ability as possible and only use motors or active controls if it's really necessary, so we can save energy in the process."

In part because it's consistent with these goals, the OSU researchers also hope to win what's called the "W" Prize, a \$200,000 award that's been promised for the first robot that can move 10 kilometers within 10,000 seconds, through and over obstacles, using less energy than it would take a human to do the same task.

Part of the OSU research are studies under way with colleagues in England to examine ostriches, and make direct comparisons to the robot behavior. Animals, including ostriches, have a great ability to respond to unexpected disturbances while running, and can provide insights into the needed robotic equivalent -- combining spring-mass models with "force control actuators." "Long term, there's no reason we shouldn't be able to build robots or robotic devices with excellent locomotion ability," Hurst said. "Clearly this might be useful for some military or police applications in highly dangerous situations. But I could also see great improvements possible with prosthetic limbs that work much better than existing technology, or even the creation of exoskeletons that might allow someone with limited motor ability to walk effectively."

BY
E.LAVANYAA.
B.LAKSHMIPRABHA.
III-C

TACHYONS – A WAY TO TRANSFER DATA FASTER THAN LIGHT

It's the time to think about the travel through light years .If it happens think a case how much time could a signal take to cross one light year distance, its one year of time *. (i.e.) a signal is sent to greet a person's marriage, but the signal reaches him when he took his new infant on his hands. It's not a joke; just imagine the nearest galaxy is 117 million light-years far away from earth and how many years would the signal take to reach earth if it's sent right now. It's a fact that how

we are lagging in technology. Tachyons is one of the best key area of research for fastest data transfer that has no end up with TBPS**. To achieve this novel system the mankind seeks some preety support from the engineers there lies our hope of future .As defined A tachyons ("elementary particle") is a hypothetical subatomic particle which moves faster than light. Tachyon is a particle with space-like four-momentum and imaginary proper time. The first description of tachyons is attributed to German physicist Arnold Sommerfeld . If tachyons were conventional, localizable particles which could be used to send signals faster than light (FTL), this would lead to violations of causality,condensation & other higher "c" problems that are born with high velocity travels.

Causality

The property of causality is a fundamental principle of theoretical particle physics. If tachyons could be used to transmit information FTL.They could also be used to violate causality.The causality can be explained in easy way .(ie) If one of the two events represents the sending of a signal from one location and the second event represents the reception of the same signal at another location, then as long as the signal is moving at the speed of light or slower, the mathematics of simultaneity ensures that all reference frames agree that the transmission-event happened before the reception-event. However, in the case of a hypothetical signal moving faster than light, there would always be some frames in which the signal was received before it was sent, so that the signal could be said to have moved backwards in time. And since one of the two fundamental postulates of special relativity says that

the laws of physics should work the same way in every inertial frame, then if it is possible for signals to move backwards in time in any one frame, it must be possible in all frames. This means that if observer A sends a signal to observer B which moves FTL in A's frame but backwards in time in B's frame, and then B sends a reply which moves

FTL in B's frame but backwards in time in A's frame, it could work out that A receives

the reply before sending the original signal, a clear violation of causality in every frame.

Condensation

A tachyon is a quantum of a field—usually a scalar field—whose squared mass

is negative, and is used to describe spontaneous symmetry breaking****: The existence of such a field implies the instability of the field vacuum; the field is at a local maximum rather than a local minimum of its potential energy, much like a ball at the top of a hill. A very small impulse (which will always happen due to quantum fluctuations) will lead

the field to roll down with exponentially increasing amplitudes: it will induce tachyon

condensation. It is important to realize that once the tachyonic field reaches the minimum

of the potential, its quanta are not tachyons any more but rather have a positive mass-squared, such as the Higgs boson****.

Technically, the squared mass is the second derivative of the effective potential, at a point where the first derivative is zero. So for a tachyonic field the second

derivative is negative, meaning that the effective potential is at a local maximum rather than a local minimum. Therefore this situation is unstable and the field will roll down to another point, stopping only at a local minimum, where its quanta have a non-negative squared mass, so that it is not tachyonic any longer. Since a tachyon's squared mass is negative, it formally has an imaginary mass. This is a special case of the general rule, where unstable massive particles are formally described as having a complex mass, with the real part being their mass in usual sense, and the imaginary part being the decay rate in natural units. These are the some of of facts that tachyon have and many other still to be unreavled.

* - calculated by the existing high end OFC technology

**TBPS-Tera Bites Per Second (1×10^{12} bits per second)

On October 7, 2008, the Royal Swedish Academy of Sciences awarded the 2008 Nobel Prize in Physics to two Japanese citizens and a Japanese-born American for their work in subatomic physics. American Yoichiro Nambu, 87, half of the prize for the discovery of the mechanism of spontaneous broken symmetry. Japanese physicists Makoto Kobayashi and Toshihide Maskawa shared the other half of the prize for discovering the origin of the broken symmetry. * Higgs boson- a hypothetical massive scalar elementary particle predicted to exist by the Standard Model in particle physics (simple unitary)

BY,

R.RAMASAMY RAJA,

C.RAM BHARATH,

III-D.

HUMAN AREA NETWORK

ABSTRACT

Here is a new concept called “RED TACTON” which makes the human body as a communication network by name HAN (Human Area Network). Focusing on the naturalness, inevitability and sense of security conveyed by touching which is called as Red Tacton which could let people transfer data to each other handhelds by means of handshake or a physical contact. Using an embedded transmitter a PDA Red Tacton sends a 5 volts pulse along the surface of body. The human body shunts most of electricity to ground resulting in weak electric field that can be modulated to carry signals. As the receiver is located on the part of the body, the components of a hands free handset or in an acquaintance PDA as it senses modulation in the electric field the receiver decodes them to recover the data. Human Area Network is an important over current wireless technologies such as Bluetooth because users can narrowly limit signal recipients rather than broadcasting to all devices within given range. Using Red Tacton enabled devices music from our digital audio player in our pocket would pass through our clothing and shoot over our body to headphones into our ears. Instead of fiddling around with the cable to connect our digital camera to our computer we could transfer pictures just by touching the PC while the camera is around your neck and since data can pass from one body to another you could also exchange electronics business cards by shaking hands.

Introduction:

Human society is entering an era of ubiquitous computing, when networks

are seamlessly interconnected and information is always accessible at our fingertips. In the world of computers, networking is the practice of linking two or more computing devices together for the purpose of sharing data. RedTacton enables the first practical Human Area Network between body-centered electronic devices and PCs or other network devices embedded in the environment via a new generation of user interface based on totally natural human actions such as touching, holding, sitting, walking, or stepping on a particular spot. By making Human Area Networks feasible, RedTacton will enable ubiquitous services based on human centered interactions and therefore more intimate and easier for people to use. RedTacton can be used for intuitive operation of computer-based systems in daily life, temporary one-to-one private networks based on personal handshaking, device personalization, security, and a host of other applications based on new behavior patterns enabled by RedTacton. At last we predicted how Redtacton changes lifestyle in future. Background:

At the end of 2002 the Japanese telecommunications group NTT had announced that it would develop a new data transmission technology that uses the conductive properties of the human body to exchange information between electronic devices. With RedTacton the company has now scarcely two and a half years later presented its first prototype of a Human Area Network (HAN). The basic principle of the RedTacton concept is the creation of minute electrostatic fields that modulated by a code-giver to carry digital information are capacitively transmitted to human skin as a conductor, thence to be transmitted upon

physical contact. To read out the data bits NTT makes use of a novel type of opto-electrical receiver equipped with a miniature laser whose light ray is reflected by a crystal. At the receiving end the electrostatic fields evoked by the RedTacton sender cause the electrical field in the environment of the crystal to change, which in turn induces changes in the polarization of the reflected laser light. These changes in polarization the opto-electrical receiver interprets as data bits, which it converts into electrical signals that can be processed by a PDA, for instance.

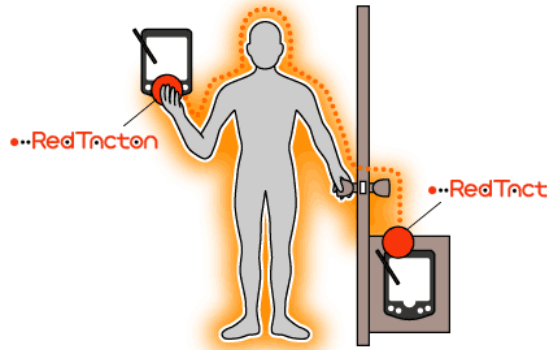
According to NTT this technology - which can also be made to work when door handles, switches or turnstiles are touched - allows bi-directional signals to be sent at rates of up to 10 Mbit/s. The strength of the electrostatic fields evoked the company puts at but a few hundred milliwatts. Prior to the stage planned for 2006 in which the HAN technology is to be ready for the market, NTT intends to conduct a number of field tests this summer, in which it is inviting companies with an interest in the technology to participate. Japanese company Nippon Telegraph and Telephone Corporation (NTT) claims to have developed the first viable Human Area Network (HAN) device, enabling fast data transfer between devices using the human body as a conduit. NTT reckons this latest advance on the wireless Personal Area Network concept - dubbed RedTacton - can transmit data over the surface of the skin at up to 2Mbps. Where it differs, though, from previous offerings, is that a RedTacton-enabled device does not have to be in direct contact with the skin - only within about 20cm.

Human society is entering an era of ubiquitous computing, where everything

is networked. The practical implementation of ubiquitous services requires three levels of connectivity: Wide Area Networks (WAN), typically via the Internet, to remotely connect all types of servers and terminals; Local Area Networks (LAN), typically via Ethernet or WiFi connectivity among all the information and communication appliances in offices and homes; and Human Area Networks (HAN) for connectivity to personal information, media and communication appliances within the much smaller sphere of ordinary daily activities-- the last one meter. RedTacton is a break-through technology that, for the first time, enables reliable high-speed HAN.

In the past, Bluetooth, infrared communications (IrDA), radio frequency ID systems (RFID), and other technologies have been proposed to solve the "last meter" connectivity problem. However, they each have various fundamental technical limitations that constrain their usage, such as the precipitous fall-off in transmission speed in multi-user environments producing network congestion. "With Bluetooth, it is difficult to rein in the signal and restrict it to the device you are trying to connect to. You usually want to communicate with one particular thing, but in a busy place there could be hundreds of Bluetooth devices within range." Furthermore, humans apparently make poor airdrops, something which is "good for security because even if you encrypt data it is still possible that it could be decoded, but if you can't pick it up it can't be cracked".

What's RedTacton ?



RedTacton is a new Human Area Networking technology that uses the surface of the human body as a safe, high speed network transmission path. RedTacton takes a different technical approach. Instead of relying on electromagnetic waves or light waves to carry data, RedTacton uses weak electric fields on the surface of the body as a transmission medium. A RedTacton transmitter couples with extremely weak electric fields on the surface of the body. The weak electric fields pass through the body to a RedTacton receiver, where the weak electric fields affects the optical properties of an electro-optic crystal. The extent to which the optical properties are changed is detected by laser light which is then converted to an electrical signal by a detector circuit. - RedTacton uses the minute electric field emitted on the surface of the human body. Technically, it is completely distinct from wireless and infrared. -A transmission path is formed at the moment a part of the human body comes in contact with a RedTacton transceiver. Physically separating ends the contact and thus ends communication. -Using RedTacton, communication starts when terminals carried by the user or embedded in devices are linked in various combinations according to the

user's natural, physical movements. - Communication is possible using any body surfaces, such as the hands, fingers, arms, feet, face, legs or torso. RedTacton works through shoes and clothing as well.

How RedTacton Works ?

RedTacton can achieve duplex communication over the human body at a maximum speed of 10 Mbps.

1. The RedTacton transmitter induces a weak electric field on the surface of the body.

2. The electric field sensor (transistor or photonic electric field sensor) detects electric field that reaches the RedTacton receiver.

3. Signals are processed in the receiver circuit and the data is downloaded.

Potential Applications -

One-to-One services: With the ability to send attribute data from personal information devices worn on the body to computers embedded in the environment, one-to-one services could be implemented that are tailored to the individual needs of the user. Intuitive operation of personal information devices: Communication is triggered by totally natural human actions and behavior, so there is no need to insert smart cards, connect cables, tune frequencies, or any of the other inconveniences usually associated with today's electronic devices.

Device personalization: Setup, registration, and configuration information for an individual user can all be uploaded to a device at the instant the device is touched, eliminating the need for the device to be registered or configured in advance.

New behavior patterns: Tables, walls, floors and chairs can all act as conductors and dielectrics, turning furniture and other architectural

elements into a new class of transmission medium. For example, a user could have instant access to the Internet merely by placing a laptop onto a conductive tabletop.

Security applications-

RedTacton could be installed on doors, cabinets and other locations calling for secure access, such that each secure access could be initiated and authenticated with a simple touch. At the same time, all the transaction details and relevant user attributes (personal identity, security clearance, etc.) could be logged by the security system. Conclusion The applications of RedTacton are enormous, medical security and data transfer are just the start. Conferencing can be enabled without the use of wiring, and walls and desks and doorknobs can be conduits for data transfer. In education the applications are endless from lesson outlines to administrative forms data could be transferred quickly and easily without pedagogical intervention. This technology seems to be a glimpse into the future. At first it will be limited in use and fairly expensive, but as it becomes more widespread we could see the extinction of the key and ID card as we know them today. Imagine walking up to your house, reaching for a doorknob which automatically unlocks the door, and walking right in. No more fumbling around for keys. That's a future we'd like to live in.

BY,
V.N.SUDHARSANAM,
II-E.

FUZZY LOGIC

INTRODUCTION:

Fuzzy logic has rapidly become one of the most successful of today's technologies for developing sophisticated control systems. The reason for which is very simple. Fuzzy logic addresses such applications perfectly as it resembles human decision making with an ability to generate precise solutions from certain or approximate information. It fills an important gap in engineering design methods left vacant by purely mathematical approaches (e.g. linear control design), and purely logic-based approaches in system design.

WHAT OFFER?	DOES	IT The first
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applications of fuzzy theory were primary industrial, such as process control for cement kilns. However, as the technology was further embraced, fuzzy logic was used in more useful applications. In 1987, the first fuzzy logic-controlled subway was opened in Sendai in northern Japan. Here, fuzzy-logic controllers make subway journeys more comfortable with smooth braking and acceleration. Best of all, all the driver has to do is push the start button! Fuzzy logic was also put to work in elevators to reduce waiting time. Since then, the applications of Fuzzy Logic technology have virtually exploded, affecting things we use everyday.

Take for example, the fuzzy washing machine . A load of clothes in it and press start, and the machine begins to churn, automatically choosing the best cycle. The fuzzy microwave, Place chili, potatoes, or etc in a fuzzy microwave and push single button, and it cooks for the right time at the proper temperature. The fuzzy car, maneuvers itself by following simple verbal instructions from its driver. It can even stop itself when there is an obstacle immediately ahead using sensors. But, practically the most exciting thing about it, is the simplicity involved in operating it.

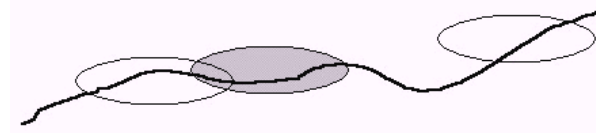
WHAT DOES FUZZY LOGIC MEAN?

Before illustrating the mechanisms which make fuzzy logic machines work, it is important to realize what fuzzy logic actually is. Fuzzy logic is a superset of conventional(Boolean) logic that has been extended to handle the concept of partial truth- truth values between "completely true" and "completely false". As its name suggests, it is the logic underlying modes of reasoning which are approximate rather than exact. The importance of fuzzy logic derives from the fact that most modes of human reasoning and especially common sense reasoning are approximate in nature.

FUZZY RULES: Human beings make descisions based on rules. Although, we may not be aware of it, all the descisions we make are all based on computer like if-then statements. If the weather is fine, then we may decide to go out. If the forecast says the weather will be bad today, but fine tommorow, then we make a descision not to go today, and postpone it till tomorrow. Rules associate ideas and relate one event to another.

Fuzzy machines, which always tend to mimic the behaviour of man, work the same way. However, the descision and the means of choosing that descision are replaced by fuzzy sets and the rules are replaced by fuzzy rules. Fuzzy rules also operate using a series of if-then statements. For instance, if X then A, if y then b, where A and B are all sets of X and Y. Fuzzy rules define fuzzy patches, which is the key idea in fuzzy logic.

A machine is made smarter using a concept designed by Bart Kosko called the Fuzzy Approximation Theorem(FAT). The FAT theorem generally states a finite number of patches can cover a curve as seen in the figure below. If the patches are large, then the rules are sloppy. If the patches are small then the rules are fine.



Fuzzy Patches

In a fuzzy system this simply means that all our rules can be seen as patches and the input and output of the machine can be associated together using these patches. Graphically, if the rule patches shrink, our fuzzy subset triangles gets narrower. Even novices can build control systems that beat the best math models of control theory. Naturally, it is math-free system.

Case Study: Fuzzy Traffic Light Controller

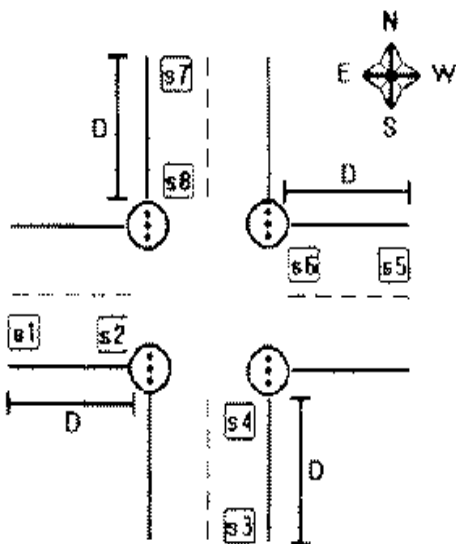


BACKGROUND:

In a conventional traffic light controller, the lights change at constant cycle time, which is clearly not the optimal solution. It would be more feasible to pass more cars at the green interval if there are fewer cars waiting behind the red lights. Obviously, a mathematical model for this decision is enormously difficult to find. However, with fuzzy logic, it is relatively much easier.

FuzzyDesign :

First, eight incremental sensors are put in specific positions as seen in the diagram below.



The first sensor behind each traffic light counts the number cars coming to the intersection and the second counts the cars passing the traffic lights. The amount of cars between the traffic lights is determined by the difference of the reading of the two sensors. For example, the number of cars behind traffic light north is $s7-s8$. The distance D , chosen to be 200ft., is used to determine the maximum density of cars allowed to wait in a very crowded situation. This is done by adding the number of cars between to paths and dividing it by the total distance. For instance, the number of cars between the East and West street is $(s1-s2)+(s5-s6)/400$. Next comes the fuzzy decision process which uses the three step mentioned Above (fuzzyfication, rule evaluation and defuzzification).

As before, firstly the inputs and outputs of the design has to be determined. Assuming red light is shown to both North and South streets and distance D is constant, the inputs of the model consist of :

- 1) Cycle Time
- 2) Cars behind red light

3) Cars behind green light
The cars behind the light is the maximum number of cars in the two directions. The corresponding output parameter is the probability of change of the current cycle time. Once this is done, the input and output parameters are divided into overlapping membership functions, each function corresponding to different levels. For inputs one and two the levels and their corresponding ranges are zero(0,1), low(0,7), medium(4,11), high(7,18), and chaos(14,20). For input 3, the levels are very short(0,14), short(0,34), medium(14,60), long(33,88), very long(65,100), limit(85,100). The levels of output are no(0), probably no(0.25), maybe(0.5), probably yes (0.75), and yes(1.0). Note: For the output, one value (singleton position) is associated to each level instead of a range of values. The corresponding graphs for each of this membership function are drawn in the similar way above.

The rules, as before are formulated using a series of if-then statements, combined with AND/OR operators'. Ex: if cycle time is medium AND Cars behind Red is low AND Cars behind Green is medium, then change is Probably Not. With three inputs, each having 5, 5, and 6 membership functions, there are a combination of 150 rules. However using the minimum or maximum criterion some rules are combined to a total of 86.

This process, also mentioned above converts the fuzzy set output to real crisp value. The method used for this system is center of gravity:

Crisp Output = $\frac{\text{Sum}(\text{Membership Degree} * \text{Singleton Position})}{\text{Membership degree}}$ For

example, if the output membership degree, after rule evaluation are: Change Probability Yes=0, Change Probability Probably Yes=0.6, Change Probability Maybe=0.9, Change Probability Probably No= 0.3, Change Probability No=0.1

then the crisp value will be: Crisp Output=(0.1*0.00)

+ (0.3*0.25)+(0.9*0.50)+(0.6*0.75)+(0*1.00)/0.1+0.3+0.9+0.6+0 =0.51

HOW FAR CAN A FUZZY LOGIC GO?

Just from the examples given Previously, it is clear that fuzzy logic can be used in numerous applications. It can appear almost in anyplace where computers and modern control theory are overly precise as well as in tasks requiring delicate human intuition and experience-based knowledge. Consider an example below which is currently undergoing intensive research in OMRON Research Center, Japan.

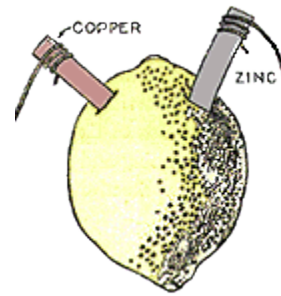
It may seem obvious that babies nowadays don't drink the way it is described in child care books. They may drink a little or a lot depending on their physical condition, mood, and other factors. But in a fuzzy-logic program can be created that would recommend how much to feed the baby, mothers would have an easier time raising the child. The basis of the research is to develop a program to determine the appropriate amount of milk to feed the child according to a knowledge base that includes the child's personality, physical condition, and some environmental factors. This can prevent the child from being fed unnecessarily. Now although adapting fuzzy logic to babies may seem silly, one can easily imagine using it to control the feeding of animals in captivity, for instance.

CONCLUSION: In conclusion, as Man gets hungry in finding new ways of

improving our way of life, new, smarter machines must be created. Fuzzy logic provides a simple and efficient way to meet these demands and the future of it is limitless.

Result: Machine beats Man!!!

Electricity from a lemon



It is possible to get electricity from a lemon (and a few other acidic fruits and vegetables) using two strips of metal. The most readily available combination is copper and zinc. The zinc piece can be taken from the casing of an old carbon "D" cell (battery); some zinc coated nails may work as well. The copper can be a coin containing a high amount of copper. (Note: some recent copper coins, including the newer U.S. pennies, contain low amounts of copper mixed with zinc. If in doubt, use a pure copper strip.) They are stuck into the lemon, and wires are attached to each one. The wires should be attached securely to form a good electrical contact. If the other ends of the wires are attached to a small light bulb (flashlight bulb) there should be sufficient current to light it. Note that the metal pieces should be clean, and that they should not touch inside the lemon.

The electric current comes from the chemical reaction between the acid in the lemon and the zinc. Positively charged hydrogen in the acid oxidizes the zinc, causing electrons to flow from the zinc to the acid. In the process, some of the hydrogen in the acid is reduced to hydrogen gas. The copper merely completes the circuit.

Troubleshooting

Judging by my e-mail, even though this project is very simple, it is especially

difficult to get working. Keep in mind that one or even several lemons in series produce only a tiny amount of voltage. The amount is unpredictable and dependent on many factors, including the acidity of the lemon and the metals used. One lemon should light a tiny bulb, but then again it may not. Try testing with a voltmeter. If the meter swings a tiny bit, you know that it is working. Wiring several lemons in series (bulb-copper-lemon-zinc-connecting wire-copper-lemon-zinc-bulb) may do the trick. If you can't get a light bulb to work, try improvising a galvanometer to show the voltage. Do this by wrapping about ten turns of thin wire (insulated) around a small magnetic compass to form a coil. Strip the free ends of the wire and touch them to the zinc and copper strips in the lemon. The current should be enough to set up a small magnetic field in the coil, causing the compass needle to divert.

By,
P.PRITHIVINATH ,
G.RAJ MOHAN,
III-D.

EXPLORING CURRENT TRANSFORMER APPLICATIONS

For a variety of applications, using current transformers is an efficient way to sense current with minimum insertion loss.

Current transformers can perform circuit control, measure current for power measurement and control, and perform roles for safety protection and current limiting. They can also cause circuit events to occur when the monitored current reaches a specified level. Current monitoring is necessary at frequencies from the 50 Hz/60 Hz power line to the higher frequencies of switchmode transformers that range into the hundreds of kilohertz.

The object with current transformers is to think in terms of current transformation

rather than voltage ratios. Current ratios are the inverse of voltage ratios. The thing to remember about transformers is that $P_{out} = (P_{in} - \text{transformer power losses})$. With this in mind, let's assume we had an ideal loss-less transformer in which $P_{out} = P_{in}$. Since power is voltage times current, this product must be the same on the output as it is on the input. This implies that a 1:10 step-up transformer with the voltage stepped up by a factor of 10 results in an output current reduced by a factor of 10. This is what happens on a current transformer. If a transformer had a one-turn primary and a ten-turn secondary, each amp in the primary results in 0.1A in the secondary, or a 10:1 current ratio. It's exactly the inverse of the voltage ratio — preserving volt times current product.

How can we use this transformer and knowledge to produce something useful? Normally, an engineer wants to produce an output on the secondary proportional to the primary current. Quite often, this output is in volts output per amp of primary current. The device that monitors this output voltage can be calibrated to produce the desired results when the voltage reaches a specified level.

A burden resistor connected across the secondary produces an output voltage proportional to the resistor value, based on the amount of current flowing through it. With our 1:10 turns ratio transformer that produces a 10:1 current ratio, a burden resistor can be selected to produce the voltage we want. If 1A on the primary produces 0.1A on the secondary, then by Ohm's law, 0.1 times the burden resistor will result in an output voltage per amp.

Many voltage transformers have adjusted ratios that produce the desired output voltage and compensate for losses. The turns-ratios or actual turns aren't the primary concern of the end-user. Only the voltage output and possibly regulation and other loss parameters may be of concern. With current transformers, the user must know the current ratio to use the transformer. The knowledge of amps in per amps out is the basis for use of the current transformer. Quite often, the

end users provide the primary with a wire through the center of the transformer. They must know what secondary turns are to determine what their output current will be. Generally, in catalogues, the turns of the transformers are provided as a specification for use.

With this knowledge, the user can choose the burden resistor to produce their desired output voltage. The output current of 0.1A for a 1A primary on the 1:10 turns ratio transformer will produce 0.1 V/A across a 1Ω burden resistor, 1V per amp across a 10Ω burden and 10V per amp across a 100Ω burden resistor.

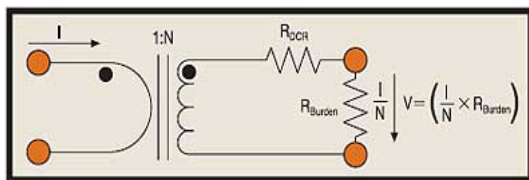


Fig.1. Ideal current transformer circuit.

Fig. 1 shows an ideal transformation ratio. In this analysis, the secondary dc resistance (R_{DCR}) doesn't become part of the calculation. When considering the secondary current, only the actual current affects V . How well that current can be determined controls the accuracy of the prediction of V . The secondary dc resistance is best analyzed by reflecting it to the primary by R_{DCR}/N^2 .

When choosing the burden resistor, the engineer can create any output voltage per amp, as long as it doesn't saturate the core. Core saturation level is an important consideration when specifying current transformers. The maximum volt-microsecond product specifies what the core can handle without saturating. The burden resistor is one of the factors controlling the output voltage. There's a limit to the amount of voltage that can be achieved at a given frequency. Since frequency = $1/\text{cycle period}$, if the frequency is too low (cycle period too long) so that voltage-time product exceeds the core's flux capacity, saturation will occur. The flux that exists in a core is proportional to the voltage times cycle period. Most specifications provide a

maximum volt-microsecond product that the current transformer can provide across the burden resistor. Exceeding this voltage with too large a burden resistor will saturate the transformer and limit the voltage.

What happens if the burden resistor is left off or opens during operation? The output voltage will rise trying to develop current until it reaches the saturation voltage of the coil at that frequency. At that point, the voltage will cease to rise and the transformer will add no additional impedance to the driving current. Therefore, without a burden resistor, the output voltage of a current transformer will be its saturation voltage at the operating frequency.

There are factors in the current transformer that affect efficiency. For complete accuracy, the output current must be the input current divided by the turns ratio. Unfortunately, not all the current is transferred. Some of the current isn't transformed to the secondary, but is instead shunted by the inductance of the transformer and the core loss resistance. Generally, it's the inductance of the transformer that contributes the majority of the current shunting that detracts from the output current. This is why it's important to use a high-permeability core to achieve the maximum inductance and minimize the inductance current. Accurate turns ratio must be maintained to produce the expected secondary current and the expected accuracy. Fig. 2 shows the current transformed is smaller than the input current by:

$$I_{\text{TRANSFORMED}} = I_{\text{INPUT}} - I_{\text{CORE}} - jI_{\text{MAG}} \quad (1)$$

What about the effect the transformer will have on the current it's monitoring? This is where the term burden enters the picture. Any measuring device alters the circuit in which it measures. For instance, connecting a voltmeter to a circuit causes the voltage to change from what it was before the meter was attached. However minuscule this effect may or may not be, the voltage you read isn't the voltage that existed before attaching the meter. This is also true with a current transformer. The burden resistor on the

secondary is reflected to the primary by $(1/N^2)$, which provides a resistance in series with the current on the primary. This usually has minimal effect and is usually only important when you are concerned about the current that would exist when the transformer isn't in the circuit, such as when it's used as a temporary measuring device.

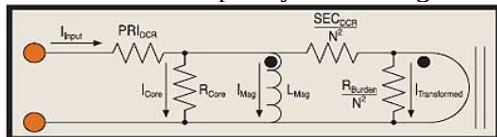


Fig. 2. Current transformer loss components.

Notice the four loss components in the circuit of Fig. 2. The resistance of the primary loop (PRI_{DCR}), the core loss resistance (R_{CORE}), the secondary DCR (R_{DCR}) is reduced by $1/N^2$, and the secondary burden resistor R_{BURDEN} is also reduced by a factor of N^2 . These are losses that affect current source (I). The resistances have an indirect effect on the current transformer accuracy. It's their effect on the circuit that they are monitoring that alters its current. The primary dc resistance (PRI_{DCR}) and the secondary DCR/N^2 (R_{DCR}/N^2) don't detract from the I_{input} that is read or is affecting the accuracy of the actual current reading. Rather, they alter the current from what it would be if the current transformer weren't in the circuit. With the exception of the burden resistor, these loss resistors are the components that contribute to the loss in the transformer and heating.

This wasted energy is usually small compared with the power in the circuit it's monitoring. Usually, the design of the transformer and choice of the burden resistor will be within the maximum energy loss the end user can allow. As battery-operated devices come into wider use and power consumption contributes to the energy crisis — even this power may be of concern. Under these circumstances, it may require special design attention to power consumption.

Current transformers are an efficient way to measure current. Since the burden resistor is reflected to the primary by $1/N^2$, the resistance seen in the circuit being

monitored can be very small. This allows a larger voltage to be created on the output with minimal effect on the circuit being measured. A simpler and lower-cost method to measure current is to use a sense resistor connected in series with the current. However, this method can only be used when power consumption is of secondary concern. With the more frequent use of battery-powered devices and the prevailing need to reduce power consumption, the extra expense of a current transformer can soon be recovered with use. Also, with high current or when a voltage of any magnitude is required, a sense resistor would be impractical.

Global Dimming:

Burning of fossil fuels is creating two effects

Two effects of fossil fuel productions are:

- Greenhouse gases that cause global warming
- By-products which are pollutants that cause global dimming

What is global dimming?

Fossil fuel use, as well as producing greenhouse gases, creates other by-products. These by-products are also pollutants, such as sulphur dioxide, soot, and ash. These pollutants however, also change the properties of clouds.

Clouds are formed when water droplets are seeded by air-borne particles, such as pollen. Polluted air results in clouds with larger number of droplets than unpolluted clouds. This then makes those clouds more reflective. More of the sun's heat and energy is therefore reflected back into space. This reduction of heat reaching the earth is known as Global Dimming.

Impacts of global dimming: millions already killed by it?

Global warming results from the greenhouse effect caused by, amongst other things, excessive amounts of greenhouse gases in the earth's atmosphere from fossil fuel burning. It would seem then, that the other by-products which cause global dimming may be an ironic savior. A deeper look at this, however, shows that unfortunately this is not the case.

Health and environmental effects

The pollutants that lead to global dimming also lead to various human and environmental problems, such as smog, respiratory problems, and acid rain. The impacts of global dimming itself, however, can be devastating.

Millions from Famines in the Sahel in the 70s and 80s

The death toll that global dimming may have already caused is thought to be massive. Climatologists studying this phenomenon believe that the reflection of heat have made waters in the northern hemisphere cooler. As a result, less rain has formed in key areas and crucial rainfall has failed to arrive over the Sahel in Northern Africa. In the 1970s and 1980s, massive famines were caused by failed rains which climatologists had never quite understood *why* they had failed.

The answers that global dimming models seemed to provide, the documentary noted, has led to a chilling conclusion: "what came out of our exhaust pipes and power stations [from Europe and North America] contributed to the deaths of a million people in Africa, and afflicted 50 million more" with hunger and starvation.

Billions are likely to be affected in Asia from similar effects

Scientists said that the impact of global dimming might not be in the

millions, but billions. The Asian monsoons bring rainfall to half the world's population. If this air pollution and global dimming has a detrimental impact on the Asian monsoons some 3 billion people could be affected.

As well as fossil fuel burning, contrails is another source

Contrails (the vapor from planes flying high in the sky) were seen as another significant cause of heat reflection. During the aftermath of the September 11, 2001 terrorist attacks in the United States, all commercial flights were grounded for the next three days. This allowed climate scientists to look at the effect on the climate when there were no contrails and no heat reflection. What scientists found was that the temperature rose by some 1 degree centigrade in that period of 3 days.

Global Dimming is hiding the true power of Global Warming

The above impacts of global dimming have led to fears that global dimming has been hiding the true power of global warming.

Currently, most climate change models predict a 5 degrees increase in temperature over the next century, which is already considered extremely grave. However, global dimming has led to an underestimation of the power of global warming. Addressing global dimming only will lead to massive global warming. Global dimming can be dealt with by cleaning up emissions. However, if global dimming problems are only addressed, then the effects of global warming will increase even more. This may be what happened to Europe in 2003.

In Europe, various measures have been taken in recent years to clean up the emissions to reduce pollutants that create smog and other problems, but without

reducing the greenhouse gas emissions in parallel. This seems to have had a few effects:

- This may have already lessened the severity of droughts and failed rains in the Sahel.
- However, it seems that it may have caused, or contributed to, the European heat wave in 2003 that killed thousands in France, saw forest fires in Portugal, and caused many other problems throughout the continent.

The documentary noted that the impacts of addressing global dimming *only* would increase global warming more rapidly. Irreversible damage would be only about 30 years away. Global level impacts would include:

- The melting of ice in Greenland, which would lead to more rising sea levels. This in turn would impact many of our major world cities
- Drying tropical rain forests would increase the risk of burning. This would release even more carbon dioxide into the atmosphere, further increasing global warming effects. (Some countries have pushed for using “carbon sinks” to count as part of their emission targets. This has already been controversial because these store carbon dioxide that can be released into the atmosphere when burnt. Global dimming worries increase these concerns even more.)

These and other effects could combine to lead to an increase of 10 degrees centigrade in temperature over the next 100 years, not the standard 5 degrees which most models currently predict.

This would be a more rapid warming than any other time in history, the

documentary noted. With such an increase,

- Vegetation will die off even more quickly
- Soil erosion will increase and food production will fail
- A Sahara type of climate could be possible in places such as England, while other parts of the world would fare even worse.
- Such an increase in temperature would also release one of the biggest stores of greenhouse gases on earth, methane hydrate, currently contained at the bottom of the earth’s oceans and known to destabilize with warming. This gas is eight times stronger than carbon dioxide in its greenhouse effect. As the documentary also added, due to the sheer amounts that would be released, by this time, whatever we would try to curb emissions, it would be too late.

“This is not a prediction,” the documentary said, “it is a warning of what will happen if we clean up the pollution while doing nothing about greenhouse gases.”

**BY,
RAMASAMY RAJA .R,
RAM BALAJI .T.R.S,
III-D.**

LIM – ELECTRONIC SPEED LIMITER:

Mercedes Benz cars apply a variable speed limiter called the LIM. All Mercedes cars available in the market today have LIM facilities unlike the older versions, which used to have the cruise controller instead.

The LIM or speed limiter is an optional facility provided to the driver, which enables him to set a maximum cruising speed between thirty kilometers per hour and the cars maximum possible speed. Once this maximum speed has been set, it gets stored and appears as an icon in the cars instrument cluster. This provides the driver to increase the accelerating velocity by a single kilometer or ten kilometers at a time to attain the desired limit.

The LIM in a Mercedes Benz can be regarded as an extra precaution against accidents or creation of risky situations. It is an innovative step taken by Mercedes Corporation to ensure better road safety. This is generally in the form of a display indicator. Any violation of this velocity is closely monitored by the speed limiter in the car, which automatically lets the driver know in the form of visual or sound alerts. This device often comes in use when there is a pre-defined speed limit to follow or while traversing an unknown route.

The LIM in Mercedes Benz car's often comes with a facility called speed limit assistance. This facility along with the LIM becomes the ultimate tool to avert accidents. While the latter is used to inform the driver of the relevant issues regarding speed limits and signs, the later can be used to fix required limits hence reducing the toil of the driver. While there are huge assortments of items devised to help control speed on cars, the LIM facility of Mercedes Benz has surely redefined all its competitors.

Speedtronic variable speed limiter is the further development of the conventional cruise control explained above, which allows the driver to set a maximum speed limit for themselves (the car) in the range between 30km/h and the vehicles maximum speed.

Pressing the "Lim" switch on the cruise control stalk sets the speed limiter. The desired maximum speed is then stored and appears on the central display in the instrument cluster. A small orange light is lit on the stalk to indicate that the speed limiting function will be in use if the stalk is moved, as opposed to the Cruise control function.

**BY,
RAMASAMYRAJA .R,
RAM KUMAR .R,
III-B.**

DANGER-SENSING DRIVER SEAT:

A car seat pokes you in the ribs when an accident looms.

Today's carmakers boast of their new vehicles' cocoon like qualities , including the latest noise -reducing features and great shocks that smooth out those pumbs on the road. Myriad warning alarms are designed to counteract the insular environment when there's trouble, but Yale university engineers are looking for a more subtle approach to signaling: vibrating motors and movable cams that send signals straight from the seat back to your back.

Such a system could be particularly useful for warning you of cars in your blind spot- an area you can't see with either the side - view or rearview mirrors. The blind spot is blamed for sideswipe accidents that occur by the hundreds of thousands every year. Today's blind-spot monitors warn you by flashing a signal in the rearview mirror or on the car's dashboard.

But according to john morel, a former seaway engineer and now an assistant professor of mechanical engineering at Yale, a visual signal is

the wrong one to send. The visual sense, he says, is already saturated in car. On top of that morel adds, an alert that appears in front of you about something that's behinds you means you have to translate the information, costing crucial response time as well as mental effort that could distract you from other dangers.

Instead, Morrell's lab has designed a car seat that acts as a tactile interface between the driver and the environment. It uses vibrating motors as well as servos that press on the driver's back to signal the location of a following car.

The ultimate goal is a "renaissance user interface", morel says of the design," using the mind and body to the fullest". The design was inspired by his work at segway, where "every part of your body is coupled into the experience."

A demonstration setup includes a modified car seat, a steering wheel, foot pedals, and a computer running an open-source driving stimulator called TORCS, short for "the open racing car stimulator." Twenty cell phone-motor tactors are arranged in a rectangular array across the back of the seat. A car coming up directly behind the driver in the stimulation activates the centre vibrators, while a car to the right or the left will activate the same-side vibrators, giving the driver a directional cue. The closer a car gets, the more intense the vibration. Two cams on each side add to the signal strength by pressing on the driver's rib cage when a car shows up on either side. In their preliminary study, the yale engineers calculated the time a driver spent with another car in its blind

spot during a "commuting exercise". Drivers were also given simultaneous math tasks to simulate distraction during driving. The results showed that the vibrotactile feedback improved drivers' performance over that attained by using the rearview mirror alone.

It's not the first time engineers have experimented with tactile signals to compensate for sensory overload during driving. Researchers at general motors and TNO, a Dutch research organization, found that test subjects could actually pick up directional cues from a vibrating seat cushion while driving. Such a system could provide direct navigation information or even help warn of imminent collisions.

If systems like these turn out to be useful, car interiors will stay peaceful but may get more informative.

By,
P.Bhuvaneswari,
III –A.

ROLE OF INTERNET IN HEALTH CARE:

Growth of the Internet and of the telecommunications sector has affected management in all economic sectors including health care. Convergence between enterprise intranets and the Internet contributes to development of services such as content distribution and access infrastructures directly offered on the Internet. Examples in the health care sector include e-mail services among health care professionals and patients, discussion and support groups, messaging and fax services, specialized

information search, portal infrastructures, and Web hosting. Management of such services on the Internet reduces the need for specialized personnel such as Webmasters and network managers within traditional health care institutions. Emergence of software applications - ASPs (Application Service Providers) - and infrastructures on the Internet reduces software licensing as well as human resources and other administrative costs associated with local acquisition and management. Database management outsourced to Internet infrastructures reduces network complexity and overload while increasing the speed, reliability, and rigor of information searching and processing. Medical errors often occur in the clinical communication space, and they may be associated with cognitive difficulties of processing complex information as well as limitations of human memory. Cognitive strategies for coping with such difficulties include development of selective schemata for retention of critical information and use of asynchronous communication channels to manage memory overload in an interruptive communication environment. The type of computational support or telecommunications infrastructure appropriate to clinical decision making may be associated with the common ground shared by human and technological agents in communication processes, as well as the context of such common ground. Stable common ground shared by actors in the clinical decision-making system may be associated with asynchronous modes (prerecorded; store-and-forward) of information transmission, and active computational modeling based on preemptive information storage. For

example, perfecting methods may be used to restore archived images to workstations in anticipation of needs for historical case comparisons, patient management, or clinical problem solving. On the other hand, shifting ground may require isochronous (real-time) telecommunications transmission, and conversation through passive computing channels. It is critical to note that real-time talk accounts for as much as 50% of clinical information transactions. This continues to be necessary in increasingly interdisciplinary and diverse medical teams despite availability of advanced information technologies for clinical decision support. Modeling clinical decision making in practice requires holistic analysis of human, software, and infrastructural components of both clinical decision making and human collaboration focused on goal achievement. Cultural and ideological assumptions underlying work practice determine the dynamics of system-wide behavior.

Diverse models of service distribution and payment promote entry of new institutional actors into the health care system, while development of advanced telecommunications infrastructures may support either centralized health care authority or free-market dynamics driven by consumer demand. Consumer participation is generally associated with system emphasis on preventive health care strategies while at the same time managerial tools are designed to control health care costs. Limited health care resources may be associated with participative rationalization of health care services by systematic gathering and integration of individual and collective consumer preferences in

methodological models for budgeting and other resource allocation.

The efficiency of rationalization in health care markets depends upon the principles of universal access to information and services and the symmetry of information flows among consumers and health care professionals. (Symmetry of information flows refers to a consequence of universal access to information: consumers have access to information as do health care professionals. In traditional health care organizations, consumers do not have such access and so are unable to participate in decision making.) Some research reports, for example, that consumer information with participative decision making and risk sharing may reduce demand for expensive medical procedures such as surgery.

**By,
V.Indumathy,
III –B.**

MEN ARE EMOTIONAL ROBOTS:



All of us have feelings and emotional reactions to certain things in life. At least in Western cultures, women typically have the luxury of expressing their feelings directly, and to seek the support of friends and family, whereas men have little choice but to hide these emotions/problems or they will usually

face social scrutiny and be labeled as weak, emo or whatever.

There are four basic emotions: sadness, fear, anger, and happiness. The only real acceptable emotions men can exhibit without being labeled as "weak" are anger and happiness. The truth is that every one, not just women experience fear and sadness. I hear it said so many times that men have it easier in relationships/breakups because they don't get attached at the same level emotionally like women do. The truth is men go through the exact same experience but have been socialized to hide it or risk being considered deviant or weak among others and I believe this makes the experience harder.

Women love confident men, I am as sure of that as I am of anything. But somehow men who actually exhibit these normal emotions are considered to lack confidence. So even though many men have fear over their future career, financial situation, life path, or the countless other things in life, we have no choice but to hide our true feelings to appear confident and masculine. These notions of desired male behavior are embodied in the cultural stereotypes of the heroic male, often represented in film and television. Fearless, resourceful, stoic and usually facing

Inflatable Women and Robot Babies

Japan is in big trouble. What's the trouble? A lack of babies is destroying the country's economy. The birthrate in Japan is about 1.25 children per woman. 2.1 is the replacement rate. So what you have are no babies and lots of old people. Not exactly a roadmap to productivity, is it? So some folks have thought of a way to get people hooked

on babies. How? By building robot babies. Yup Robot babies. How sad is it that so many babies are killed that the only way folks can figure out to teach people that babies are great is by building robot babies? This is a very very strange world. I can't help but think that this will have the same effect as inflatable women. Blow up dolls don't really make men like women more, do they? I mean, I've never heard of a great husband saying he owes his great relationship to his human wife to what he learned from his inflatable spouse. Having an inflatable date kinda' guarantees you'll never actually get a real live woman, doesn't it? I fear that the robot babies will have a similar effect, which is to say none at all in actually producing real babies.



BY,
R.DIVYA,
A.JAYARANI,
III-B.

MICRO ELECTRO MECHANICAL SYSTEMS (MEMS)

Imagine a machine so small that it is imperceptible to the human

eye. Imagine working machines no bigger than a grain of pollen. Imagine thousands of these machines batch fabricated on a single piece of silicon, for just a few pennies each. Imagine a world where gravity and inertia are no longer important, but atomic forces and surface science dominates. Imagine a silicon chip with thousands of microscopic mirrors working in unison, enabling the all optical network and removing the bottlenecks from the global telecommunications infrastructure. You are now entering the micro domain, a world occupied by an explosive technology known as MEMS. A world of challenge and opportunity, where traditional engineering concepts are turned upside down, and the realm of the "possible" is totally redefined.

Micro-Electro-Mechanical Systems (MEMS) is the integration of mechanical elements, sensors, actuators, and electronics on a common silicon substrate through micro fabrication technology.

While the electronics are fabricated using integrated circuit process sequences the micromechanical components are fabricated using compatible micromachining processes that selectively etch away parts of the silicon wafer or add new structural layers to form the mechanical and electromechanical devices.

Bulk micromachining:

The whole thickness of a silicon wafer is used for building the micro-mechanical structures. Silicon is machined using various etching process. Anodic bonding of glass plates or additional silicon wafers is used for adding features in the

third dimension and for hermetic encapsulation. Bulk micromachining has been essential in enabling high performance pressure accelerometer sensors and that have changed the shape of the sensor industry in the 80's and 90's.

Surface micromachining:

Surface micromachining uses layers deposited on the surface of a substrate as the structural materials, rather than using the substrate itself. Surface micromachining was created in the late 1980s to render micromachining of silicon more compatible with planar integrated circuit technology, with the goal of combining MEMS and integrated circuits on the same silicon wafer. The original surface micromachining concept was based on thin polycrystalline silicon layers patterned as movable mechanical structures and released by sacrificial etching of the underlying oxide layer. Inter-digital comb electrodes were used to produce in-plane forces and to detect in-plane movement capacitively. This MEMS paradigm has enabled the manufacturing of low cost accelerometers.

MEMS Products

Accelerometer:

An accelerometer is a device that measures proper acceleration, the acceleration experienced relative to freefall. Single and multi-axis models are available to detect magnitude and direction of the acceleration as a vector quantity, and can be used to sense orientation, acceleration, vibration shock, and falling. Micro machined accelerometers are increasingly present in portable electronic devices and video game controllers, to detect the position of the device or provide for game input.

A pressure sensor measures pressure, typically of gases or liquids. Pressure is

an expression of the force required to stop a fluid from expanding, and is usually stated in terms of force per unit area. A pressure sensor usually acts as a transducer; it generates a signal as a function of the pressure imposed. Pressure sensors are used for control and monitoring in thousands of everyday applications

Interferometric Modulator :

The Interferometric Modulator (iMoD) is a technology used in electronic displays that can create various colours through the interference of reflected light. An iMoD based reflective flat panel display can include hundreds of thousands of individually addressable iMoD elements. iMoD displays represent one of the foremost examples of a (MEMS) based device

**BY,
Allen Antony,
III- A.**

MICROCONTROLLER BASED POWER THEFT IDENTIFICATION:

ABSTRACT

Science and technology with all its miraculous advancements has fascinated human life to a great extent that imagining a world without these innovations is hardly possible. While technology is on the raising slope, we should also note the increasing immoral activities. With a technical view, "Power Theft" is a non-ignorable crime that is highly prevalent, and at the same time it directly affects the economy of a nation. Data collected over Tirunelveli District,

Bell Trichy proves the necessity of this project.

Detecting and eradicating such crimes with the assistance of the developing scientific field is the "Need of the Hour". With these views was this paper conceived and designed. Our paper provides a complete and comprehensive tool to prevent power theft which is very simple to understand and easy to implement (Accepted by T.N.E.B officials). It includes four sections - transmitting, receiving, counter display and processing sections. **DESCRIPTION OF OUR IMPLEMENTATION IDEAS:** The disc revolutions are sensed into pulses by optical slot sensor. These pulses are shaped and given as control signal to the CMOS switch which bypasses carrier wave generated by PLL provides as input to receiving section where transmitted signal is selected by the Intermediate frequency transformer. For each lock a pulse is sent out. The counter section is designed to send out pulse for every six input pulse from the receiver section. This count is parallel distributed in a 7-segment display and then to uc for further processing. uc performs the function of indication and identification. Pin details, features, connections and software employed for uc89c51 are described in detail. We believe our implementation ideas are a boon to the electricity board offering them a chance to detect accurately the location and amount of power theft. Logical view for a digital meter is also included in our presentation.

INTRODUCTION:

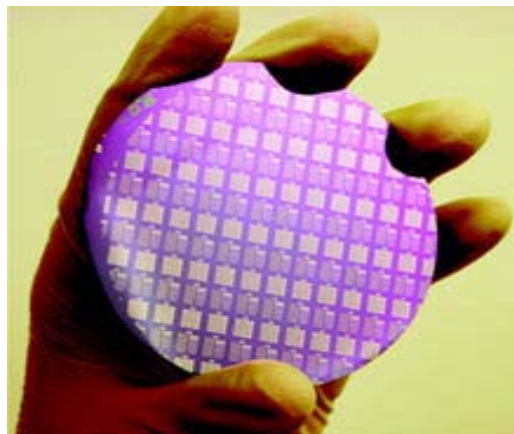
"TODAY'S TECHNICIANS ARE SO FOCUSSED ON THE TREES OF TECHNOLOGICAL CHANGE THAT THEY FAIL TO SEE THE FOREST;

THE UNDERLYING ECONOMIC FORCES THAT DETERMINE SUCCESS AND FAILURE..."
"TECHNOLOGY CHANGES ECONOMY LAWS DO NOT"

Electricity is the modern man's most convenient and useful form of energy without which the present social infrastructure would not be feasible. The increase in per capita production is the reflection of the increase in the living standard of people. When importance of electricity is on the increasing side, then how much should theft of this energy or illegal consumption of power from the transmission lines be averted? Power theft has become a great challenge to the electricity board. The dailies report that Electricity Board suffers a total loss of 8 % in revenue due to power theft every year, which has to be controlled. Our paper identifies the Power theft and indicates it to the Electricity board through Power line. We had also dealt about the remote monitoring of an energy meter.

MICROCONTROLLER BASED AUTOMATION:

Embedded systems - a combination of software, hardware and additional mechanical parts that together forms a component of a larger system, to perform a specific function. Automation has made the art of living comfortable



and easy. Embedded systems have made the process of automation a most successful one. Here, we have focused on automotive, an area of embedded controllers, in which we have dealt with the Power theft identification and also about the remote monitoring of an energy meter.

"Technology has taken the world by storm

performance ratings and exceptionally value for money prices"

The microcontroller chip is preprogrammed to perform a dedicated or a narrow range of functions as a part of a larger system, usually with minimal user or

operator intervention. Our paper throws light on automated monitoring of theft identification, which is an application of embedded controllers.

MODES OF THEFT:

It has been seen that there are 4 common methods of power theft as given below:-

Bogus seals and tampering of seals. Meter tampering, meter tilting, meter interface and meter bypassing. Changing connection.

Direct tapping from line. Due to introduction of modern electronic metering equipments, power thieves are utilizing more technological methods. Recent cases of power theft discovered by British inspectors included customers tunneling out to roadside mains cables and splicing into the supply, a garage taking its night time power supply from the nearest lamp post and domestic customers drilling holes into meter boxes and attempting to stop the counter wheels from turning. Another method of Power theft is by keeping a strong

magnet in front of the disc in the energy meter and thus arresting the rotation of the disc, connecting the load directly to the power line bypassing the energy meter. But, it can be avoided easily by providing a non magnetic enclosure.

Besides spotting out the line where power theft is suspected to occur, it also detects the amount of energy stolen. Compact size, lightweight for quick and high accuracy make the system more effective.

Power theft identification, in this paper, is done by converting the disc revolutions of each consumer's energy meter and distribution transformer into pulses. These pulses are frequency division multiplexed and transmitted

PRACTICAL NANOTUBE ELECTRONICS:

Researchers develop a new method for making efficient nanotube transistor arrays.

Carbon nanotubes are a promising material for making display control circuits because they're more efficient than silicon and can be arrayed on flexible surfaces. Until recently, though, making nanotubes into transistors has been a painstaking process. Now researchers at the University of Southern California have demonstrated large, functional arrays of transistors made using simple methods from batches of carbon nanotubes that are relatively impure.

The pixels in a computer or television screen, whether it's an LCD or plasma, are each controlled by several transistors. In today's devices, these transistors are made from silicon. Arrays of these transistors need to be made at high temperatures and in a vacuum, so

they're very expensive, says Chongwu Zhou, an associate professor of electrical engineering at USC and researcher on the nanotubes project.

Transistors have also been made from carbon nanotubes, but that, too, presents challenges. "Many people use one nanotube to make a very small, high-performance transistor" for computer chips, says Zhou. But that one-to-one ratio won't work for displays, in which a large surface must be covered in transistors. "If we use one nanotube for one transistor, the yield will never be high enough" to work for large-scale manufacturing of big screens, he says. Zhou believes his approach will solve this problem by making larger transistors from mats of nanotubes.

The USC researchers make large arrays of carbon nanotube transistors using solution-processing techniques at room temperature. They start by placing a silicon wafer in a chemical bath to coat its surface with a nanotube-attracting chemical, then rinse off the residue. The treated wafer is then immersed in a solution of semiconducting carbon nanotubes, which are attracted to its surface. The wafer, now coated with a carpet of nanotubes, is rinsed clean again. To make transistors from this tangled mess, the researchers put down metal electrodes at selected locations. The electrodes define where each transistor is and carry electrons into and out of the nanotubes that lie between them. Areas of silicon underlying each device act as the transistors' gates. So far, they've built a prototype device on a four-inch silicon wafer and used it to control a simple organic light-emitting diode display. This work is described online in the journal *Nano Letters*.

Other researchers have made transistors from nanotube carpets using solution-

processing, but these projects started from mixtures of conducting and semiconducting nanotubes, leading to very poor performance. And late last year, researchers at IBM and Northwestern University used highly purified semiconducting nanotubes to make higher-performance transistor arrays in which all the nanotubes line up in nearly straight lines, which improves their electrical properties.

"This is the first time someone has shown solution-deposited, purified semiconducting tubes for high-quality transistors," says John Rogers, professor of materials science and engineering at the University of Illinois at Urbana-Champaign. "The accomplishment is in the integration of several promising approaches to demonstrate a full sequence for the fabrication of electronics."

Now that his group has demonstrated the feasibility of these techniques, Zhou says, it's working to build a truly integrated organic LED display that is flexible and transparent. Such a display might be rolled up to fit in a pocket, or mounted on a car windshield to display information to the driver. The first step is eliminating the rigid silicon. Because the nanotubes may be laid down at room temperature, the USC researchers can build them on electrically active plastic sheets that can't tolerate high temperatures. They're also working to replace the stiff metal electrodes with a coating of indium tin oxide, a commonly used, flexible, transparent electrode material. In their prototype, the organic LED pixels are connected to the transistor array by wires; to integrate them they'll need to come up with methods for building the LEDs on top of the control circuit.

Zhou says he is talking with display companies about commercializing these methods. Korean display giant LG has demonstrated interest in carbon nanotube electronics, and IBM researchers have been publishing on the topic. However, the only company to come out with a nanotube electronic product so far is the Menlo Park, CA startup Unidym, which makes electrodes from the material.

Researchers in the field have been talking about nanotube displays for years, and the holdup, says Mark Hersam of Northwestern University, has been the lack of a big enough supply of semiconducting carbon nanotubes. In 2006, materials science professor Hersam developed a simple method for purifying nanotubes based on their properties by centrifuging them in a soapy solution.

By,
RAM PRASAD .D,
RAM BALAJI .T.R.S,
III –D.

INDIA WORKING ON UAV ANTI-COLLISION SYSTEM:

Indian aerospace scientists have developed an in-flight collision avert system that prevents unmanned aerial vehicles from crashing into enemy aircraft or other objects.

The model predictive static programming algorithm protection system, developed at the Indian Institute of Science, uses a series of installed collision guidance algorithms as instructions that allow the UAVs to detect objects, especially if they are flying low.

This includes tall buildings, towers and other aircraft, including commercial passenger planes.

The MPSP Algorithm can also be used in medium- and long-range missiles to ensure they don't crash into objects such as anti-missile missiles as they approach their own target. MPSP can redirect the missiles back on course to their target without loss of accuracy.

The developer, Radhakant Padhi, 37, said he has been working on algorithms for aerospace for more than a decade and perfected the algorithm technology during his project related to advanced missile technology at the INDIAN Institute of Science in Bangalore.

In 2005, Padhi developed an advanced version of the algorithm, called the MPSP algorithm, while working on one of India's missile guidance systems.

Padhi also said he received \$80,000 of funding from Air Force Research Lab in the United States to further develop the MPSP Algorithm.

AFRL, operated by the U.S. Air Force Materiel Command, controls the Air Force science and technology research budget. The laboratory was formed in 1997 at Wright-Patterson Air Force Base. It was a consolidation of the four Air Force laboratory facilities of Wright, Phillips, Rome, and Armstrong as well as the Air Force Office of Scientific Research.

It has worked with NASA, Department of Energy National Laboratories, Defense Advanced Research Projects Agency and other research organizations within the U.S. Department of Defense. Projects include the X-37, X-40, X-53, and HTV-3 X, YAL-1A, Advanced Tactical Laser and the Tactical Satellite Program.

AFRL, as with similar technology research establishments in the United

States, is facing a staffing problem as 40 percent of its workers are set to retire over the next two decades. The country also isn't producing enough scientists to keep up with job vacancies.

One reason for the personnel shortage is a large percentage of science and engineering graduates in the United States are foreign citizens who aren't eligible for work because of security clearances needed for many of the jobs. Government statistics show that 60 percent of all doctoral candidates in the sciences are foreign-born, a report in The Boston Globe newspaper said last year.

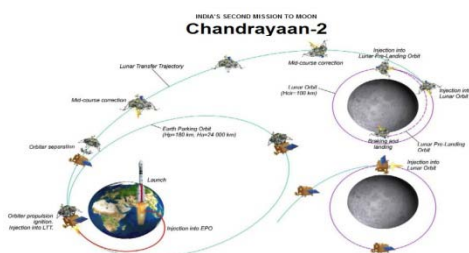
"If the requirement is you have to be a U.S. citizen, then you have a large pool that simply isn't eligible," said Mark Regets, a senior analyst at the National Science Foundation who tracks science and engineering graduates and workforce.

Air Force Materiel Command is looking to fill more than 5,000 positions by October 2011, many of them in chemistry, physics, and electrical, aeronautical and environmental engineering. Jobs include researching cleaner fuels, laser-guided weapons, UAVs and cyber protection.

**BY,
RAM BALAJI T.R.S
RAM PRASAD .D**

III –D.

CHANDRAYAAN:



Chandrayaan-2 will consist of the spacecraft and a landing platform with

the moon rover.

The platform with the rover will detach itself off after the N: spacecraft reaches its orbit above the moon, and land on lunar. A motorised rover will be released on the moon's surface from the Lander. The location for the Lander will be identified using data from Chandrayaan-1 payload MIP .

In Chandrayaan-1, MIP will detach itself from the spacecraft and it will impact on the moon's surface. The MIP will have three instruments. Annadurai, its mass spectrometer will sense the moon's atmospheric constituents as it keeps falling for 18 minutes and crashes on the moon. Its altimeter will measure the instantaneous altitude during its descent. Its video-imaging system will look at the moon from close proximity in order that ISRO scientists may take decisions on the terrain where it will land. The rover will weigh between 30 kg and 100 kg, depending on whether it is to do a semi-hard landing or soft landing. The rover will have an operating life-span of a month. It will run predominantly on solar power.

If ISRO wants to operate the rover for two or three months, its engineers will configure the vehicle and its instruments including a battery back-up to go into a low-power mode, with the rover waking up when sunlight streams through. When the sunlight comes, the solar-powered battery cells will be re-charged and the equipment will be switched on one by one for the rover to function for another two weeks. The batteries will be re-charged every two weeks.

**BY,
T.MATHUMITHA,
III-C.**

INDIA AND SATELLITE COMMUNICATION SYSTEMS:

India first experimented with geosynchronous telecommunications relays in 1981 and now has three active spacecraft in GEO. Moreover, the launch of INSAT 2A in July, 1992, marked the debut of India's first domestically built operational GEO space-craft. In a departure from most nations, India's GEO platforms combine a communications mission with that of Earth observation.

- APPLE
- INSAT 1
- INSAT 2
- INSAT 3
- ASC Network

India undertook the Satellite Instructional Television Experiment (SITE) in 1975-76 to telecast a series of educational programs on health, family planning, agriculture and the like to over 2,500 Indian villages via the US satellite, ATS-6. It was the largest sociological experiment ever carried out in the world. The Satellite Telecommunication Experimental Project (STEP), conducted using Franco-German Symphonie satellite during 1977-79, was another major demonstration of communication applications of space.

India has registered an impressive growth in the telecom sector. Over the years the country has developed a vast telecom network comprising over 25000 telephone exchanges and 21.5 million working connections. There is a large network of optical fibre cables, digital microwave and satellite communication systems. A very strong industrial base has been built in the telecom sector with a large number of national and multinational telecom companies.

A number of policy changes have been made in the recent past which, if implemented, is bound to have a significant impact on the telecom scenario. The most significant among the changes is the announcement of a New Telecom Policy (NTP) 1999. The policy envisages development of telecom facilities in remote, rural and tribal areas of the country and their availability to the masses at affordable costs. The NTP 1999, which has come into effect from April 1, 1999, aims at making telephones available on demand by the year 2002 and to achieve teledensity of seven per hundred persons by the year 2005. In case of rural areas, the current teledensity is proposed to be raised from 0.4 to 4 by the year 2010. The policy document of NTP outlines rapid growth in the telecom sector in India with a projected teledensity of 15 by the year 2010. This will require a massive investment of over 23 billion dollars in the next five.

**By,
A.Nithya,
T.Nishevitha,
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III-D.**

SENSOR LESS MOTOR CONTROL SIMPLIFIES WASHER DRIVES:

While clutch and gearbox assemblies in washing machines have been replaced with belt-driven induction motors with tachometer feedback, digital-control algorithms will enable direct drive from sensor less permanent-magnet motors.

Washing controls have used power inverters to control motor speeds for years. Variable-speed control enables a

significant simplification of the mechanical system by replacing a clutch and gearbox with a belt drive. Washers in North America commonly use a three-phase induction motor with tachometer feedback. The drive typically uses slip torque control during the washing cycle and field weakening for high-speed spinning. Compared to drives that do not provide field weakening, this simple mechanical system enables a higher spin speed that extracts more water from the clothes, which reduces the energy the dryer consumes.

To gain further energy savings, appliance engineers are now introducing laundry algorithms that minimize the hot water that the washing cycle consumes. However, the new algorithms require a higher dynamic response from the drive train so manufacturers seek to replace the belt-drive system with direct-drive permanent-magnet (PM) motors.

Compared to a three-phase induction motor, a direct-drive PM motor uses a larger number of motor poles to generate higher torque at lower speeds for the same power input. The ratio of the drum speed to the motor electrical frequency remains almost the same, but instead of stepping down the motor speed using, for example, a 10-to-1 pulley ratio, increasing the motor-pole count by a factor of 10 achieves the same effect.

Induction motors require a relatively large number of slots per pole and a small air gap that makes it difficult to build motors with high pole counts. PM motors provide much more construction flexibility and are the best choice for the direct drive. A direct-drive motor with an external rotor that connects directly to the washer drum. Concentrated stator windings make it possible to design the motor with a “pancake” construction that

easily fits into the washing machine's available space.

Though PM ac motors provide advantages over induction motors for washer applications, one critical issue to solve is rotor position sensing. Transformer action generates the rotor field in the induction motor, so the field is always synchronous to the stator current. However, the motor-drive circuit must measure the rotor position in the PM machine to synchronize the stator current with the rotor field. Given the rotor position, it is possible to drive the PM ac motor efficiently because the drive circuit can align the stator current to the optimum angle relative to the rotor field.

In the direct-drive motor shown, Hall Effect sensors detect the rotor-magnet position. However, designs that eliminate these sensors alleviate the reliability issues that result from mounting the sensor electronics assembly within the motor housing.

One popular sensorless algorithm measures the rotor-flux position based on the stator winding's back EMF. This algorithm is suitable when driving the PM motor in six-step mode with rectangular winding currents. The algorithm is easy to implement but produces high torque ripple because of delays in current commutation due to the winding inductance.

One of the major drawbacks of this scheme is that the torque ripple causes unacceptably high acoustic-noise levels in the direct-drive motor, as the rotor acts like a loud speaker. Another important problem with the back-EMF-sensing algorithm is that it does not allow high-speed field weakening, which is required for high-speed spinning.

Sensorless Control

A sensor less control algorithm can use motor-winding-current measurements to indirectly determine the rotor-flux position. This algorithm allows for sinusoidal currents to drive the motor. These sinusoidal waveforms produce low torque ripple, which minimizes the acoustic noise. The control method also easily provides for field-weakening operation for high-speed spinning.

Typically, sensor less control algorithms determine the rotor-flux position based on the winding's back EMF — an indirect measurement that derives from the motor circuit model. The advantage of this approach is that it is applicable when sinusoidal-current waveforms drive the motor, which produce smooth, glitch-free torque.

The control diagram describes the field-oriented-control (FOC) algorithm for the permanent magnet synchronous motor (PMSM). A feature of this algorithm is that it transforms the three-phase ac currents in the stator windings into a rotating reference frame where they appear as two dc-current components representing the stator current's torque and flux components.

The reference transformations occur in two stages. The first stage calculates the equivalent stator currents in a two-phase machine model. The second performs a vector rotation to calculate the stator currents' effect in a reference frame synchronous with the rotor shaft.

BY,
S.RAJESWARI,
J.RAJA RAJESWARI,
III-D.

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SMS BASED INDUSTRIAL AUTOMATION USING PERSONAL COMPUTER:

ABSTRACT:

The short message system (SMS) of a standard mobile phone can be used for much more than just exchanging cryptic message. This application finds a humble mobile working in a remote site monitoring and controlling external equipments. The SMS service provides by the service providers are comparatively low cost. Hence the system is highly efficient and low cost. It can be implemented for a variety of industrial applications. In this SMS techniques together with the micro controller technology are used in a wide variety of applications in industry, including computer peripherals, business machines, motion control, and robotics, which are, included in process control and machine tool applications. The automation system is designed by keeping in mind that it should be user friendly and it should be possible to operate it from anywhere in the world. This system is designed in such a way to reduce the wiring complexity and manpower. Thus the system provides an excellent hold as per industrial motion control systems.

Mobile phones have become a widespread means of communication. It becomes a part of everyday life with ever more people enjoying the service and extra freedom they provide. It works on the basis of Global System for Mobile Communication (GSM). A subscriber from any systems can access telecommunication services by using a Subscriber Identify Module (SIM) card in a handset suitable for the network on

the visited system The Short Message Service allows text messages to be sent and received to and from mobile telephones. The text can comprise words or numbers or an alphanumeric combination. Because Simple person - to - person messaging is such an important component of total SMS traffic volumes, anything that simplifies message generation as well as extended utility of the SMS being sent is an important enabler of Short Message Service. Such extended utility of SMS fulfills certain important requirements. Keeping this in mind the research work "SMS BASED INDUSTRIAL AUTOMATION USING PERSONAL COMPUTER " is developed. This system is developed to control the functions of a device from a remote area through the SMS of a mobile phone using personal computer and Micro controller.

The short message service (SMS) of a standard mobile phone can be used for much more than just exchanging cryptic message. This application finds a humble mobile working in a remote site monitoring and controlling external equipment. This system accommodates the SMS techniques together with the micro controller technology that are used in a wide variety of applications in industry via computer peripherals, business machines and robotics that are related to process control and machine tool applications. This Automation system is designed in such a way that it has to be user friendly and it must be possible to operate it from anywhere is the world.



DEFINING NEXT GENERATION RADIO WITH SDR:

Software-defined radio is set to pave the way for the next generation of military communications equipment. For its first 100 years or so, the improvement of radio was driven by the development of better hardware, including improved components, more sophisticated circuits and more precise manufacturing.

The arrival of software-defined radio (SDR) changed all that SDR takes advantage of the processing power of modern computer technology to emulate the behaviour of a radio circuit. The software defines how the radio performs, making it feasible for a single radio to emulate and communicate with many different types. Unlike its hardware-based counterparts, these devices can be improved and updated and given new capabilities by changing the software.

Theoretically, it is possible to reduce SDR to a computer that decodes everything arriving at an antenna that's connected to it. In practice, however, this is not feasible: antenna voltages are tiny, way below anything a computer's analogue-to-digital converters (ADC) can handle, so at the very least an SDR has to have a low-noise amplifier between the antenna and the ADC.

"SDR takes advantage of the processing power of modern computer technology to emulate the behaviour of a radio circuit."

Amplification can create problems, though, because spurious signals from other equipment and even background radiation are also amplified and can distort the desired signals, or even block them completely. One solution is to put band-pass filters between the antenna and the amplifier, but these reduce the radio's flexibility. Some SDRs have several switch-able channels, each with its own filters and amplifiers, increasing discrimination but reducing flexibility and adding complexity to the circuit.

SOFTWARE DESIGN:

With SCA standardising the way software interacts with radio hardware, it is no longer necessary or desirable for each hardware manufacturer to create unique software. Hence, manufacturers now view software as just another component that can be bought, like semiconductors, batteries and switches. Specialist software companies such as Prism Technologies and Wind River exist to supply this market. Prism's Spectra SDR development suite allows SDR manufacturers to create their own compliant waveforms, either new or to match legacy equipment. Wind River produces operating systems for SDR, and the two companies have joined forces to provide a complete, high-performance, commercial off-the-shelf SDR solution for a wide array of hardware.

NEXT STEP - COGNITIVE RADIO:

"The term cognitive radio identifies the point at which wireless personal digital assistants (PDAs) and the related networks are sufficiently computationally intelligent about radio

resources and related computer-to-computer communications to: (a) detect user communications needs as a function of use context, and (b) provide radio resources and wireless services most appropriate to those needs."

What applies to a PDA can apply equally well to military SDR systems. The idea is that radios should be smart enough to find space in a crowded spectrum and use it to join whatever network the operator requests, breaking the link between network, geography and a particular location on the electromagnetic spectrum.

The objective of cognitive radio is to improve interoperability during joint and combined operations, and coexistence with commercial and civil systems. The US DARPA XG (next Generation) programme exists to develop this technology.

Software Defined Radio - Benefits:

- A family of radio "products" to be implemented using a common platform architecture, allowing new products to be more quickly introduced into the market.
- Software to be reused across radio "products", reducing development costs dramatically.
- Over-the-air reducing the time and costs associated with operation and maintenance.

For Radio Service Providers, SDR Enables:

- New features and capabilities to be added to existing infrastructure without requiring major new capital expenditures, allowing service providers to quasi-future proof their networks.
- The use of a common radio platform for multiple markets, significantly

reducing logistical support and operating expenditures.

- Remote software downloads, through which capacity can be increased, capability upgrades can be activated and new revenue generating features can be inserted.

For End Users - from business travelers to soldiers on the battlefield, SDR

Technology aims to:

- Reduce costs in providing end-users with access to ubiquitous wireless

Communications – enabling them to communicate with whomever they need, whenever they need to and in whatever manner is appropriate.

**BY,
RAM BALAJI .T.R.S,
RAM PRASAD .D,
III –D.**

DRIVER LESS CAR

A robotic Volkswagen Passat shown at Stanford University



A driverless car is an autonomous vehicle that can drive itself from one point to another without assistance from a driver; in other words, with an autopilot system. Some believe that autonomous vehicles have the potential to transform the transportation industry

while virtually eliminating accidents, and cleaning up the environment. According to urban designer and futurist Michael E. Arch, driverless electric vehicles—in conjunction with the increased use of virtual reality for work, travel, and pleasure—could reduce the world's 800,000,000 vehicles to a fraction of that number within a few decades.^{[1][2]} Arch claims that this would be possible if almost all private cars requiring drivers, which are not in use and parked 90% of the time, would be traded for public self-driving taxis that would be in near constant use. This would also allow for getting the appropriate vehicle for the particular need—a bus could come for a group of people, a limousine could come for a special night out, and a Segway could come for a short trip down the street for one person. Children could be chauffeured in supervised safety, DUIs would no longer exist, and 41,000 lives could be saved each year in the U.S. alone.

Driverless passenger car programs include the 800 million EC EUREKA Prometheus Project on autonomous vehicles (1987–1995), the passenger vehicles (using the FROG-navigation technology) from the Netherlands, the ARGO research project from Italy, and the DARPA Grand Challenge from the USA.

Navigation

The ability to plot a route from where the vehicle is to where the user wants to be has been available for several years. These systems, based on the US military's Global Positioning System are now available as standard car fittings, and use satellite transmissions to ascertain the current location, and an on-board street database to derive a route to the target. The more sophisticated

systems also receive radio updates on road blockages, and adapt accordingly. There are also sensors that greatly affect the whole nature of it.

Control of vehicle:

As automotive technology matures, more and more functions of the underlying engine, gearbox etc. are no longer directly controlled by the driver by mechanical means, but rather via a computer, which receives instructions from the driver as inputs and delivers the desired effect by means of electronic throttle control, and other drive-by-wire elements. Therefore, the technology for a computer to control all aspects of a vehicle is well understood.

Work done in simulation

While developing control systems for real cars is very costly in terms of both time and money, much work can be done in simulations of various complexity. Systems developed using simpler simulators can gradually be transferred to more complex simulators, and in the end to real vehicles. Some approaches that rely on learning requires starting in a simulation to be viable at all, for example evolutionary robotics approaches.

Autonomous robots:

Autonomous robots are robots which can perform desired tasks in unstructured environments without continuous human guidance. Many kinds of robots have some degree of autonomy. Different robots can be autonomous in different ways. A high degree of autonomy is particularly desirable in fields such as space exploration, cleaning floors, mowing lawns, and waste water treatment.

Examples of progress towards commercial autonomous robots

Self-maintenance



Exteroceptive sensors: 1. blue laser rangefinder senses up to 360 distance readings in a 180-degree slice; 2. 24 round golden ultrasonic sensors sample range readings in a 15-degree cone; 3. ten touch panels along the bottom detect shoes and other low-lying objects. 4. break beams between the lower and upper segments sense tables and other mid-level obstacles.

The first requirement for complete physical autonomy is the ability for a robot to take care of itself. Many of the battery powered robots on the market today can find and connect to a charging station, and some toys like Sony's Aibo are capable of self-docking to charge their batteries.

Self maintenance is based on "proprioception", or sensing one's own internal status. In the battery charging example, the robot can tell proprioceptively that its batteries are low and it then seeks the charger. Another common proprioceptive sensor is for heat monitoring. Increased proprioception will be required for robots to work autonomously near people and in harsh environments

Outdoor autonomous position-sensing and navigation

Outdoor autonomy is most easily achieved in the air, since obstacles are rare. Cruise missiles are rather dangerous highly autonomous robots. Pilotless drone aircraft are increasingly

used for reconnaissance. Some of these unmanned aerial vehicles (UAVs) are capable of flying their entire mission without any human interaction at all except possibly for the landing where a person intervenes using radio remote control. But some drone aircraft are capable of a safe, automatic landing also. Outdoor autonomy is the most difficult for ground vehicles, due to: a) 3-dimensional terrain; b) great disparities in surface density; c) weather exigencies and d) instability of the sensed environment.



The Seekur and MDARS robots demonstrate their autonomous navigation and security capabilities at an airbase. (Courtesy of MobileRobots Inc) In the US, the MDARS project, which defined and built a prototype outdoor surveillance robot in the 1990s, is now moving into production and will be implemented in 2006? The General Dynamics MDARS robot can navigate semi-autonomously and detect intruders, using the MRHA software architecture planned for all unmanned military vehicles. The Seekur robot was the first commercially available robot to demonstrate MDARS-like capabilities for general use by airports, utility plants,

corrections facilities and Homeland Security.^[3]

The Mars rovers MER-A and MER-B (now known as Spirit rover and Opportunity rover) can find the position of the sun and navigate their own routes to destinations on the fly by:

- mapping the surface with 3-D vision
- computing safe and unsafe areas on the surface within that field of vision
- computing optimal paths across the safe area towards the desired destination
- driving along the calculated route;
- repeating this cycle until either the destination is reached, or there is no known path to the destination

**BY,
S.STEPHEN PONSINGH,
II –E.**

SUPER BUG

Summary

Gram-negative Enterobacteriaceae with resistance to carbapenem conferred by New Delhi metallo- β -lactamase 1 (NDM-1) are potentially a major global health problem. We investigated the prevalence of NDM-1, in multidrug-resistant Enterobacteriaceae in India, Pakistan, and the UK.

Methods Enterobacteriaceae isolates were studied from two major centres in India—Chennai (south India), Haryana (north India)—and those referred to the UK's national reference laboratory. Antibiotic susceptibilities were assessed, and the presence of the carbapenem resistance gene bla_{NDM-1} was established by PCR. Isolates were typed by pulsed-field gel electrophoresis of

XbaI-restricted genomic DNA. Plasmids were analysed by S1 nucleasedigestion and PCR typing. Case data for UK patients were reviewed for evidence of travel and recent admission to hospitals in India or Pakistan.

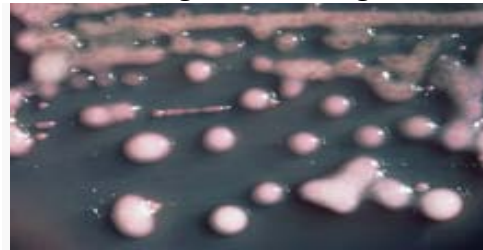
Findings We identified 44 isolates with NDM-1 in Chennai, 26 in Haryana, 37 in the UK, and 73 in other sites in India and Pakistan. NDM-1 was mostly found among *Escherichia coli* (36) and *Klebsiella pneumonia* (111), which were highly resistant to all antibiotics except to tigecycline and colistin. *K. pneumonia* isolates from Haryana were clonally but NDM-1 producers from the UK and Chennai were clonally diverse. Most isolates carried the NDM-1 gene on plasmids:

those from UK and Chennai were readily transferable whereas those from Haryana were not conjugative. Many of the UK NDM-1 positive patients had travelled to India or Pakistan within the past year, or had links with these countries.

Interpretation The potential of NDM-1 to be a worldwide public health problem is great, and co-ordinated international surveillance is needed.

New Delhi Metalloid-beta-lactamase (NDM-1) is an enzyme which makes bacteria resistant to a broad range of beta-lactam antibiotics. This includes antibiotics of the carbapenem family, which are a mainstay for the treatment of antibiotic-resistant bacteria. The gene for NDM-1 is one member of a large gene family that encodes beta-lactamase enzymes called carbapenemases. Bacteria which carry such genes are often referred to in the news media as "superbugs", since infections with these bacteria are very hard to treat successfully. Indeed, the United Kingdom Health Protection Agency has stated that "most isolates with NDM-1

enzyme are resistant to all standard intravenous antibiotics for treatment of severe infections." NDM-1 was first identified in December 2009 in a patient hospitalised in New Delhi with a *Klebsiella pneumoniae* infection. It has since been detected in bacteria in India, Pakistan, the United Kingdom, the United States, and Canada. The most common organisms expressing this enzyme are the Gram-negative bacteria *Escherichia coli* and *Klebsiella pneumonia*, but the gene for NDM-1 can spread from one strain of bacteria to another through horizontal gene transfer.



NDM-1 BACTERIA

Enzyme function :

Carbapenems are a class of beta-lactic antibiotics which are capable of killing most bacteria by inhibiting the synthesis of one of their cell wall layers. The carbapenems were developed to overcome antibiotic resistance mediated by bacterial beta-lactamase enzymes. However, the *bla*_{NDM-1} gene produces NDM-1, which is a carbapenemase beta-lactamase -

an enzyme that hydrolyzes and inactivates these carbapenem antibiotics. Carbapenemases are particularly dangerous resistance mechanisms, since they can inactivate a wide range of different antibiotics.^[4] The NDM-1 enzyme is one of the class B metalloid-beta-lactamase; other types of carbapenemase are class A or class D beta-lactamases. (The class A *Klebsiella pneumonia* carbapenemase (KPC) is

currently the most common carbapenemase, which was first detected in North Carolina, USA, in 1996 and has since spread worldwide.^[6] A later publication indicated that Enterobacteriaceae that produce KPC were becoming common in the United States.^[7]

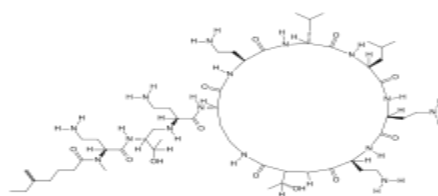
The resistance conferred by this gene ($\text{bla}_{\text{NDM-1}}$) therefore aids the expansion of bacteria that carry it throughout a human host, since they will face less opposition from populations of antibiotic-sensitive bacteria, which will be diminished by the original antibacterial treatment.

Origin spread: The NDM-1 enzyme was named after New Delhi, the capital city of India, as it was first described by Yong et al. in December 2009 in a Swedish national who fell ill with an antibiotic-resistant bacterial infection that he acquired in India. The infection was unsuccessfully treated in a New Delhi hospital and after the patient's repatriation to Sweden, a carbapenem-resistant *Klebsiella pneumoniae* strain bearing the novel gene was identified. The authors concluded that the new resistance mechanism "clearly arose in India, but there are few data arising from India to suggest how widespread it is." In March 2010 a study in a hospital in Mumbai found that the majority of carbapenem-resistant bacteria isolated from patients carried the $\text{bla}_{\text{NDM-1}}$ gene.

In May 2010 a case of infection with *E. coli* expressing NDM-1 was reported in Coventry in the United Kingdom. The patient was a man of Indian origin who had visited India 18 months previously, where he had undergone dialysis. In initial assays the bacteria was fully resistant to all antibiotics tested, while later tests found that it was susceptible to tigecycline and colistin. The authors

warned that international travel and patients' use of multiple countries' healthcare systems could lead to the "rapid spread of NDM-1 with potentially serious consequences".

As of June 2010, there were three reported cases of Enterobacteriaceae isolates bearing this newly described resistance mechanism in the US, the CDC stated that "All three U.S. isolates were from patients who received recent medical care in India." However, US experts have stated that it is unclear if this strain is any more dangerous than existing antibiotic-resistant bacteria such as methicillin-resistant *Staphylococcus aureus*, which are already common in the USA.



Structure of colistin, one of the few antibiotics able to treat NDM-1 positive bacterial infections.

In July 2010 a team in New Delhi reported a cluster of three cases of *Acinetobacter baumannii* bearing $\text{bla}_{\text{NDM-1}}$ that were found in the intensive care unit of a hospital in Chennai, India in April 2010. As previously, the bacteria were fully resistant to all the aminoglycoside β -lactam and quinolone antibiotics, but were susceptible to tigecycline and colistin. This particularly broad spectrum of antibiotic resistance was heightened by the strain bearing expressing several different resistance genes in addition to $\text{bla}_{\text{NDM-1}}$.

A study by a multi-national team was published in the August 2010 issue of

the journal *The Lancet Infectious Diseases*. This examined the emergence and spread of bacteria carrying the bla_{NDM-1} gene. This reported on 37 cases in the United Kingdom, 44 isolates with NDM-1 in Chennai, 26 in Haryana and 73 in various other sites in Pakistan and India. The authors' analysis of the strains showed that many carried bla_{NDM-1} on plasmids, which will allow the gene to be readily transferred between different strains of bacteria by horizontal gene transfer. All the isolates were resistant to multiple different classes of antibiotics, including beta-lactam antibiotics, fluoroquinolones, and aminoglycosides, but most were still susceptible to the polymyxin antibiotic colistin.

On August 21, 2010 Ontario, Canada had its first confirmed case of the "superbug" in Brampton. There have been other confirmed cases in British Columbia and Alberta.

In early August 2010 a chemical compound GSK 299423, was found to significantly fight against antibiotic-resistant bacteria by making such bacteria unable to reproduce, citing a likely treatment to the NDM-1 strain.

Indian response

The Indian health ministry has disputed the conclusion of the August 2010 *Lancet* study that the gene originated in India or Pakistan, describing this conclusion as "unfair" and stating that Indian hospitals are perfectly safe for treatment. Indian politicians have described linking this new drug resistance gene to India as "malicious propaganda" and blamed multinational corporations for what they describe as selective malignancy. A Bharatiya Janata

Party politician has instead argued that the journal article is bogus and represented an attempt to scare medical tourists away from India. The Indian Ministry of Health released a statement "strongly refute [in]" naming the enzyme "New Delhi". The primary author of the 2010 *Lancet* study, who is based in the University of Madras, has stated that he does not agree with the part of the article that advises people to avoid elective surgeries in India.

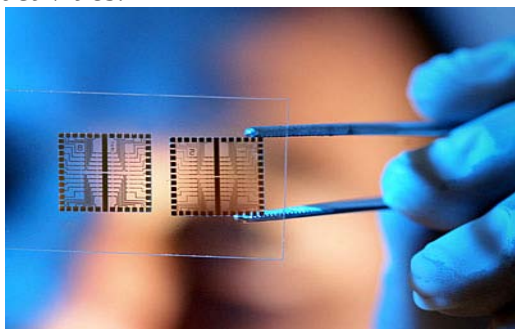
In contrast, an editorial in the March 2010 issue of the *Journal of Association of Physicians of India* blamed the emergence of this gene on the widespread misuse of antibiotics in the Indian healthcare system, stating that Indian doctors have "not yet taken the issue of antibiotic resistance seriously" and noting little control over the prescription of antibiotics by doctors and even pharmacists. The *Times of India* states that there is general agreement among experts that India needs both an improved policy to control the use of antibiotics and a central registry of antibiotic-resistant infections. Patient zero:

In August 2010, the first reported death due to bacteria expressing the NDM-1 enzyme was recorded as a Belgian man who had become infected, while being treated in a hospital in Pakistan. He died despite being administered colistin, a powerful antibiotic. A doctor involved in his treatment said, "He was involved in a car accident during a trip to Pakistan. He was hospitalized with a major leg injury and then repatriated to Belgium

**BY,
R.GOWSALIYA,
II-B.**

THE DEVELOPMENT OF BIO CHIPS :

The development of biochips is a major thrust of the rapidly growing biotechnology industry, which encompasses a very diverse range of research efforts including genomics, proteomics, and pharmaceuticals, among other activities.



Advances in these areas are giving scientists new methods for unraveling the complex biochemical processes occurring inside cells, with the larger goal of understanding and treating human diseases. At the same time, the semiconductor industry has been steadily perfecting the science of micro-miniaturization. The merging of these two fields in recent years has enabled biotechnologists to begin packing their traditionally bulky sensing tools into smaller and smaller spaces, onto so-called biochips. These chips are essentially miniaturized laboratories that can perform hundreds or thousands of simultaneous biochemical reactions. Biochips enable researchers to quickly screen large numbers of biological analyses for a variety of purposes, from disease diagnosis to detection of bioterrorism agents.

BY,
R.Manikandapriyan,
III-C.

TINY PROBES MEASURE SIGNALS INSIDE CELLS:

Nanowire transistors could make better connections between the body and electronic devices.

Researchers at Harvard University have made biocompatible, nanometer-scaled transistors that can be used to take highly precise electrical and chemical readings inside cells. The bioprobes are much more sensitive than the passive electrodes that have been used to make intracellular measurements in the past.



Cell probe: This V-shaped nanowire, shown near a cluster of cells, has a transistor at its tip. When inserted into a cell, the transistor can send and receive electrical signals. The probe is attached to electrical contacts on a flexible backing.

The Harvard group, led by chemistry professor Charles Lieber, is now developing more sophisticated bioelectronics that will take advantage of transistors' ability to send as well as receive electrical signals. They're also working with a tissue-engineering group to develop implantable bioelectronics that could make better connections between the body and neural prosthetics such as those that control some artificial limbs. The probes, which are based on silicon nanowires, can be grouped in large arrays, so the researchers also hope to use them to get a picture of biochemical and electrical networks in the large groups of cells that make up

tissues. Such measurements are difficult to make today.

Conventional metal electrodes have been used to take electrical and chemical readings in single cells, but they are invasive, and they can't achieve good electrical performance unless they are relatively large compared to the cells themselves. They irritate tissues, and they can't amplify or process signals. They limit the sophistication of neural prosthetics because they don't have a very good connection to the nervous system at the single-cell level--the level at which the body processes information. The Harvard cell probes, described today in the journal *Science*, are three-dimensional, V-shaped silicon nanowires with transistors at their tips. They're flexible and coated with two layers of lipid molecules, just as a cell is. When the transistor tip, which is about the size of a virus, encounters a cell, the cell pulls it inside. Lieber group found that the tips can also be removed gently, with no ill effects to the cell. They've used the transistor probes to take electrical measurements in single cells and are now using them to measure electrical activity in the groups of adjacent cells that form tissues.

"They've demonstrated very impressive intracellular signal detection," says Yi Cui, a professor of materials science and engineering at Stanford University and a former member of Lieber lab. What makes this possible, says Cui, is the innovative structure of the bioprocess. Lieber makes millions of the free-standing nanowires at once using a three-step growth process. First he grows one arm of the V from a silicon-containing gas. Then he creates a kink in the wire using a technique he developed last year. Next, he chemically treats the kink to create a transistor and then

induces the nanowire to start growing again. The completed wire turns back 60 degrees at the kink. Electrical contacts on a variety of substrates can be connected to the arms of the V, turning the nanowires into three-dimensional electronic probes.

"These could be used for electrophysiology experiments to study the nervous system in a detailed way over long periods of time, or for cell-based drug screens, especially for cardiac drugs," says Lieber. Silicon-nanowire transistors have also been used as chemical sensors: their resistance changes measurably when a biomolecule such as mRNA or a protein attaches to a binding molecule at the surface. Lieber is interested in extending this capability to the probes.

The Harvard researchers are now collaborating with a group at MIT to incorporate the nanoprobe into medical devices, including scaffolds used to make artificial tissues. Circuits of nanowires could "innervate" an artificial tissue so that it could measure and respond to electrical signals propagating through the heart or brain.

**BY,
M.ASHWINI,
III-A.**

TOUCH SCREEN TECHNOLOGY:

Touch screen technology has the potential to replace most functions of the mouse and keyboard. The touch screen interface is being used in a wide variety of applications to improve human-computer interaction. As the technology advances, people may be able to operate computers without mice and keyboards.

Because of its convenience, touch screen technology solutions has been applied more and more to industries, applications, products and services, such as Kiosks, POS (Point-of-Sale), consumer electronics, tablet PC, moderate to harsh Machine Control, Process Control, System Control/Office Automation and Car PC, etc.

The touch panels themselves are based around four basic screen technologies: Resistive, Capacitive, Infrared (IR), and Surface Acoustical Wave (SAW). Each of those designs has distinct advantages and disadvantages.

Resistive Touch Screen Technology

Offering excellent durability and resolution, resistive technology is used in a variety of applications and environments. The Analog Resistive touch screen is a sensor consisting of two opposing layers, each coated with a transparent resistive material called indium tin oxide (ITO). The ITO used has a typical sheet resistivity between 100 and 500 ohms per square. The layers are separated by a pattern of very small transparent insulating dots. Silver ink bus bars (~50mW/sq) make an electrical connection to the surface of the ITO at the outside edges, spanning the desired axis of the given layer. Silver ink traces (~50mW/sq) connect the bus bars to an electromechanical connector used for interfacing to the sensor. The cover sheet has a hard, durable coating on the outer side, and a conductive coating on the inner side. When touched, the conductive coating makes electrical contact with the coating on the glass, and a touch is registered by the analog controller.

Resistive touch screens deliver cost-effective, consistent and durable performance in environments where

equipment must stand up to contaminants and liquids, such as in restaurants, factories, and hospitals. Disadvantages of Resistive technology include only 75% optical transparency and the fact that a sharp object can damage the resistive layers. The Analog Resistive technology is perfect for PDAs, web phones, and other handheld consumer applications.

Capacitive Touch Screen Technology

The touchpad contains a two-layer grid of electrodes that are connected to a sophisticated full-custom mixed signal integrated circuit (IC) mounted on the reverse side of the pad. The upper layer contains vertical electrode strips while the lower layer is composed of horizontal electrode strips. The IC measures "Mutual capacitance" from each of the horizontal electrodes to each of the vertical electrodes. A human finger near the intersection of two electrodes modifies the mutual capacitance between them, since a finger has very different dielectric properties than air. When a user touches the screen, some of the charge is transferred to the user, and makes the potential difference on the screen. After the panel controller recognizes that, the controller will send the X-Y axis information to the PC port. The advantage is that capacitive technology transmits almost 90% percent of the light from the screen. The superior efficiency gives capacitive better than resistive technology.

Surface Acoustic Wave (SAW) Technology

The Surface Acoustic Wave (SAW) technology is one of the most advanced touch screen types. The technology is based on two transducers (transmitting and receiving) placed for the both of X and Y axis on the touch panel. The other important element of SAW is placed on

the glass, called reflector. The controller sends electrical signal to the transmitting transducer, and transducer converts the signal into ultrasonic waves and emits to reflectors that are lined up along the edge of the panel. After reflectors refract waves to the receiving transducers, the receiving transducer converts the waves into an electrical signal and sends back to the controller. When a finger touches the screen, the waves are absorbed, causing a touch event to be detected at that point.

Compared to Resistive and Capacitive technologies, SAW technology provides superior image clarity, resolution, and higher light transmission. Because the panel is all glass, there are no layers that can be worn, giving this technology the highest durability factor and also the highest clarity. Disadvantages of Surface Acoustic Wave (SAW) technology include the facts that the touch screen must be touched by finger, gloved hand, or soft-tip stylus (something hard like a pen won't work) and that the touch screen is not completely sealable, can be affected by large amounts of dirt, dust, and / or water in the environment.

The Surface Acoustic Wave technology is recommended for ATMs, Amusement Parks, Banking and Financial Applications, public information kiosks, computer based training, or other high traffic indoor environments.

Touch Screen Functional Description

Touching the top surface compresses the flexible top layer to the supported bottom layer causing electrical contact of the two layers between the span of insulating dots. Determining a touch location requires two measurements, one to obtain an X-axis coordinate and one to obtain a Y-axis coordinate. A single axis measurement is taken by applying a drive voltage across the ITO of one layer

via the silver ink bus bar and trace connections. The voltage applied to this layer produces a voltage gradient across the ITO. The voltage linearly changes from the minimum drive voltage at one end to the maximum drive voltage at the other end. The opposing layer, via a path through its ITO and silver ink connections, is used to measure the voltage at the point of contact on the voltage driven layer. This process is repeated, alternating functions of the two layers to obtain a measurement on the other axis.

**BY,
RAMASAMYRAJA .R
RAM PRASAD .D**

III -D

CELL PHONE SECURITY USING VOICE RECOGNISATION:

The word cell phone has become the greatest buzz word in the cellular communication. Still the protection systems in cell phones are not good enough to give true security. Our article tells about a new system of cell phone protection using voice recognition.

SUPER STAR EXAMPLE:

“HI BUDDY I’M SHIVAJI , COOL....”

The great exposure given to this system was by these great words, by a great man. The biometric security system can be understood very easily by this film.

WHY VOICE RECOGNITION:

Biometric is the science of measuring an individual’s property. By determining an individuals physical features in an

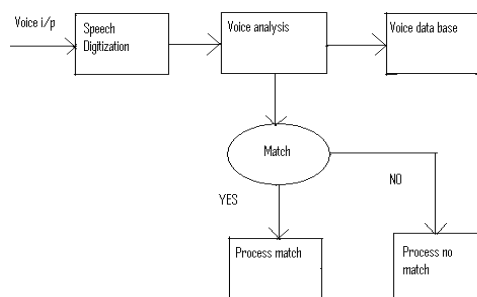
authentication inquiry and comparing this data with stored biometric reference data, identification for a specific user can be determined and authentication for access can be granted.

Although biometrics are effective and virtually , fraud-proof , an individuals speech pattern has more unique qualities and is often recognized as the most natural.

SIMPLE PROCESS:

Voice biometrics works by digitizing a profile of a person's speech to produce a stored model voice print, rather like a template, which is referred to each time that person attempts to access secure data. Biometrics technology reduces each spoken word into segments: sub-word like syllables, phonemes, triphones or similar units of sound, composed of several dominant frequencies called formants which remain relatively constant over the segment. Each segment has three or four dominant tones that can be captured in digital form and plotted on a table or spectrum.

CELL PHONE PROTECTION USING VOICE RECOGNISATION:



In our article the main idea is that when a user buys a cell phone he has to record his voice (only some portion) in the cell phone. For each person the phoneme (The smallest unit of speech that differentiates one utterance from another in any spoken language) is unique. The frequency spectrum of speech signal is

encoded by means of vocoder. This coded signal is stored in the internal system memory the flash ROM(8M).

This spectrum is encoded by means of vocoder and stored in flash ROM.

Then the speech signal of the user is compared with this original spectrum of the user voice. If the spectrum matches then the user is authorized user. If not the transmission will be abruptly cut off

FREQUENCY AND AMPPLITUDE:

Here we are taking frequency and not amplitude because the amplitude spectrum varies with amplitude of speech signal. If the person speaks in shrill voice or in a angry mood then the amplitude varies. Hence we go for frequency spectrum since it will be constant irrespective of amplitude.

SUPERMACYVOICE RECOGNITION

In biometrics there are varieties of technologies available for security reasons like finger print recognition , iris recognition and facial recognition etc. yet we are going for voice recognition because of its low cost and some of its advantages it possess.

UNIQUENESS: An identical trait won't appear in two people.

UNIVERSALITY: Occur in as many people as a possible

MEASURABILITY: Are measurable with simple technical instruments

USER FRIENDLY: Are easy and comfortable to measure.

CONCLUSION:

We can't think of a communication world with out cell phones. The future is going to be more advanced and sophisticated cell phones. Since PREVENTION IS BETTER THAN

CURE, it is better than to prevent cell phone theft than being stolen..

The above protection system proposed by our paper has its unique advantages than any other existing systems. This method can be adopted universally to protect cell phones.

BY,
S.PRASATH,
II-c.

CONTENT BASED IMAGE RETRIVAL:

The growth of digital imaging:

The use of images in human communication is hardly new – our cave-dwelling ancestors painted pictures on the walls of their caves, and the use of maps and building plans to convey information almost certainly dates back to pre-Roman times. But the twentieth century has witnessed unparalleled growth in the number, availability and importance of images in all walks of life. Images now play a crucial role in fields as diverse as medicine, journalism, advertising, design, education and entertainment.

Technology, in the form of inventions such as photography and television, has played a major role in facilitating the capture and communication of image data. But the real engine of the imaging revolution has been the computer, bringing with it a range of techniques for digital image capture, processing, storage and transmission. The involvement of computers in imaging can be dated back to 1965, with Ivan Sutherland's Sketchpad project, which demonstrated the feasibility of computerised creation, manipulation and storage of images, though the high cost of hardware limited their use until the mid-1980s. Once computerised imaging became affordable (thanks largely to the development of a mass market for computer games), it soon penetrated into areas traditionally depending heavily on images for communication, such as

engineering, architecture and medicine. Photograph libraries, art galleries and museums, too, began to see the advantages of making their collections available in electronic form. The creation of the World-Wide Web in the early 1990s, enabling users to access data in a variety of media from anywhere on the planet, has provided a further massive stimulus to the exploitation of digital images. The number of images available on the Web was recently estimated to be between 10 and 30 million.

The need for image data management:

The process of digitisation does not in itself make image collections easier to manage. Some form of cataloguing and indexing is still necessary – the only difference being that much of the required information can now potentially be derived automatically from the images themselves. The extent to which this potential is currently being realized is discussed below.

The need for efficient storage and retrieval of images – recognized by managers of large image collections such as picture libraries and design archives for many years – was reinforced by a workshop sponsored by the USA's National Science Foundation in 1992. After examining the issues involved in managing visual information in some depth, the participants concluded that images were indeed likely to play an increasingly important role in electronically-mediated communication. However, significant research advances, involving collaboration between a number of disciplines, would be needed before image providers could take full advantage of the opportunities offered. They identified a number of critical areas where research was needed, including data representation, feature extractions and indexing, image query matching and user interfacing.

One of the main problems they highlighted was the difficulty of locating a desired image in a large and varied collection. While it is perfectly feasible to identify a desired image from a small collection simply by browsing, more effective techniques are

needed with collections containing thousands of items. Journalists requesting photographs of a particular type of event, designers looking for materials with a particular colour or texture, and engineers looking for drawings of a particular type of part, all need some form of access by image content.

The earliest use of the term content-based image retrieval in the literature seems to have been by Kato, to describe his experiments into automatic retrieval of images from a database by colour and shape feature. The term has since been widely used to describe the process of retrieving desired images from a large collection on the basis of features (such as colour, texture and shape) that can be automatically extracted from the images themselves. The features used for retrieval can be either primitive or semantic, but the extraction process must be predominantly automatic. Retrieval of images by manually-assigned keywords is definitely not CBIR as the term is generally understood – even if the keywords describe image content.

CBIR differs from classical information retrieval in that image databases are essentially unstructured, since digitized images consist purely of arrays of pixel intensities, with no inherent meaning. One of the key issues with any kind of image processing is the need to extract useful information from the raw data (such as recognizing the presence of particular shapes or textures) before any kind of reasoning about the image's contents is possible. Image databases thus differ fundamentally from text databases, where the raw material (words stored as ASCII character strings) has already been logically structured by the author. There is no equivalent of level 1 retrieval in a text database.

CBIR draws many of its methods from the field of image processing and computer vision, and is regarded by some as a subset of that field. It differs from these fields

principally through its emphasis on the retrieval of images with desired characteristics from a collection of significant size. Image processing covers a much wider field, including image enhancement, compression, transmission, and interpretation. While there are grey areas (such as object recognition by feature analysis), the distinction between mainstream image analysis and CBIR is usually fairly clear-cut. An example may make this clear. Many police forces now use automatic face recognition systems. Such systems may be used in one of two ways. Firstly, the image in front of the camera may be compared with a single individual's database record to verify his or her identity. In this case, only two images are matched, a process few observers would call CBIR. Secondly, the entire database may be searched to find the most closely matching images. This is a genuine example of CBIR. Research and development issues in CBIR cover a range of topics, many shared with mainstream image processing and information retrieval. Some of the most important are:

- understanding image users' needs and information-seeking behaviour
- identification of suitable ways of describing image content
- extracting such features from raw images
- providing compact storage for large image databases
- matching query and stored images in a way that reflects human similarity judgments
- efficiently accessing stored images by content

Conclusions:

Several reviews of the literature on image retrieval have been published, from a variety of different viewpoints. Ensor reviews methods for providing subject access to pictorial data, developing a four-category framework to classify different approaches. He discusses the strengths and limitations

both of conventional methods based on linguistic cues for both indexing and search, and experimental systems using visual cues for one or both of these. His conclusions are that, while there are serious limitations in current text-based techniques for subject access to image data, significant research advances will be needed before visually-based methods are adequate for this task. He also notes, as does Cawkell [1993] in an earlier study, that more dialogue between researchers into image analysis and information retrieval is needed.

Aigrain discuss the main principles of automatic image similarity matching for database retrieval, emphasizing the difficulty of expressing this in terms of automatically generated features. They review a selection of current techniques for both still image retrieval and video data management, including video parsing, shot detection, key frame extraction and video skimming. They conclude that the field is expanding rapidly, but that many major research challenges remain, including the difficulty of expressing semantic information in terms of primitive image features, and the need for significantly improved user interfaces. CBIR techniques are likely to be of most use in restricted subject domains, and where synergies with other types of data (particularly text and speech) can be exploited.

BY,

A.ALICE,

III-A.

WIRELESS ROCKET LAUNCH CONTROL SYSTEM:

Overview of Hobby Rocketry and Launch Systems:

Hobby rocketry has grown over the past few decades to include some very large and Powerful rockets that weigh over 100 pounds and are over 12 feet long. These larger rockets require

1000 feet or more of distance between the launch pad and people, cars and buildings for proper safety. This requires some form of launch system capable of Working over this distance. Most launch control systems use some combination of batteries, wires, switches, and relays. Many hobby rocket club launch systems include a launch console with a bank of switches to allow the Launch Control Officer (LCO) to select which rocket(s) are to be launched, a master safety key switch, and a “big red button” to launch the rocket. At the launch pads, there are batteries and relays. The coils of the relays are driven through wires that are connected to the launch console. The relays connect to the rocket motor igniter through a pair of wires and alligator clips. These wires are problematic for a number of reasons but there are two primary ones. First, people often trip over them and knock over some part of the launch system, sometimes damaging it. Obviously, people can fall and get hurt too. Second, even when treated with respect and care, the wires fatigue and break resulting in a frustrating day at the launch field debugging a broken launch system.

My solution: I have developed a launch control system that is safe, effective and robust that solves all of the problems with typical launch control systems available today. There are two types of components in my system: the launch console where the system is operated from, and the pad unit which goes at the launch pads and supports 4 launch pads Sper pad unit. It primarily uses a wireless network but supports wired connections to save money in specific instances where the wires are not a problem.

For example, the wireless network can be used from the launch console to the first pad

Unit to support the first 4 pads. But then, if there are 4 more pads very close to the first 4, then a second wired pad unit could be connected to the first pad unit with a short wire. This saves the cost of the radio transceiver in the second pad unit. Due to the possibility of injury and property damage with these larger rockets, safety..

The most important design criteria in this system. A number of safety features have been incorporated into this design and they work well. The flexibility and rich peripheral set of the Atmel ATmega processors allowed me to achieve all of the design goals of the project. In addition, I was able to combine two designs onto one board to save design effort and inventory costs.

Summary

This launch control system met all of the design goals and has been a useful addition to the launch equipment of our rocket club. To date, the system has a perfect safety record and there has not been a rocket launched that was unintended. The system has been robust and reliable. Given the flexibility of a wireless system that can be expanded at will and inexpensively augmented with some wired connections, the system will meet all future needs of the club's launch control system.

ZIGBEE

ZigBee is the name of a specification for a suite of high level communication protocols using small, low-power, low data rate digital radios based on the IEEE 802.15.4 standard for wireless personal area networks (WPANs), such as wireless headphones connecting with cell phones via short-range radio. The technology is intended

to be simpler and cheaper than other WPANs, such as Bluetooth. ZigBee is targeted at radio-frequency (RF) applications which require a low data rate, long battery life, and secure networking.

ZigBee is a low data rate, two-way standard for home automation and data networks. The standard specification for up to 254 nodes including one master, managed from a single remote control. Real usage examples of ZigBee includes home automation tasks such as turning lights on, setting the home security system, or starting the VCR. With ZigBee all these tasks can be done from anywhere in the home at the touch of a button. ZigBee also allows for dial-in access via the Internet for automation control.

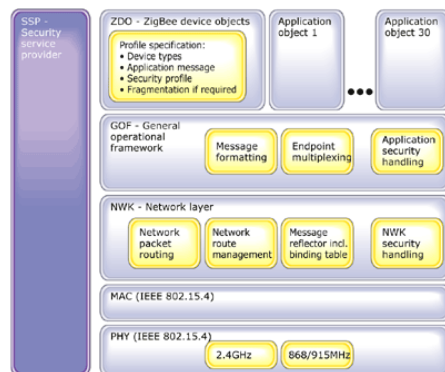
The ZigBee standard uses small very low-power devices to connect together to form a wireless control web. A ZigBee network is capable of supporting up to 254 client nodes plus one full functional device (master). ZigBee protocol is optimized for very long battery life measured in months to years from inexpensive, off-the-shelf non-rechargeable batteries, and can control lighting, air conditioning and heating, smoke and fire alarms, and other security devices. The standard supports 2.4 GHz (worldwide), 868 MHz (Europe) and 915 MHz (Americas) unlicensed radio bands with range up to 75 meters.

ZigBee Applications:

ZigBee enables broad-based deployment of wireless networks with low-cost, low-power solutions. It provides the ability to run for years on inexpensive batteries for a host of monitoring applications: Lighting

controls, AMR (Automatic Meter Reading), smoke and CO detectors, wireless telemetry, HVAC control, heating control, home security, Environmental controls and shade controls,

ZIGBEE STACK ARCHITECTURE



It may be helpful to think of IEEE 802.15.4 as the physical radio and ZigBee as the logical network and application software, as Figure 1 illustrates. Following the standard Open Systems Interconnection (OSI) reference model, ZigBee protocol stack is structured in layers. The first two layers, physical (PHY) and media access (MAC), are defined by the IEEE 802.15.4 standard. The layers above them are defined by the ZigBee Alliance. The IEEE working group passed the first draft of PHY and MAC in 2003. A final version of the network (NWK) layer is expected sometime this year.

use direct-sequence spread spectrum but with binary-phase-shift keying modulation.

Power consumption:

Ultra-low power consumption is how ZigBee technology promotes a long lifetime for devices with non rechargeable batteries. To minimize power consumption and promote long battery life in battery-powered devices,

end devices can spend most of their time asleep, waking up only when they need to communicate and then going immediately back to sleep. ZigBee envisions that routers and the coordinator will be mains-powered and will not go to sleep.

Zigbee benefits:

In all of its uses, ZigBee offers four inherent, beneficial characteristics:

- Low cost
- Range and obstruction issues avoidance
- Multi-source products
- Low power consumption

**BY,
P.PRITHIVINATH ,
M.PRAKASH,
III-D.**

10 WAYS TO REDUCE FEAR



Fear often interferes with our ability to live more fulfilling lives. Fear is the critical voice in our head saying: *"You'll never succeed, so why try?"*. Fear makes us listen to those voices and give up before we've even started.

When we let fear rule our lives, we miss out on opportunities and leads us to dismiss people and situations that could help us reach our goals. Fear comes in different guises and different forms - procrastination for example. Here are 10 ways to reduce and break through fear.

1. **Write down your fears** and how they affect your life.
2. **Are there things that you're holding back on**, because you're frightened that you are not good enough or of making changes generally? You have a limited time on this earth. *Deadlines are called deadlines for a reason!*
3. **Listen to your self-talk.** Many of us don't so much talk to ourselves as make statements ie *"I'll never be able to get that job"*. This cuts your brain dead and gives it nothing to work on. It's gone for a prolonged holiday!

4. **Start asking yourself questions.** For example, ask *"how can I make giving this speech an enjoyable experience?"*. Now your brain has something to do! Don't expect the answer straight away (but be aware of opportunities and situations that provide clues) - but you're going in the right direction. Get into the habit of asking yourself questions rather than making blanket (negative) statements.

5. **Redefine "mistakes" and learn from them.** Don't beat yourself up over mistakes you've made in the past and being fearful of starting new relationships, jobs etc. That gets you nowhere. Call a mistake an *"opportunity for learning"* and seek the **learning** in every "mistake".

6. **Seek the company of supportive people** and reduce contact with those who foster feelings of negativity and have a "fearful" approach to life.

7. **Read books about people who have succeeded** despite the obstacles and look at the principles and qualities behind what they did, their thinking and **attitude**. Closer to home - seek out role models who represent what you want to become. Learn from their approach and **attitude**.

8. **Know that whatever quality you like or admire** in someone is lying dormant in you - waiting to be developed. How else would you recognise it? Make a list of the qualities you most admire in someone else and ask yourself how they can be acknowledged and developed in you.

9. **Your past doesn't have to define your future.** Be aware of your past and

willing to let go of it. Each day brings a new opportunity to do and think about things differently.

10. List your goals and the actions you need to take to achieve them every day. Every time you do something that brings you a little closer to achieving your goals, you will feel better about yourself and reduce your unnecessary fears.

and

get yourself a Coach! Or someone to (in) formally support and encourage you to break through your fear and achieve your goals.

“YOUR LOW CONFIDENCE WILL DESTROY THE REST OF THE LIFE”.

**KANAGALAKSHMI.S
ECE-B
3rd YEAR
9908005120**

ASTRONAUTS AND COSMONAUTS

The word “astronaut” was first used in 1880 by the British writer Percy Greg. It was the name he gave to a spaceship in his novel “across the zodiac” .by the 1950s it was the word used for a space voyager. The Russian word is cosmonaut.

- 1.12 April 1961 first person in space—soviet cosmonaut Yuri Gagarin, Vostok1
2. 5 may 1961 first US astronaut—Alan B. Shepard Jr, mercury 3
3. 6 august 1961 first flight of over 24 hours – Gherman S. Titov(USSR),vostok2

4. 20 February 1962 first US orbit--- John H. glennjr ,friendship 7
- 5.16 June 1963 first woman in space – Valentina V.Tereshkova (USSR) ,vostok6
- 6.18 march 1965 first space walk --- Aleskeileonov(USSR) v0skhod 2
- 7.23 march 1965 first two man US mission---- John young and Virgil gus groom Gemini 3.
- 8.3 June 1965 first us spacewalk--- Edward H.white Gemini 4
9. 24 April 1967 first space death --- Vladimir M. Komarov (USSR), soyuz1
10. 20 July 1969 first moon landing --- Neil Armstrong and Edwin E. Buzz Aldrin., Apollo
11. 18 June 1983 first US woman in space ---Sally Ride , Challenger
12. 18 may 1991 first British Astronaut ---Helen Sharman, Mir
- 13.4 December 1998 first inter national space station established.
14. 2 November 2000 first crew on international space station
- 15.28 April - 6 may 2001 first space tourist
- 16.15-16 October first astronaut launched by china
- 17.2010 Atlantis, Discovery ,Endeavour retires and been replaced by NASA’S Orion.

**KALEESHWARI.P
ECE-B
3RD YEAR**

SUCCESS MANTRA OF BILL GATES

“SOME PEOPLE ARE BORN GREAT, SOME PEOPLE ACHIEVE GREATNESS, BUT GREATNESS IS TRUST UPON SOME”

This is a wonderful quote that suits for **MR. BILL GATES**; he who reduced the globe into a small town.

Life of Mr. Bill Gates:

On a beautiful day of October 28, 1955, William Henry Gates III was born at Seattle with his two sisters. As a child Bill enjoyed rocking back and forth; today he still has a habit of rocking when he is thinking about something. He attended the most exclusive preparatory school which had tuition fees of about \$5,000.

At the age of 13, his father brought him a computer. Within a week, he had surpassed the knowledge of computer teacher. He started learning BASIC Programming Language, which was a breeze for Bill and started writing his own programmes. His great passion created an intense love for computers. His school rented a time on DEC PDP-10, which Bill was able to use to pursue an interest in computers a rare opportunity at that time. He successfully completed the SAT Examination and he started his own company.

At Harvard, Bill was a way ahead of his peers so he spent most of his time in programming and playing poker. He promised his teacher “I would become a Millionaire by age of 30”. But, he underestimated himself, he was **Billionaire** when he was 31 and he devoted his life to the MICROSOFT.

BILL GATES - THE SECRET OF HIS SUCCESS

The world has been talking about Gate’s secret to success for so many years. He is passionate, very hard working and at the same time benevolent. When a businessman asked the Bill Gates the secret of success the great tycoon presented a five – point master formula for success:

- PASSION
- INTELLIGENCE
- INTEGRITY
- A GOOD TEAM
- LEADERSHIP QUALITY

FIND YOUR PASSION

No mind boggling success story has been written till now without a great passion. Passion generates focus – so necessary to get in any field. Bill Gates himself has immense passion for computer and software when he was admitted in a lakeside school he got hooked to the computer and surpassed the knowledge of his teacher within a week. According to great scientist Albert Einstein “Genius is one percent inspiration and ninety nine percent perspiration”. But without passion, it is difficult to work hard. When you recognize, you are real hidden talent and work to develop you become passionate and self-motivated to work hard. When you have passion the work becomes a fun for you and you want to enjoy it more and more.

MAKE YOURSELF INTELLIGENT

Gates said that success also depends on your intelligence. Different people have

different level of intelligence. Like physical strength no person can try to develop his intelligence. You can develop intelligence by following steps:

1. Read good books
2. Lap-up to the information in the Internet regarding General Knowledge
3. Arrange debates related to a field of interest
4. Practice memorizing i.e. start from simple things like memorizing phone numbers or some quotations related to your field
5. Keep yourself physically fit and mentally relaxed

BE SINCERE IN YOUR EFFORTS

Integrity is always useful whether it is business or family life. Deceiving causes friction in human relations and hampers your progress. Try to be sincere in everything – in your efforts, relations, and business policies. ‘Honesty is the best policy’ is really wise saying. It is the quality of integrity that Bill Gates always got long term associates in business. He also got a life long friend and associate in his wife Melinda. Research has also proved that integrity is also good for health as it prolog one’s life span.

CREATE A GOOD TEAM

Creating a team of talented people is necessary to rise fast in business. The motto of modern business world is ‘combine and succeed’. Teams are of two types; partnership team and assistants’ team. Even the big companies are joining hands in modern world to become stronger to face competition. For a single person, it becomes very difficult to survive or

succeed in a big business jungle. What is a good team?

T → Together
E → Everybody
A → Achieve
M → More

LEAD YOUR PEOPLE TO VICTORY

Your team will work better if you are able to provide good leadership to them. For a businessman, his workforce is like his family. He shares its joys and sorrows and motivates them time to time better and achieves the targets. By proper rewards and punishment methods we should keep workforce disciplined and united leading them to big success. We should always share our gains with employees or assistants by giving special allowances or other rewards. “Motivation and implementation is a key to success”. This is an important quality required for a good LEADER.

SUCCESS TIPS IN VIEW OF BILL GATES

1. Motivation 
 - Giving gifts
 - Strict
 - Self
2. Visualization
3. Self Chanting Mantra i.e. “I am a successful person”, “I am becoming better day by day
4. Success formula – $\text{SUCCESS} = \text{Number of times I fail} + 1$

N.HARINI
2ND YEAR, ECE-B

TYPES OF BOOKS

Almanac

This is an annual calendar of dates and events. It originally contained astronomical facts and figures, but now includes useful statistics and other information. Almanac is sometimes spelled Almanack.

Anthology

A collection of writings by one author or on a theme, such as an anthology of animal poems.

Atlas

A book of maps.

Autobiography

An account of a person's life written by himself or herself. Sometimes a "ghost" writer may help someone write an autobiography.

Bibliography

A list of books or sources on a particular subject.

Biography

A book about someone's life written by another person.

Chronology

A dictionary of dated events.

Dictionary

An alphabetical list of words and their meanings. There are also dictionaries on a single subject. for example Dictionary of art.

Dictionary of quotations

A listing of interesting remarks or extracts from the writings of famous people.

Directory

An alphabetical list of names and addresses of people or organizations.

Encyclopedia

A book of articles on many subjects, arranged alphabetically. Encyclopedia are often published in many volumes or, today, on CD-ROM or on the internet.

Gazetteer

A book that lists and describes places(countries,cities, etc.,)

Glossary

A list of words and phrases used in a particular subject area. For instance , A glossary of computer terms.

Language dictionary

A dictionary that gives translations of words and phrases to and from other languages. For example, a French-english/ English-ferench dictionary.

Monogarth

A book on a specialized subject, such as the work of a particular artist.

Novel

A work of fiction ,usually involving imaginary people. A historical novel may be about real historical characters.

Novella

A short of fiction

Phrase book

A book of travelers, with translations of words and phrases into and from a foreign language.

Pop-up book

A book, usually for children, made so that pictures stand up or move when the pages are opened.

Thesaurus

A book organized to show words with similar meanings.

Travel guide

A book of informations about a particular country or area.

Who's who or biographical dictionary

Alphabetical summaries of the lives of famous people , living or dead, sometimes by their subjects(such as a dictionary fo scientists)

Year book

An annual reference book with information about the events of the previous year. Yearbooks containing statistical information about a country are often called abstracts

A.ALICE
3RD YEAR, ECE-A

THE CHALLENGE

Let others lead small lives,
But not you.
Let others argue over small things,
But not you.
Let others cry over small hurts,
But not you.
Let others leave their future
In someone else's hands,
But not you.

BY,

P.NIRANJANA
3RD YEAR, ECE-D

CHILDHOOD FRIENDS

As childhood friends, we grew up
together,
Swearing to be friends forever and ever.
Sometimes we would argue and fight,
Other times we would laugh and stay up
all night.

We went from playing with games and
toys,
To talking and dreaming about different
boys.
My thoughts and feelings, to you I
would confide,

Never having anything to hide.

Friends we do remain,
Things changing, and things staying the
same.
To each other we still listen and share,
About each other, we will always care.

BY
S.NITHYA DEVI
3rd, ECE-'D'

CULTURE OF INDIA



A dancer performing Kathak, one of the eight major Indian classical dances. Originated during 11th century BC Vedic age, Kathak is a Hindu mythology storytelling style and encompasses several forms of Hindu mudrās.

The culture of India has been shaped not only by its long history, unique geography and diverse demography, but also by its ancient heritages, which were formed during the Indus Valley Civilization and evolved further during the Vedic age, rise and decline of Buddhism, the Golden age, invasions from Central Asia, European colonization and the Indian independence movement.

India's languages, religions, dance, music, architecture and customs differ from place to place within the country, but nevertheless possess a commonality. The culture of India is an amalgamation of these diverse sub-cultures spread all over the Indian subcontinent and traditions that are several millennia old.

Regarded by some historians as the "oldest living civilization of Earth", the Indian tradition dates back to 8,000 BC and has a continuous recorded history for over 2,500 years.^[4] Several elements of India's diverse culture — such as Indian religions, yóga and Indian cuisine — have had a profound impact across the world.

Religions and spirituality:



Close-up of a statue depicting Maitreya at the Thikse Monastery in Ladakh, India. Dharmic religions, such as Hinduism and Buddhism, are indigenous to India.

Main articles: Religion in India and Indian religions

India is the birth place of Hinduism, Buddhism, Jainism and Sikhism, collectively known as Indian religions. Indian religions, also known as Dharmic religions are a major form of world

religions along with Abrahamic ones. Today, Hinduism and Buddhism are the world's third- and fourth-largest religions respectively, with over 2 billion followers altogether, and possibly as many as 2.5 or 2.6 billion followers. India is also the birthplace for the Sikh, Jain, Lingayat, and Ahmadiyya faiths

India is one of the most religiously diverse nations in the world, with some of the most deeply religious societies and cultures. Religion still plays a central and definitive role in the life of most of its people.

The religion of 80% of the people is Hinduism. Islam is practiced by around 13% of all Indians. Sikhism, Jainism and especially Buddhism are influential not only in India but across the world. Christianity, Zoroastrianism, Judaism and the Bahá'í Faith are also influential but their numbers are smaller. Despite the strong role of religion in Indian life, atheism and agnostics also have visible influence along with a self-ascribed tolerance to other people.

Festivals:



Radha and gopikas celebrating Holi, known as the "festival of colors".

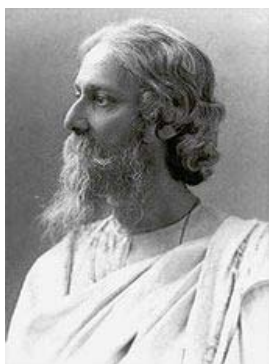
India, being a multi-cultural and multi-religious society, celebrates holidays and festivals of various religions. In addition, many states and regions have local

festivals depending on prevalent religious and linguistic demographics. Popular religious festivals include the Hindu festivals of Navratri, Diwali, Ganesh Chaturthi, Durga puja, Holi, Rakshabandhan and Dussehra. Several harvest festivals, such as Sankranti, Pongal and Onam, "Nuakhai" are also fairly popular.

Certain festivals in India are celebrated by multiple religions. Notable examples include Diwali, which is celebrated by Hindus, Sikhs and Jains, and Buddh Purnima, celebrated by Buddhists and Hindus. Islamic festivals, such as Eid-ul-Fitr, Eid al-Adha and Ramadan, are celebrated by Muslims across India. Sikh Festivals, such as Guru Nanak Jayanti, Baisakhi are celebrated with full fanfare by Sikhs and hindu.

Languages and literature

History



Rabindranath Tagore, a prominent Bengali novelist and poet and Asia's first Nobel laureate.

With its oldest core dating back to as early as 1500 BC, the Rigvedic Sanskrit is one of the oldest attestations of any Indo-Iranian language, and one of the earliest attested members of the Indo-

European language family, the family which includes English and most European languages. Sanskrit has had a profound impact on the languages and literature of India. Hindi, India's most spoken language, is a "Sanskritized register" of the Khariboli dialect. In addition, all modern Indo-Aryan languages, Munda languages and Dravidian languages, have borrowed many words either directly from Sanskrit (tatsama words), or indirectly via middle Indo-Aryan languages (tadbhava words). Words originating in Sanskrit are estimated to constitute roughly fifty percent of the vocabulary of modern Indo-Aryan languages, and the literary forms of (Dravidian) Telugu, Malayalam and Kannada. Part of the Eastern Indo-Aryan languages, the Bengali language arose from the eastern Middle Indic languages and its roots are traced to the 5th century BC Ardhamagadhi language.

Tamil, one of India's major classical languages, descends from Proto-Dravidian languages which was spoken around the third millennium BC in peninsular India. Tamil literature has existed for over two thousand years and the earliest epigraphic records found date from around the third century BC. Another major Dravidian language, Kannada is attested epigraphically from the mid-1st millennium AD, and literary Old Kannada flourished in the 9th to 10th century Rashtrakuta Dynasty. Pre-old Kannada (or Purava Hale Gannada) was the language of Banavasi in the early Common Era, the Satavahana and Kadamba periods and hence has a history of over 2000 years. The Ashoka rock edict found at Brahmagiri (dated to 230 BC) has been suggested to contain a word in identifiable Kannada. .

Epics:

The Rāmāyaṇa and the Mahābhārata are the oldest preserved and well-known epics of India. Versions have been adopted as the epics of Southeast Asian countries like Thailand, Malaysia and Indonesia. The Ramayana consists of 24,000 verses in seven books (kāṇḍas) and 500 cantos (sargas), and tells the story of Rama (an incarnation of the Hindu preserver-god Vishnu), whose wife Sita is abducted by the demon king of Lanka, Ravana. This epic played a pivotal role in establishing the role of dhārma as a principal ideal guiding force for Hindu way of life. The earliest parts of the Mahabharata text date to 400 BC and is estimated to have reached its final form by the early Gupta period (ca. 4th c. AD). Other regional variations of these, as well as unrelated epics include the Tamil Ramavataram, Kannada Pampa Bharata, Hindi Ramacharitamanasa, and Malayalam Adhyathmaramayanam. In addition to these two great Indian epics, there are five major epics in the classical Tamil language — Silappatikaram, Manimekalai, Civaka-cintamani, Valayapathi and Kundalakesi.

Performing arts

Dance:



A kuchipudi performer from Andhra Pradesh.

Indian dance too has diverse folk and classical forms. Among the well-known folk dances are the bhangra of the Punjab, the bihu of Assam, the chhau of Jharkhand and Orissa, the ghoomar of Rajasthan, the dandiya and garba of Gujarat, the Yakshagana of Karnataka and lavani of Maharashtra and Dekhnni of Goa. Eight dance forms, many with narrative forms and mythological elements, have been accorded classical dance status by India's National Academy of Music, Dance, and Drama. These are: bharatanatyam of the state of Tamil Nadu, kathak of Uttar Pradesh, kathakali and mohiniattam of Kerala, kuchipudi of Andhra Pradesh, manipuri of Manipur, odissi of the state of Odisha and the sattriya of Assam. .

Music:

The music of India includes multiple varieties of religious, folk, popular, pop, and classical music. The oldest preserved examples of Indian music are the melodies of the Samaveda that are still sung in certain Vedic Śrauta sacrifices. India's classical music tradition is heavily influenced by Hindu texts. It includes two distinct styles: Carnatic and Hindustani music.



Panchavadyam temple music in Kerala.

The earliest account of Indian musical hymns is found in the 1000 BC Sāmaveda.^[59] It proposed a tonal structure consisting of seven notes, which were named, in descending order, as Krusht, Pratham, Dwitiya, Tritiya, Chaturth, Mandra and Atiswār. These refer to the notes of a flute, which was the only fixed frequency instrument. Both the Hindustani classical music and the Carnatic classical music systems are based on the melodic mode (known as Rāga), sung to a rhythmic cycle or tala. These principles were refined in the nāṭyaśāstra (200 BC) and the dattilam (300 AD).

Purandaradasa is considered the "father of carnatic music" (Karnataka sangeetapitamaha). He concluded his songs with a salutation to Lord Purandara Vittala and is believed to have composed as many as 475,000 songs in the Kannada language. However, only about 1000 are known today.

Prominent contemporary Indian musical forms include filmi and Indipop. Filmi refers to the wide range of music written and performed for mainstream Indian cinema, primarily Bollywood, and accounts for more than 70 percent of all music sales in the country. Indipop is one of the most popular contemporary styles of Indian music which is either a fusion of Indian folk, classical or Sufi music with Western musical traditions.

Architecture:



Considered to be an "unrivaled architectural wonder",^[69] the Taj Mahal in Agra is a prime example of Indo-Islamic architecture.

Indian architecture encompasses a multitude of expressions over space and time, constantly absorbing new ideas. The result is an evolving range of architectural production that nonetheless retains a certain amount of continuity across history. Some of its earliest production are found in the Indus Valley Civilization (2600–1900 BC) which is characterised by well planned cities and houses. Religion and kingship do not seem to have played an important role in the planning and layout of these towns.

During the period of the Mauryan and Gupta empires and their successors, several Buddhist architectural complexes, such as the caves of Ajanta and Ellora and the monumental Sanchi Stupa were built. Later on, South India produced several Hindu temples like Chennakesava Temple at Belur, the Hoysaleswara Temple at Halebidu, and the Kesava Temple at Somanathapura, Brihadeeswara Temple, Thanjavur, the Sun Temple, Konark, Sri Ranganathaswamy Temple at Srirangam, and the Buddhastupa (ChinnaLanjadibba and Vikramarkakotadibba) at Bhattiprolu. Angkor Wat, Borobudur

and other Buddhist and Hindu temples indicate strong Indian influence on South East Asian architecture, as they are built in styles almost identical to traditional Indian religious buildings.



The Umaid Bhawan Palace in Rajasthan, one of the largest private residences in the world

The traditional system of VaastuShastra serves as India's version of FengShui, influencing town planning, architecture, and ergonomics. It is unclear which system is older, but they contain certain similarities. FengShui is more commonly used throughout the world. Though Vastu is conceptually similar to FengShui in that it also tries to harmonize the flow of energy, (also called life-force or Prana in Sanskrit and Chi/Ki in Chinese/Japanese), through the house, it differs in the details, such as the exact directions in which various objects, rooms, materials, etc. are to be placed.

With the advent of Islamic influence from the west, Indian architecture was adapted to allow the traditions of the new religion. Fatehpur Sikri, Taj Mahal, Gol Gumbaz, Qutub Minar, Red Fort of Delhi are creations of this era, and are often used as the stereotypical symbols of India. The colonial rule of the British Empire saw the development of Indo-Saracenic style, and mixing of several other styles, such as European Gothic.

The Victoria Memorial or the Chhatrapati Shivaji Terminus are notable examples.

Indian architecture has influenced eastern and southeastern Asia, due to the spread of Buddhism. A number of Indian architectural features such as the temple mound or stupa, temple spire or shikhara, temple tower or pagoda and temple gate or torana, have become famous symbols of Asian culture, used extensively in East Asia and South East Asia. The central spire is also sometimes called a vimanam. The southern temple gate, or gopuram is noted for its intricacy and majesty.

By,

A.SUBHA

PRIYADHARSHINI

ECE-E, 2nd YEAR

MY PARENTS

You are both special in every way,
Encouraging me more & more each
passing day.

You both are the reason why I'm so
strong,
With you two at the helm not a thing
could go wrong.

You've both helped me through many
trials & tribulations,
You've made things better in every
situation.

Thank you both for always being there,
And showing me that you truly care.

Words could never explain how I feel

about you,
But I hope you know that I truly love
you two!

P.DIVYA
II year ECE – B

SOME INTERESTING FACTS:

All the planets in our solar system rotate anticlockwise, except Venus. It is the only planet that rotates clockwise.

Hummingbirds are the only animal that can also fly backwards.

Insects do not make noises with their voices. The noise of bees, mosquitoes and other buzzing insects is caused by rapidly moving their wings.

The cockroach is the fastest animal on 6 legs covering a meter a second.

The word "listen" contains the same letters as the word "silent".

The only 2 animals that can see behind itself without turning its head are the rabbit and the parrot.

A 'jiffy' is an actual unit of time for 1/100th of a second.

India invented the Number System. Zero was invented by Aryabhatta.

The whip makes a cracking sound because its tip moves faster than the speed of sound.

A hippopotamus can run faster than a man.

India never invaded any country in her last 10000 years of history.

'Hippopotomonstrosesquippedaliophobia' is the fear of long words.

Didaskaleinophobia is the fear of going to school.

Phobatrivaphobia is a fear of trivia about phobias!!

It is impossible to lick your elbow. (We know you gonna try this!!!)

A snail can sleep for 3 years. (Wow, lucky chap eh?)

The names of the continents all end with the same letter with which they start

In 1883 the explosion of the volcano Krakatoa put so much dust into the earth's atmosphere that sunsets appeared green and the moon appeared blue around the world for almost two years.

"Almost" is the longest word in the English language with all the letters in alphabetical order.

Twenty-Four-Karat Gold is not pure gold since there is a small amount of copper in it. Absolutely pure gold is so soft that it can be molded with the hands.

Electricity doesn't move through a wire but through a field around the wire.

Do you know the names of the three wise monkeys? They are: Mizaru (See no evil), Mikazaru (Hear no evil), and Mazaru (Say no evil).

55 per cent of people yawn within 5 minutes of seeing someone else yawn.

Reading about yawning makes most people yawn. Hello, zzzzz zzzz?

-ANUSHYA.R

ECE-A (III YR)

FAMILY

A family
is made of love and tears,
laughter and years.
It grows stronger
with the passing of time.
More precious
with the making of memories.

sometimes a family is made of ones
you don't like for a while.....
But you love for a life time.
It's a gift whose value is found
Not in numbers but in its capacity
to love.

It's the place you find
someone to encourage you.
believe in you.
Celebrate with you and comfort you.
A family is where your heart
feels most at home because you're
Always wanted, always welcomed,
Always needed,

BY

PRANAVI

ECE-D

FAMOUS LAST WORDS

JOHN F.KENNDY-1963

Moments before he was shot, Kennedy replied to “you certainly can’t say that the people of Dallas haven’t given you a nice welcome, mr president” with his final words, “no, you certainly can’t”.

QUEEN VICTORIA-1901

British queen: “oh ,that peace may come.” (a reference to the war at the time in south Africa). “Bertie!”(her husband, Prince Albert).

THOMAS ALVA EDISON

It is very beautiful over there.

SOCRATES

Crito,I owe a cock to Asclepius. Will you remember to pay the debt?

PABLO PICASSO

Drink to me!

GEORGE WASHINGTON

‘Tis well’

O HENRY

Turn up the lights. I do not want to go home in the dark.

OSCAR WILDE

I am dying as I have lived; beyond my means.

PA.DIVYA

ECE-B

FRIENDSHIP POEM

DEAR O DEAR !

You entered into my life and
Vanished the darkness of ignorance ,
And made my life BRIGHTER !

DEAR O DEAR !

Whenever I get FEAR,
You'll be there to wipe my TEAR;
Though the time that we
Spend TOGETHER is MEAGRE,
You are always CLOSER !

DEAR O DEAR !

Now you are not NEAR;
But the time that we spend TOGETHER,
And your memories dwell in my heart
FOREVER !

DEAR O DEAR !

Live with CHEER,
No more TEAR and FEAR;
Hope you'll achieve HIGHER,
Our friendship will live for many
YEAR(s)
And You are my FRIEND FOREVER
!!!

M.SUGANYA @ DHARANI
ECE-E
2ND YEAR

HELPING THE HOMELESS

There are more than 11 abandoned children in India, where a growing number of newborn babies are being dumped anonymously in cots placed outside, orphanages is an initiative to deter infanticide.

By helping the homeless, I see more than my share of the down-and-out. We find them in cities, in suburbs, under bridges, in parks, and on vacant industrial property. They live in tents, lean-tos, plywood and tarp contraptions that defy definition. Some of the homeless do work. Some have substance-abuse problems. Some suffer from forms of mental illness. Every one of them wants something the rest of us want as well: love and understanding.

Once a famous psychologist was leaving the Las Vegas Airport in a rental car when he saw a beggar, Disheveled and limping, he plopped a backpack on the median at a highway intersection, opened a zipper, and pulled out a sign.

As the psychologist drew closer, he unfolded the cardboard and it read:

The sign read "Why lie?" "I want beer."

The psychologist was struck by the honesty of the sign . . . and touched by the reality of this man's plight. I recalled someone once telling me, "If a homeless guy has a funny sign, he probably hasn't been homeless very long."

The Tendency to Judge:

For some, the homeless offer an opportunity for charity and for self-reflection. Many people think the homeless choose that way of life. In some cases, this is partly true. Let me offer one illustration.

A woman has been living in the woods for 14 years. She grew up in a moderate-income home in North Carolina. She worked as a nurses' assistant, had an apartment with her boyfriend. He

decided to move to another town and she went with him.

When they arrived, neither of them could find work. They pitched a tent in the woods to keep expenses down. She got a low-paying job and he started drinking. He became physically weak. The woman occasionally had seizures that interfered with her work and caused her to call in sick too often. She lost her job, her insurance, and the medication that held her seizures in check. She turned to alcohol to self-medicate.

The woman was finally beaten by her boyfriend to the extent she needed to be hospitalized for broken bones in her face and ribs. When she got out, local police helped build her a shelter in the woods. She remains there today, still self-medicating and caring for several stray dogs and cats.

In a relatively small area around this woman's camp there are 50 or 60 homeless, each with a unique story of why they are homeless.

- Some are illegal immigrants.
- Some are veterans.
- Some are people who found out a low-paying job doesn't come close to paying for rent, electricity, food, and medicine.

Free those who are wrongly imprisoned and stop oppressing those who work for you. Treat them fairly and give them what they earn. Give clothes to those who need them, and do not hide from relatives who need your help. If you do these things, your salvation will come like the dawn. Yes, your healing will

come quickly. Your godliness will lead you forward, and the glory of the LORD will protect you from behind".

So many people ignore the homeless, partly because they believe there is little they can do to "fix" them. And they are absolutely right. In fact, it is not God's intention that we "fix" them any more than it is His intention that they "fix" us. Everyone -- rich, poor, every race, every age -- has a legitimate role to play as a single body.

We may go to a homeless camp or an orphanage or a rest home with the intention of helping someone else, but ultimately, we will be helping each other grow together into the this healthy body god envisioned from the beginning of time. We live in temporary shelters, however modest or grandiose, waiting to take our places in god's mansion.



Ignoring the homeless will not make them go away, the strife continues....

S.Anjuyoga,

III yr-ECE 'A'

THE LIFE AFTER DEATH

The ancient Egyptians believed that they could enjoy life after death. This belief in an afterlife sometimes led to much preparation for death and burial. It resulted, for example, in the construction of the pyramids and other great tombs for kings and queens. Other Egyptians had smaller tombs.



The Egyptians believed that the bodies of the dead had to be preserved for the next life, and so they mummified (embalmed and dried) corpses to prevent them from decaying. After a body was mummified, it was wrapped in layers of linen strips and placed in a coffin. The mummy was then put in a tomb. Some Egyptians mummified pets, including cats and monkeys. A number of Egyptian mummies have survived to the present day.

The Egyptians filled their tombs with items for use in the after life. These items included clothing, wigs, food, cosmetics, and jewellery. The tombs of rich Egyptians also had statues representing servants who

would care for them in the next world. Scenes of daily life were painted on walls inside the tombs. The Egyptians believed that certain prayers said by priests would make Osiris bring the scenes as well as the dead to life.

The first pyramid in ancient Egypt was built for King Zoser about 2650 B.C. It rises about 60 meters and is called the Step Pyramid.

BY

S.JEYASREE
ECE-B (III Year)

HOW TO REACH YOUR GOALS



Have you ever wondered why can't I accomplish anything or why don't I earn more? Are there things that you wished you had or could do. Here are six steps to take that will lead you to achieving your goals

Steps:

1. First look inside yourself and decide what you really want and what you

need to do to achieve that. If you are having trouble consider sitting down with your boss or with a close friend to talk it out until it becomes clear.

2. Focus on improving your skills in areas that will help you achieve what you really want (your primary goals). One of the biggest uses and wastes of time is focusing on and working to improve your skills to do things that don't need to be done. Are you an accountant at Pep Boys? Do you really need to know how to change a transmission or do you need to know how to balance books, negotiate deals, and handle finances. If you are in business for yourself think about what your business niche is and how to be better within that niche. Look at other small businesses such as Greendura (www.greendura.com). This company decided that they wanted to be the best most customer friendly medical equipment out there so their business is focused on that. They have a website designed for ease of use. They offer a money back guarantee, and they even offer same day mailing of some items. They decided what they wanted and all decisions are based on that goal. What do you want? Base all your decisions on achieving that goal.
3. Next write down your goals. The process of writing them will help solidify them in your mind. It will also be something you can pull out when making decisions to remind you that your decisions must be geared towards achieving those goals.
4. Now set a deadline to each of your goals and write that down. This will give you a timeline and force you to act instead of procrastinate. You

can't procrastinate because you can see the deadline and know that if you do not do something you will run out of time.

5. Next make a list of all the things that you need to do to achieve your goals. As you look at these tasks visualize doing them and how they connect to your bigger goals. If you can not connect something to your bigger goal delete it from the list. You only do what is needed to achieve your goals and you do not need to waste your time doing things that don't help you reach your goals.
6. Now organize these lists into a single plan with all your goals and all your tasks. Add deadlines for each task. Then break the tasks into subtasks. This will make them easier to do because they are small bite size items, instead of larger overwhelming tasks that encourage procrastination.



S.KANAGALAKSHMI

ECE-B , 3RD YEAR

HERE'S SOMETHING INTERESTING...

Why are most of us right handed?

It is estimated that , 96% of all people are right –handed.This is due to the asymmetry of our brain . The left half of the brain controls the right side of the body and the right half of the brain controls the left side of the body. And the left half, or the left brain , outweighs the right. That is why the right side of our body is more skilled than the left side and most of us are right-handed.

Our body , too, is not symmetrical. We find our one foot larger than the other. The heart is on the left side and the liver on the right. But ,then , how are some people left- handed ? And how can some others use both hands equally well?

It is all due to certain differences in the structure of the brain. The brain consists of the cerebrum(the two cerebral hemispheres), the cerebellum and the medulla oblongata. The cerebral hemispheres are connected by a band of nerve fibres called 'corpus callosum'. Normally , the logical functions are performed by the left hemisphere and the abstract by the right. It is the changes in this arrangement which make the left-handed and people using both hands. It has been found that the corpus callosum is larger in these two groups than in the right-handed people. The greatest sculptors and painters of all times, Leonardo da Vinci and Michaelangelo, were both left-handed.

Why does the smoke rise in the atmosphere in a curl?

It is due to atmospheric resistance. Hot and light gases , emitted from the chimney, have to break atmospheric resistance which prevents them from going straight upwards. Thus the smoky curls facilitate their path upwards.

BY

M.GURU LAKSHMI

ECE-B (III year)

THE INDIAN PHYSICIST WHO WORKED WITH ALBERT EINSTEIN



Early life and career:

Bose was born in Calcutta, India, the eldest of seven children. His father, Surendranath Bose, worked in the Engineering Department of the East Indian Railway Company. Bose attended Hindu School in Calcutta, and later attended Presidency College, also in Calcutta, earning the highest marks at

each institution. He came in contact with teachers such as Jagadish Chandra Bose and Prafulla Chandra Roy who provided inspiration to aim high in life. From 1916 to 1921 he was a lecturer in the physics department of the University of Calcutta. In 1921, he joined the department of Physics of the then recently founded Dhaka University. In 1924, while working as a Reader at the Physics Department of the University of Dhaka, Bose wrote a paper deriving Planck's quantum radiation law without any reference to classical physics and using a novel way of counting states with identical particles. This paper was seminal in creating the very important field of quantum statistics. After initial setbacks to his efforts to publish, he sent the article directly to Albert Einstein in Germany. Einstein, recognizing the importance of the paper, translated it into German himself and submitted it on Bose's behalf to the prestigious

Zeitschrift für Physik.

As a result of this recognition, Bose was able to leave India for the first time and spent two years in Europe, during which he worked with Louis de Broglie, Marie Curie, and Einstein.

After his European stay S N Bose returned to Dhaka in 1926. He was also Dean of the Faculty of Science at Dhaka University for a long period. When the partition of India became imminent, he returned to Calcutta and taught at Calcutta University until 1956, when he retired and was made professor emeritus.

Bose published a paper on Max Planck's black body law, Planck's Law and the Hypothesis of Light Quanta (1924). This paper led him into

work with Albert Einstein. Bose also worked on statistical mechanics, the electromagnetic properties of the ionosphere, theories of X-ray crystallography, and unified field theory. He was named National Professor of India in 1958.

BY

P.GOMATHI PRIYA

ECE-B(III year)

HOW THEY CAME TO BE CALLED

There are many companies / brands / products whose names were derived from strange circumstances.

ADOBE

This came from the name of the river Adobe Creek that ran behind the house of the founder John Warnock.

APPLE COMPUTERS

It was the favourite fruit of the founder Steve Jobs.

COMPAQ

This name was formed by using COMP, for computer, and PAQ to denote a small integral object.

GOOGLE

The name started as a joke boasting about the amount of information the search engine would be able to search. It was originally named 'Googol', a word for the number represented by 1 followed by 100 zeros. After founders – Stanford

graduate students Sergey Brin and Larry Page presented their project to an angel investor, they received a cheque made out to 'Google'.

HOTMAIL

Founder Jack Smith got the idea of accessing e-mail via the web from a computer anywhere in the world. When Sabeer Bhatia came up with the business plan for the mail service, he tried all kinds of names ending in 'mail' and finally settled for hotmail as it included the letters 'html' - the programming language used to write web pages.

INTEL

Bob Noyce and Gordon Moore wanted to name their new company 'Moore Noyce' but that was already trademarked by a hotel chain so they had to settle for an acronym of INTEgrated Electronics.

MOTOROLA

Founder Paul Galvin came up with this name when his company started manufacturing radios for cars. The popular radio company at the time was called Victrola.

MICROSOFT

Coined by Bill Gates to represent the company that was devoted to MICRO computer software.

SUN

Founded by 4 Stanford University buddies, SUN is the acronym for Stanford University Network.

SONY

It originated from the latin word 'sonus' meaning sound, and 'sonny' a

slang used by Americans to refer to a bright youngster.

YAHOO !

The word was invented by Jonathan Swift and used in his book 'Gulliver's Travels'. It represents a person who is repulsive in appearance and action and is barely human.

Jerry Yang and David Filo selected the name because they considered themselves yahoos.

BY

**P.KALEESWARI
ECE-B (III Year)**

MOTHER'S LOVE

A Mother's love is something
that no one can explain,
It is made of deep devotion
and of sacrifice and pain,
It is endless and unselfish
and enduring come what may
For nothing can destroy it
or take that love away...
It is patient and forgiving
when all others are forsaking,
And it never fails or falters
even though the heart is breaking...
It believes beyond believing
when the world around condemns,
And it glows with all the beauty
of the rarest, brightest gems...
It is far beyond defining,
it defies all explanation,
And it still remains a secret
like the mysteries of creation...
A many splendour-ed miracle
man cannot understand
And another wondrous evidence
of God's tender guiding hand.

By

P.Pranavi,ECE-D,III yr

“MY DAD MY STRENGTH”

Here is a mighty father
For a cute little daughter
Always there to hold his precious girl
high off the ground
Where the sunlight sings aloud the
soothing song!
Like being the deep music
That tells her all is right
When she awakens frantic with
The terrors of the night.

That rises in her heart
And shows her how she might get home
When everything else falls apart.
The guiding light, in a pitch dark forest.

He is just there like the gospel of god
Giving her the love
That is her sea and air
So diving deep or soaring high
She'll always find him there.



©/thtstuc

By-S.Anju Yoga
III year - ECE 'A'

MY DEAR FRIEND

My friend when I think of you.
I think of all that we've been through.
All the times we argue and fight,
I know deep inside that it isn't right.

I, then feel bad and a lot of pain.
It feels like I've fallen from the sky like
the rain.

I love you dear friend with all of my
heart.

But now that you're gone I've fallen
apart.

I'm getting better as the days go by.
I wish sometimes this was all a big lie.

I pray to you every night.
It's like you're my fire, a burning light.

My dear friend, I miss you a lot.
I still wonder why you were put in that
spot.

I know you're in a place much better
than here.

Watching and helping me with all of my
fear.

Our friendship my dear friend,
we will have to the end.

Friends till the end is what we will be.

Someday we'll be together,
together you and me.

P.DIVYA

2ND YEAR ECE-B

GONE ARE THE DAYS....

Gone are the days...

When we got into our class together

Enjoying the teaching,

Going by their preaching...

Gone are the days...

When we spent time together

Sitting over the bench,

Sharing our views with a munch...

Gone are days...

When we promised each other,

To be together, forever...

Now...

I haven't seen you for years,

Oh my friend, oh my dear-

I need you with me, here

Please be near...

Waiting,

Waiting for you,

Forever.

Farhanah Mansur

II year, ECE B

JUST ONE

One song can spark a moment,
One flower can wake the dream
One tree can start a forest,
One bird can herald spring.

One smile begins a friendship,
One handclasp lifts a soul.
One star can guide a ship at sea,
One word can frame the goal

One vote can change a nation,
One sunbeam lights a room
One candle wipes out darkness,
One laugh will conquer gloom.

One step must start each journey.
One word must start each prayer.
One hope will raise our spirits,
One touch can show you care.

One voice can speak with wisdom,
One heart can know what's true,

One life can make a difference,
You see, it's up to you!

BY

NIRANJANA

3RD YEAR,ECE-D

☺ PRANKY CORNER ☺

This puzzle is called Lateral Thinking.
Scroll down slowly and be honest to yourself.

Think like a wizard . . .

man

Q1. -----

board

Ans. = man overboard
Okay, let's see if you've got the hang of it.

stand

Q2. -----

i

Ans. = I understand

OK . . .

Got the drift ?

Let's try a few now and see how you fare ?

Q3. /r/e/a/d/i/n/ g/

Ans. = reading between the lines

Q4. r
road
a
d

Ans. = cross road

Not having a good day now, are you
?Redeem yourself.

Q5. cycle
cycle
cycle

Ans. = tricycle
Not easy to figure out ha!

0
Q6. -----
M.D.
Ph.D.

Ans. = two degrees below zero

C'mon give it a little thought! !

knee
Q7. -----
light

Ans. = neon light
(knee - on - light)

U can prove u r smart by getting this
one.

ground
Q8. -----

feet feetfeetfeetfeetfeet

Ans. = six feet underground

Oh no, not again ! !

Q9. he's / himself

Ans. = he's by himself

Now u messing up big time.

Q10. ecnalg

Ans. = backward glance

Not even close! !

Q11. death life

Ans. = life after death

Okay last chance

Q12. **THINK**

Ans. = think big ! !

And the last one is real fundoo - - -

Q13.ababaaabbbbbaaabbbababaabbbaa
bbbb. ..

Ans. = long time no 'C'

K.ASHOK KUMAR

3RD YEAR,ECE-A



Written with a pen, Sealed with a kiss.
If you are my friend, please answer this:

Are we friends Or are we not?
You told me once But I forgot.

So tell me now And tell me true.
So I can say.... "I'm here for you."

Of all the friends I've ever met,
You're the one I won't forget.



And if I die Before you do,
I'll go to heaven And wait for you.

And come the day you don't come home,
I'll know you went the other way.

I'll give the angels Back their wings
And risk the loss Of everything.

Just to prove my friendship is true..
to have a friend like you!



Quotes to Remember...

1.The best preparation for good work tomorrow is to do good work today.
Elbert Hubbard

2.Action is the foundational key to all success.
Pablo Picasso

3.Failure is success if we learn from it.
Malcolm Forbes

4.Intelligence is really a kind of taste:
taste in ideas.
Susan Sontag

5.Ignorance is no excuse, it's the real thing.
Irene Peter

6.Good judgment comes from experience and experience comes from bad judgment.
Fred Brooks

7.Facts are the air of scientists. Without them you can never fly.
Linus Pauling

8.If you can dream it you can do it.

9.Everything comes to him who hurttles he waits.

10. It is difficult to what is impossible for the dream of yesterday is the hope of today and the reality of tomorrow.

11. What we are is god's gift to us
What we become is A gift to god.

12. Three sentences for getting success:
a; know more than other
b; work more than other
c; expect less than other.

William Shakespeare

RIDDLES

1. I am the fountain from which no one can drink. For many I am considered a necessary link. Like gold to all I am sought for, but my continued death brings wealth for all to want more. Who am i?

ANS> OIL

2. A box without hinges, lock or key, yet golden treasure lies within. Who am i?

ANS> AN EGG

3. Stanley goes to a park on his new bicycle that his Dad gifted him for Christmas. When he reached the park, he saw that there were 9 bicycles and tricycles in the parking lot. If you count all the wheels of the bicycles and tricycles comes to 23, how many tricycles would you say there were?
ANS>5

4. On your clock, the minute hand has covered the hour hand completely. It is somewhere between 8:00 and 9:00 on the clock. What time is it?

ANS> About 8:43

5. There are 5 dogs whose weights are in question here. The first dog, Jumbo is heavier than the second dog, Spike. Cuddles weighs more than Dodo, but Cuddles weighs less than Pumpkin. Bruno weighs more than Jumbo. And Pumpkin weighs less than Jumbo. Can you find the order of their weights, starting from the heaviest?

ANS> Jumbo, Pumpkin, Cuddles, Bruno & Dodo

6. What does a policeman have in his sandwich?

ANS> traffic jam

7. What asks no question but demands an answer?

ANS> a door bell

8. What belongs to you, but is used more by others?

ANS> your name

9. Where did Pilgrims land when they arrived in America?

ANS> on their feet

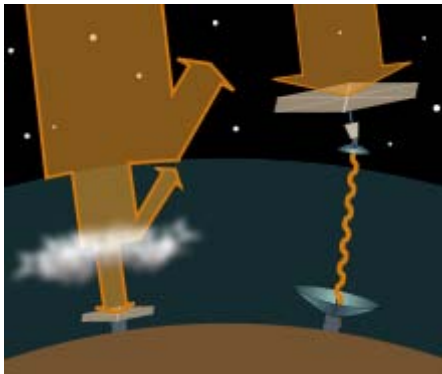
10. If you wrote all of the numbers from 300 to 400 on a piece of paper, how many times would you have written the number 3?

ANS> 120 times

SPACE-BASED SOLAR POWER

Space-based solar power (SBSP) is a system for the collection of [solar power](#) in [space](#), for use on [Earth](#). SBSP differs

from the usual method of solar power collection in that the **solar panels** used to collect the energy would reside on a **satellite** in **orbit**, often referred to as a solar power satellite (SPS), rather than on Earth's surface. In space, collection of the Sun's energy is unaffected by the various difficulties which reduce efficiency or capacities of Earth surface solar power collection.



Design

Space-based solar power essentially consists of three parts:

1. a means of collecting solar power in space, for example via **solar cells** or a **heat engine**
2. a means of transmitting power to earth, for example via microwave or laser
3. a means of receiving power on earth, for example via a microwave antennas .



The space-based portion will be in a **freefall**, **vacuum** environment and will not need to support itself against gravity other than relatively weak tidal stresses. It needs no protection from terrestrial wind or weather, but will have to cope with space-based hazards such as micrometeors and solar storms.

Solar energy conversion (solar photons to DC current)

Two basic methods of converting sunlight to electricity have been studied: **photovoltaic** (PV) conversion, and **solar dynamic** (SD) conversion.

Most analyses of solar power satellites have focused on photovoltaic conversion (commonly known as “**solar cells**”). Photovoltaic conversion uses **semiconductor** cells (e.g., **silicon** or **gallium arsenide**) to directly convert photons into electrical power via a quantum mechanical mechanism. Photovoltaic cells are not perfect in practice, as material purity and processing issues during production affect performance; each has been progressively improved for some decades. Some new, thin-film In an SPS implementation, photovoltaic cells will likely be rather different from the glass-pane protected solar cell panels familiar to many in current terrestrial use, since they will be optimized for weight, and will be designed to be tolerant of the space radiation environment, but will not need to be encapsulated against corrosion from environmental exposure or biological deterioration. They do not require the structural support required for terrestrial use, where the considerable gravity and wind loading imposes structural requirements on terrestrial implementations.

Wireless power transmission to the Earth

Wireless power transmission was proposed early on as a means to transfer energy from collection to the Earth's surface. The power could be transmitted as either microwave or laser radiation at a variety of frequencies depending on system design. Whichever choice is made, the transmitting radiation would have to be non-ionizing to avoid potential disturbances either ecologically or biologically. This established an upper limit for the frequency used, as energy per photon increases with frequency.

Location

GEO

The main advantage of locating a space power station in geostationary orbit is that the antenna geometry stays constant, and so keeping the antennas lined up is simpler. Another advantage is that nearly continuous power transmission is immediately available as soon as the first space power station is placed in orbit; other space-based power stations have much longer start-up times before they are producing nearly continuous power.

LEO/MEO instead of GEO

A collection of LEO (Low Earth Orbit) space power stations has been proposed as a precursor to GEO (Geostationary Orbit) space-based solar power. There would be both advantages and disadvantages. It might be possible to

deploy LEO systems sooner than GEO because the antenna development would take less time, but it may take longer to prepare and launch the number of required satellites.

Earth-based infrastructure

The Earth-based receiver antenna is a critical part of the original SPS concept. It would probably consist of many short dipole antennas, connected via diodes. Microwaves broadcast from the SPS will be received in the dipoles with about 85% efficiency. With a conventional microwave antenna, the reception efficiency is still better, but the cost and complexity is also considerably greater, almost certainly prohibitively so. Rectennas would be multiple kilometers across. Crops and farm animals may be raised underneath a rectenna, as the thin wires used for support and for the dipoles will only slightly reduce sunlight, or non arable land could be used, so such a rectenna would not be as expensive in terms of land use as might be supposed.

BY

P.GNANA SOUNDARYA

ECE-B

SELF APPRAISAL

A little boy went to a telephone booth which was at the cash counter of a store and dialed a number.

The store-owner observed and listened to the conversation:

Boy: "Lady, Can you give me the job of cutting your lawn?"

Woman: (at the other end of the phone line) "I already have someone to cut my lawn."

Boy: "Lady, I will cut your lawn for half the price than the person who cuts your lawn now."

Woman: I'm very satisfied with the person who is presently cutting my lawn.

Boy: (with more perseverance) "Lady, I'll even sweep the floor and the stairs of your house for free.

Woman: No, thank you. With a smile on his face, the little boy replaced the receiver. The store-owner, who was listening to all this, walked over to the boy. Store Owner : "Son... I like your attitude; I like that positive spirit and would like to offer you a job.."
Boy: "No thanks,

Store Owner: But you were really pleading for one.

Boy: No Sir, I was just checking my performance at the job I already have. I am the one who is working for that lady I was talking to!"
This is what we call "Self Appraisal"

BY
V.INDHUMATHI,
3RD YEAR
,ECE-B

STOCK MARKET

The stock market is actually an avenue from which the stock of companies is purchased and sold.

Some people still believe that the stock market and Wall Street are one and the same. However, Wall Street in New York is just a single example of a stock market.

When you review the history of the stock market, Wall Street is quite significant. The concept of the stock market was actually born and developed on Wall Street.

In 1653, an establishment was formally built where Wall Street currently exists. The purpose of this establishment was for defense rather than commerce. Dutch settlers built a 12 foot stockade fence to keep out the British and Native Americans.

By 1685, this defensive establishment was torn down and replaced. The new street was named Wall Street by the British.

What About The Stock Exchange?

Two powerful stock exchanges emerged from Wall Street, which is why it is so famous. As a result of these historical developments, chaotic trading developed and turned into the hectic financial markets we all know today.

In 1790, the first United States stock exchange was founded in Philadelphia. A group of New York traders got together two years later to consider setting up a security business. This group of 24 men ultimately founded

what is known as the New York Stock Exchange.

By 1817, New York merchants were upset by the poor state of their stock exchange. One member went to Philadelphia to check out their trading and discovered they were doing quite well in their exchange. The merchant came back to New York and shared how things were done in Philadelphia to improve operations. Within a short period of time, the New York Stock and Exchange Board were formally organized.

This exchange center was inaugurated on world famous Wall Street and the rest was history. Despite its difficult start, the New York Stock Exchange turned into the place where billions of dollars in stocks and bonds are bought and sold every day.

The Wall Street success story did not occur overnight, much like any other major enterprise. During the early 1900s, the New York Stock Exchange was on the rise. However, this incredible financial boom could not be sustained because the stock market crashed in 1929. The world was absolutely shocked and the unexpected stock market crash was the cause of the Great Depression.

Slowly the economy recovered over time. Despite this fact, the mistakes of the Great Depression came back to haunt the exchange. The stock market crashed again in 1987. It was such a crippling crash that the Dow Jones actually suffered the biggest loss in a single day in the history of the stock market.

Over the past two decades, the stock market industry and the government

have been trying to set up measures to prevent another major crash. Today the stock market is an essential part of the world economy. Thus, it is of the utmost importance to reduce or prevent another stock market crash to protect international economic interests.



Success Steps...

College - a new environment

Most of us who shined very well during school days are found to be silent and left unseen during our college days. Rather, a few people who passed out with very nominal marks during their school days give an outstanding performance in college. The reason behind may not only be enhancing the inherent potential and not knowing the way to do so. Here are some tips to shine.....

Personal qualities that support academic success

Students who have the following personal qualities are much more likely to thrive in college.

- Self-awareness and self-acceptance
- Pro activity as opposed to reactivity
- Perseverance
- Skill in setting short and long-range goals
- Use of effective support systems

- Strong emotional coping strategies
- Optimum use of people around

Steps a student should take to build a solid college support:

- Think through what kind of support you'd like to have from your parents and friends and express your needs before you go to college
- Think about the kind of academic support you'll need (for example, will you need tutoring) and make plans to set this up
- **Don't suffer in silence—speak out, reach out when you need to**
- Get professional, trained help when you need it: tutors, doctors, etc.

Self-advocacy steps

Successful students understand themselves well. They know their strengths and they have developed ways to minimize the effects of their weaknesses. They also have a clear idea of their short-range and long-term goals, and are committed to meeting these goals.

These self-advocacy steps will help students obtain the supports they need, not only from others but from themselves as well!

- Have a clear plan to graduate in a certain time frame and set your schedule to realistically accomplish this plan

- Don't wait until things get bad to see you need a different approach; if it isn't working—try another approach
- Remember: resistance and avoidance stunt maturity; meet challenges head on and don't be afraid to make mistakes
- Evaluate and think through setbacks—they are the teachers of success!
- Success is a consistent mindset that says "I can do this, I will do this!"
- Everything goes better when you get enough sleep and eat more healthy foods
- Plan ahead on how to manage stress, loneliness, and change
- Seek out stabilizing forces (people, classes, work experiences, living arrangements, etc.)
- As soon as you identify a problem surfacing in a class, figure out how to remove it from your path
- Make better time estimates; after you figure out how long you think it will take you to do what you need to do, multiply that by at least 150%
- Keep your long-term, personal goals front and center in your mind, guiding you through the tough times!
- **Reward yourself for meeting your deadlines and achieving your goals!**

Study Skills

Students who master study skills and use their strengths to learn succeed in college. Try these tips:

- Set up a realistic study schedule and keep to it!!!! Don't let spontaneity ruin your completion of college
- Frequently review and critique your study approaches to find what is working and emphasize that
- Don't put off the "boring" or unpleasant tasks; doing so will leave you unprepared and can destroy your grade; find a way to see the value it brings to you as a person or as a professional
- Take frequent, short breaks when your attention fades
- Keep a calendar and an assignment book; record all due dates, test and quiz dates, etc. and schedule study/work sessions by backward planning
- Work with classmates who are strong students and stay on track
- While reading, stop frequently to retell the main idea and details in your own words

- Read the material before you go to the lecture; bring notes or an outline and fill in added material as you listen

**YOU BECOME WHAT YOU
THINK, SO THINK BIG...**

BY

KARTHEEK SAI

&

CHANDRA PRADEEPAN

III yr ECE 'A'

TEACHERS

When we stride or stroll across the frozen lake,
We place our feet where they have never been.
We walk upon the unwalked places, But we are uneasy.
Who is down there but our old teachers?
Water that once could take no human weight-
We were students then- holds up our feet,
And goes on ahead of us for a mile.
Beneath us the teachers, and around us the stillness

KANAGALAKSHMI.S
3rd year, ECE-B

THE FIRE INSIDE

When all is lost and hope has fled
When fear is strong and strength is dead
When love and joy abandon you
When mental anguish grows in you

When the last of efforts fail to save
When your fate is ill, your mind enslaved
And when your head hangs low in misery
This is when you'll find the key

A single ember from deep within
Burns hotter and hotter, as flames begin
The fire of truth will light the way
And help you fight, this lonely day

The battle is long, the struggle is rough
Never regret not giving enough
For when we offer our very best,
Our very soul is put to the test

Stand tall and true and you'll prevail
Just hold on tight and never bail
You will survive if you don't quit
Victory is there, if you reach for it

One day in the future, you will look to the past,
And know you had what it takes to last
So never give up and good things will come,
Not just honor and pride, but a job well done.

JOHNSY MARY JOHNSON
ECE F,2ND YR

TO MY DEAR FRIEND

I got your back
You got mine,
I'll help you out
Anytime.

To see you hurt
To see you cry,
Makes me weep
And wanna die.

And if you agree
To never fight,
It wouldn't matter
Who's wrong or right.

If a broken heart
Needs a mend,
I'll be right there
To the end.

If your cheeks are wet
From drops of tears,
Don't you worry,
Let go of your fears.

Hand in hand
Love is sent,
We'll be friends
Till the end.

**By,
V.Indumathy,
III yr ECE-B,**

TONGUE TWISTERS

1. If you understand, say "understand". If you don't understand, say "don't understand". But if you understand and say "don't understand". How do I understand that you understand? Understand!
2. I wish to wish the wish you wish to wish, but if you wish the wish the witch wishes, I won't wish the wish you wish to wish.
3. Sounding by sound is a sound method of sounding sounds.
4. If two witches were watching two watches, which witch would watch which watch?
5. I thought a thought. But the thought I thought wasn't the thought I thought I thought. If the thought I thought I thought had been the thought I thought, I wouldn't have thought so much.
6. Once a fellow met a fellow In a field of beans. Said a fellow to a fellow, "If a fellow asks a fellow, Can a fellow tell a fellow What a fellow means?"
7. . Mr. Inside went over to see Mr. Outside. Mr. Inside stood outside and called to Mr. Outside inside. Mr. Outside answered Mr. Inside from inside and Told Mr. Inside to come inside. Mr. Inside said "NO", and told Mr. Outside to come outside. Mr. Outside and Mr. Inside argued from inside
8. 2and outside about going outside or coming inside. Finally, Mr. Outside coaxed Mr. Inside to come inside, then both Mr. Outside and Mr. Inside went outside to the riverside.
9. The owner of the inside inn was inside his inside inn with his inside outside his inside inn.
10. We surely shall see the sun shine shortly. Whether the weather be fine, Or whether the weather be not, Whether the weather be cold Or whether the weather be hot,... We'll weather the weather Whatever the weather, Whether we like it or not. Watch? Whether the weather is hot. Whether the weather is cold. Whether the weather is either or not. It is whether we like it or not.
11. If you tell Tom to tell a tongue-twister his tongue will be twisted as tongue-twister twists tongues.

By

P.Pranavi

ECE-D,III yr

AUTISM !!!!!

What Does Autism Mean?

Autism causes kids to experience the world differently from the way most other kids do. It's hard for kids with autism to talk with other people and express themselves using words. Kids who have autism usually keep to themselves and many can't communicate without special help. They also may react to what's going on around them in unusual ways. Normal sounds may really bother someone with autism — so much so that the person covers his or her ears. Being touched, even in a gentle way, may feel uncomfortable.

Kids with autism often can't make connections that other kids make easily. For example, when someone smiles, you know the smiling person is happy or being friendly. But a kid with autism may have trouble connecting that smile with the person's happy feelings. A kid who has autism also has trouble linking words to their meanings. Imagine trying to understand what your mom is saying if you didn't know what her words really mean. It is doubly frustrating then if a kid can't come up with the right words to express his or her own thoughts.

Autism causes kids to act in unusual ways. They might flap their hands, say certain words over and over, have temper tantrums, or play only with one particular toy. Most kids with autism don't like changes in routines. They like to stay on a schedule that is always the same. They also may insist that their toys or other objects be arranged a certain way and get upset if these items are moved or disturbed.

What Causes Autism?

Autism affects about 1 in every 150 kids, but no one knows what causes it. Some scientists think that some kids might be more likely to get autism because it or similar disorders run in their families. Knowing the exact cause of autism is hard because the human brain is very complicated.

What Do Doctors Do?

Figuring out if a kid has autism can be difficult. A parent is usually the first to suspect that something is wrong. Maybe the kid is old enough to speak but doesn't, seem interested in people, or behaves in other unusual ways. But autism isn't the only problem that can cause these kinds of symptoms. For example, kids who have hearing problems might have trouble speaking, too. Usually, the results of lab tests and other medical tests are normal in kids with autism, but doctors may do them to make sure the kid doesn't have other

problems. These medical tests can include blood and urine tests, a hearing exam, an EEG (a test to measure brain waves), and an MRI (a picture that shows the structure of the brain). Intelligence (IQ) tests also might be done.

How Is Autism Treated?

There is no cure for autism, but doctors, therapists, and special teachers can help kids with autism overcome or adjust to many difficulties. The earlier a kid starts treatment for autism, the better. Most understand words better by seeing them, so therapists teach them how to communicate by pointing or using pictures or sign language. That makes learning other things easier, and eventually, many kids with autism learn to talk.

Therapists also help kids learn social skills, such as how to greet people, wait for a turn, and follow directions. Some kids need special help with living skills (like brushing teeth or making a bed). Others have trouble sitting still or controlling their tempers and need therapy to help them control their behavior. Some kids take medications to help their moods and behavior, but there's no medicine that will make a kid's autism go away. Students with mild autism sometimes can go to regular school. But most kids with autism need calmer, more orderly surroundings. They also need teachers trained to understand

the problems they have with communicating and learning.

S.Nirmalkumar

ECE-D III Year

AWARDS IN THE RUN

Awards are presented in almost every area of achievement the ones that receive the most attention in newspapers and on television are those given to famous film stars, writers, artist and singers .Among the best known are the following...

ACADEMY AWARDS (OSCARS)

These are the most famous of all the film awards .The Oscar presentations are now watched on television by more than a billion people worldwide. The academy awards also called as Oscars after the statuette that winners receive were first presented in 1929.Walt Disney has won the most, a total of 20.

BAFTA AWARDS

The British Academy Of Film And Television arts awards began in 1947.British Actress Judi Dench has won 8 awards for her television and film roles.

BOOKER PRIZE

This British book price was set up in 1968 and is awarded each year for a novel published in the previous twelve months. It is now known as The Man Booker Prize.

BRIT AWARDS

The BRITs are popular music awards, which started in 1977. Robbie Williams has been the most successful act by far, with 15 wins (including four as a member of TAKE THAT).

CANNES FILM FESTIVAL AWARD

This annual international film festival is held in CANNES in south of France, it's most important award is the PALMED' Or (Golden Palm) ,which dates back to 1955.

GRAMMY AWARDS

The US music awards have been presented since 1959.The classical awards have been dominated by conductor sir Georg Solti, who won 38 Grammys. Stevie Wonder gained an unrivalled 22 awards for popular music, while Beyonc'e won in 5 categories in 2003.

NOBEL PRIZE

These international awards are named after Alfred Nobel the inventor of

Dynamite, and were first presented in 1901.there are now six categories:

Physics, Chemistry, Medicine, Literature, Peace and Economics.

Famous winners include WINSTON CHURCHILL (literature 1953) and MOTHER THERESA (Peace, 1959).

TURNER PRIZE

This British Art Prize Is named after Landscape Painter J.M.W.Turner and was established in 1984.The Winning Works Are Often Controversial, And usually not conventional paintings. Damien Hirst is a notorious winner.

S.CHITRA
III YEAR, ECE-A

BORN TO WIN

BORN ARE YOU IN JANUARY
YOU BELITTLE CRISIS WITH BRAVERY;
BORN ARE YOU IN FEBRUARY
YOUR COUNTENANCE IS EXTRA-ORDINARY;
BORN ARE YOU IN MARCH
SKY ISN'T BEYOND YOUR REACH
BORN ARE YOU IN APRIL,
YOUR ETIQUETTE IS CIVIL;
BORN ARE YOU IN MAY
LAURELS COME YOUR WAY;
BORN ARE YOU IN JUNE
YOU ALLURE HUMANS IN TUNE;
BORN ARE YOU IN JULY
YOUR PERFORMANCE IS SO TRULY;
BORN ARE YOU IN AUGUST,
YOUR STYLE IS ROBUST.
BORN ARE YOU IN SEPTEMBER,
YOU ARE EXUBERANT IN ENDEAVOURS.
BORN ARE YOU IN OCTOBER,
INDELIBLE IMPRESSIONS YOU RENDER
BORN ARE YOU IN NOVEMBER,
YOUR CALIBER IS AS HARD AS TIMBER;
BORN ARE YOU IN DECEMBER,
YOU MANIFEST AS COOL AS CUCUMBER;
BOTHER NOT YOUR DATE OF BIRTH,
STIVE WITH ZEAL TO REACH THE ZENITH;
BORN ARE YOU TO WIN,

TO FALL IS NOT A SIN,
RISE LIKE A PHOENIX,
WORK CONFIDE AND RELAX
TO BE WINNER
LET NOT YOUR CONFIDENCE THINNER

BY,
T.NIVETHITHA,
II YEAR
ECE-C.

FUN FUNNY FACTS

1. Donald Duck comics were banned in Finland because he doesn't wear pants.



2. The cigarette lighter was invented before the match.



3. 40 percent of McDonald's profits come from the sales of Happy Meals.



4. TYPEWRITER is the longest word that can be made using the letters only on one row of the keyboard.



5. Butterflies taste with their feet.



6. Coca-Cola was originally green.



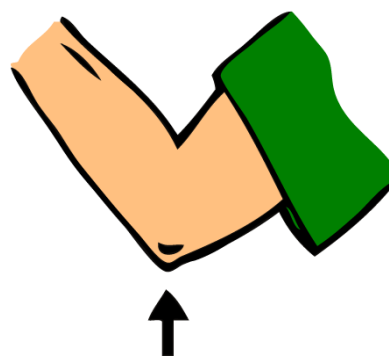
7. If you yelled for 8 years, 7 months and 6 days, you would have produced enough sound energy to heat one cup of coffee.



8. Every day more money is printed for Monopoly than the US Treasury.



9. It is impossible to lick your elbow.



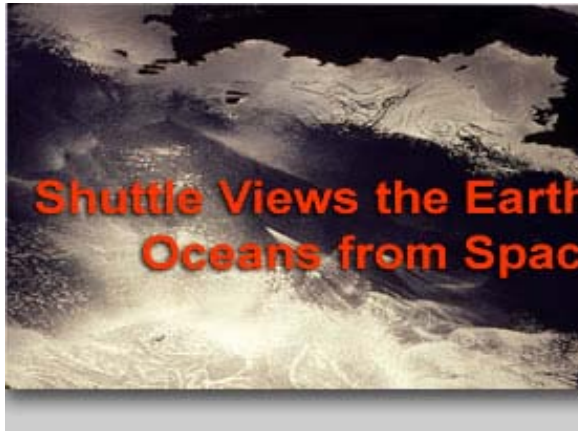
10. Intelligent people have more zinc and copper in their hair.



Kanagalakshmi.s

ECE-B

3rd year



Looking at Gemini photographs of the Gulf of Mexico, oceanographers realized that color photographs could be used to identify the spawning grounds and migration habits of the Gulf shrimp. This information was rapidly passed on to the fishing fleet. As a result of this discovery in August 1966, oceanographers have been involved in briefing the crews for every subsequent U.S. manned spaceflight.

Exciting results were returned by the Apollo and Skylab programs. Anticipating an increased spaceflight rate with the advent of the shuttle program, NASA assigned a small group of Earth scientists to the Space Shuttle Earth Observations Project (SSEOP) to train each shuttle crew and instruct them on the latest advances in various disciplines. Two oceanographers have worked with this team throughout the shuttle program.

Important oceanographic discoveries were made right from the start. On STS-1, the first test flight of Columbia in

1981, John Young and Robert Crippen photographed spiral eddies in the Gulf of Oman. At the time, this was thought to be an isolated example, but the very next mission revealed another such eddy in the Caribbean, which raised questions as to how widespread these features might be in the world's oceans.

Astronauts became fascinated with how much they could observe in the oceans that cover more than 70% of the surface of planet Earth. STS-8 photographs provided the first indication that spiral eddies might be interconnected with pictures of a field of spiral eddies throughout the Mozambique Channel in the Indian Ocean. Most intriguing was the fact that, like the circular systems of cyclones and hurricanes, spirals in the southern hemisphere appeared to rotate in the opposite direction from those seen in the northern hemisphere.

In May 1984, a change in the vehicle assigned to STS-17 (which was to become 41-G), from Columbia to Challenger, meant, because of slight differences in the onboard configuration, that an extra seat became available. No mission had carried a seven-member crew, but the mission, with its high-inclination (57°) orbit, was ideal for ocean viewing, and oceanographer Paul Scully-Power was assigned to fly.

By
T.MATHUMITHA
 3RD YEAR, ECE-C

FRIENDS APART

THIS IS TO MY DEAR FRIEND
 MY DAYS WITH YOU WERE HAY DAYS
 ALWAYS GREAT TO REMEMBER
 NOTHING BUT ONLY SMILING DAYS LIVING HAPPILY IN THIS
 MUCH TO MY HEARTS FAVOUR. RLDLY PLACE.
 EVERYTHING WAS SWEET,
 TILL WE FELL APART,
 WAITING LONG FOR OUR MEET
 TEARS ROLLED DOWN MY HEART.
 SOMETHING HAPPENED DRASTICALLY
 INNOCENT ME
 HOPING THE CHANGE SLOWLY
 VICTIMISE ME.
 IT'S ALL DESTINED TO HAPPEN
 IT'S ALL GOD'S PLAY
 HE KNOWS ALL THAT HAPPENED
 HE'LL REVEAL THE PLAY.
 I DON'T HAVE ANY PROOF
 TO GUARD AGAINST THE FAULTS SHOWERED ON ME
 GOD HAS TO PROVE
 THE INNOCENCE IN ME.
 I WAS SHATTERED TO PIECES
 WHEN YOU SAID SO
 ALES! NOT AGAIN! LET'S LIVE IN PEACE
 CO'S I NEED YOU SO.
 FUNERAL IS NOT THE MEETING PLACE
 AS YOU SAID
 LIVING HAPPILY IN THIS WORLDLY PLACE.
 WITH YOU
 ALL MY
 DEAR FRIENDS.....

BY
M.PRATIBHA II YEAR,
ECE-'D'

I LOVE MY DAD

AT 4 YEARS
 MY DADDY IS GREAT.
 AT 6 YEARS
 MY DADDY KNOWS EVERYBODY.
 AT 10 YEARS

MY DADDY IS GOOD BUT IS SHORT TEMPERED.
 AT 12 YEARS
 MY DADDY WAS VERY NICE TO ME WHEN I WAS YOUNG
 AT 14 YEARS
 MY DADDY IS GETTING FASTIDIOUS.
 AT 16 YEARS
 MY DADDY IS NOT IN LINE WITH THE CURRENT TIMES.
 AT 18 YEARS
 MY DADDY IS NOT IN LINE WITH CURRENT TIMES.
 AT 20 YEARS
 OH!IT'S BECOMING DIFFICULT TO TOLERATE
 DADDY.WONDER HOW MOTHER PUTS UP WITH HIM.
 AT 25 YEARS
 DADDY IS OBJECTING TO EVERYTHING.
 AT 30 YEARS
 IT'S BECOMING DIFFICULT TO MANAGE MY SON.I WAS SO SCARED OF MY
 FATHER WHEN I WAS YOUNG/
 AT 40 YEARS
 DADDY BROUGHT ME UP WITH SO MUCH DISCIPLINE.EVEN I SHOULD DO
 THE SAME.
 AT 45 YEARS
 I AM BAFFLED AS TO HOW MY DADDY BROUGHT US UP.
 AT 50 YEARS
 MY DADDY FACED SO MANY HARDSHIPS TO BRING US UP.I AM UNABLE TO
 MANAGE A SINGLE SON.
 AT 55 YEARS
 MY DADDY WAS SO FAR SIGHTED AND PLENED SO MANY THINGS FOR
 US.
 HE IS ONE OF HIS KIND AND UNIQUE.
 AT 60 YEARS
 THUS,IT TOOK 56 YEARS TO COMPLETE THE CYCLE AND COME BACK TO
 THE 1ST STAGE
 REALIZE THE TRUE VALUE OF YOUR PARENTS BEFORE IT TOO LATE.

BY
VETRI SELVI

IN SEARCH OF A FRIEND

MY FRIEND SHOULD BE MY MEDICINE
 WHEN I AM IN PAIN
 MY FRIEND SHOULD BE MY LETTER
 WHEN I AM FAR
 MY FRIEND SHOULD BE MY SMILE
 WHEN I AM SAD
 MY FRIEND SHOULD BE MY CHAIN
 WHEN I AM MAD
 MY FRIEND SHOULD BE MY CONCENTRATION
 WHEN I WORK
 MY FRIEND SHOULD BE MY SEASHORE
 WHEN I SOAK
 MY FRIEND SHOULD BE MY HAND-KERCHIEF
 WHEN I CRY
 MY FRIEND SHOULD BE MY LIFE
 WHEN I DIE
 WILL I GET SUCH A FRIEND?

BY,
M.PRATIBHA,II YEAR,
ECE-‘D’.

MAAA! TUJEE SALAAM

WHAT MAKES THE MOTHER SO SPECIAL
 IT'S THE LOVE SHE ALWAYS SHARED WITH ME
 THE LITTLE THOUGHT FULL ACTIONS OF HERS
 THAT SHOW HOW MUCH SHE CARED FOR ME

WHEN I WAS LITTLE SHE PROTECTED ME
 BROUGHT ME UP AND ME GROW BRIGHT
 WHEN AT TIMES, I FELT UNHAPPY
 SHE WAS THE ONE WHO HUGGED ME TIGHT

WITH GENTLENESS CARE AND PATIENCE
 SHE THOUGHT ME THE WAYS TO WORK AND PLAY
 HELPED ME TO SOLVE THE GROWN UP PROBLEMS
 AND MADE ME GREAT DAY BY DAY

AND WHEN SHE KNEW I WAS READY
SHE STEPPED ASIDE WITH TEARS OF JOY
SAW ME GONE WITH THE MAN OF HER CHOICE
PROVIDING EVERYTHING FOR ME TO ENJOY

HOW SHALL I THANK YOU OH! MY DEAR MOTHER
YES I SHALL DO THAT, BY BEING A GOOD MOTHER

BY,T.NIVETHITHA,II YEAR,ECE-‘C’

MY FRIEND

Believe me, I am there to stay
As I always stay.
I am there to stand by you,
You laugh or you cry,
But just let me feel, you are mine.
Believe me, I care a lot,
I talk to you in my thoughts,
I am there to listen to you.
But please accept me as a friend,
And let us share love with no end.
Believe me, you are my inspiration,
For you I have lots of admiration,
I am there to believe in you.
Whenever I feel lonely,
I want to meet you only.
Believe me and show your trust,
As in friendship this is must.
I am there to pray, for differences to arise never
And our friendship to flourish forever.

Khushboo
ECE-C, IInd Year

TOP 10 OCEAN

1. BERMUDA TRIANGLE:-DOES
THIS AREA OF THE ATLANTIC
PUER TO RICO,BERMUDA AND
FLORIDA REALLY SUCK SHIPS
AND AIRPLANES TO THEIR
DESTRUCTION AS THE LEGEND
TELLS?

2. SARGASSO SEA:- YOU DON'T
WANT TO TANGLE WITH THIS
FLOATING MASS OF TANGLED
VEGETATION.

3. SITE OF TITANIC SINKING THE MOST FAMOUS ICEBERGS IN HISTORY!

4. PANAMA CANAL OPENED IN 1904,THIS 50KMSCANAL JOINED THE ATLANTIC AND THE PACIFIC OCEANS AND SAVED TRIP OF THOUSANDS OF MILES AROUND CAPE HORN,AT THE TIP OF SOYTH AMERICA

5. CAPE HORN:-THIS CAPE,WITH A COMBINATION OF STRONG WINDS REACHEROUS COASTLINE,SENT MANY A SAILOR TO AN UNTIMELY DEATH.

6. STRAIT OF GIBRALTAR:-NARROW OPENING THAT JOINSVTHE MADITERRANEAN SEA TO THE ATLANTIC OCEAN.

7. GREAT BARRIER REGION:-THIS CORAL REGION HAS ONE OF THE

RICHEST ECOSYSTEMS ON THE PLANET .

8. MARIANA TRENCH:-IT IS THE DEEPEST SPART ON THE SURFACE OF EARTH.

9. HAWAII:-THESE ISLANDS ARE REALLY SEA MOUNTS.WHEN YU COMPUTE THE DISTANCE THEY RISE FROM THE OCEAN FLOOR,THE HIGEST MOUTAIN ;MOUNA KEA BECOMES THE TALLEST MOUTAIN ON THE EARTH

10. BERING STRAIT:-MANY PEOPLE BELIVE THAT UNTIL ABOUT 1200 YEARS AGO THEIR WAS A LAND BRIDGE HERE THAT PROVIDED THE ROUTE FOR PEOPLE MIGRATING FROM ASIA TO NORTH AMERICA

**BY,
M.PRATIBA,
II YEAR-ECE D**

Some facts to lead a success life

Be a source of peace and a joy for ever.

Be a happy with what you are and what you have

Be cheerful enthusiastic and active within yourself

Have determination and perservence if you do the best

God will do the best

**P.S .Janani
2nd year
ECE-B**

TENSION

You Will lose your Attention
 The moment are you in tension
 Then you are in total confusion
 And you'll feel Irritation
 Then you'll spoil personal Relation
 Ultimately, you won't get co-operation
 Then you'll make things complication
 Then your BP may also rise caution
 And you may have to make medication
 Instead understand the situation
 And try to think about the solution
 Many problems will be solved by
 Discussion
 Which will work out better in your
 Profession
 Don't think its my free suggestion
 It is only for your prevention
 If you understand my intention
 You'll never come again to Tension.....

A MINUTE OF ANGER MAY CAUSE
 A LIFE TIME AGONY TO YOU AND ALL.

R.Krishnaprasanth
N.S.Koushik
 3rd year
 ECE-C

THE TEARS WISH OF A HEART

Many words to say,
 but i am dumb;
 many lirickstolisten,
 but i am deaf;
 many pictures to see,
 but i am blind;
 but now ,
 i can see,
 every thing with my heart,
 which others may not be see;
 keep heart be precious,
 love every heart with your heart;

--
K.MANOJKUMAR
 ECE-C

THEOREM OF ABILITY.

statements:

The students of klu are very brilliant and sharp

To prove:

The students of klu are very brilliant and sharp

Proof:

In the alphatic order numbers are assigned

A=1; b=2; c=3;z=26;

l.h.s.

the = $20+8+5 = 33$

students= $19+20+21+4+5+14+20+19 = 22$

of= $15+6=21$

klu engg

$19+18+2+1+11+4+4+18 = 77$

total =253

r.h.s

very brilliant and sharp.

Very = $22+5+18+25 = 50$

Brilliant

= $2+18+9+12+12+9+1+14+20 = 122$

And = $1+14+4 = 19$

Sharp = $19+8+1+18+16 = 62$

Total=253

Total of l.h.s =253

Total of r.h.s=253

HENCE IT IS PROVED THAT

“THE STUDENTS OF KLU ENGG ARE VERY
BRILLIANT AND SHARP”

P.S.JANANI

2ND YEAR-ECE-B

GOES ON, FAREWELL...

Today, I gave my heart a hundred try
It just couldn't respond and gave a grieved cry

"Try!" Just try on me, but in vain...
For, today's the day when I'm in great big pain.

Hearing such a cry, my eyes were in tears...
Yes, today I'm leaving my 'Alma Matter', which sheltered me for years!

Today... My tongue doesn't know what to say...
But how I wish...!! God didn't show me this rainy day!

Nothing more to pen...

-Vivek Kumar,
III Yr, ECE-F.

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0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 1040

« Đ SÁÚ... þ Đ SÁÚ....

±î°jìì " Âj, pÔôÀÐ
 S, i " Æò¾É Ôõ pø " Ä
 « Äð°ÂÁj, pÔôÀÐ
 "¾jÂÔõ pø " Ä

!..i, Â

ÁüÈÀ÷, û Šà;ø pŎì, ÁüÈÀ÷, û pŎì, Èj÷, û!

! °ø....¿!ø...!

$\frac{3}{4}i\bar{E}$ " Á - í , " Ç - ĀSĀ
 Ĭ , ĭ ñ Ĭ Ĭ °øÖö;
 - Ē ĭ ø ĭ ü Ĭ $\frac{1}{2}ö \frac{3}{4}i \acute{y}$
 - ĩ , " C « í š
 - ð , ĭ ð ò $\frac{3}{4}i$ " Ā Ĭ ĩ ö!

ÍÂÎ, ù!.....

« Í ð¾Â ÷ Ç!ý
« Ê Î Í Â ð Ê Ñ Ñ Ñ Â
¿¾Ô Ð !, ì ñ Î
þ Ò ó¾î ð, Í, ù
« Ê Î Í Â Î ÷ Ç
' Ò ¿î Ù ð
À¾ì, Ò Ê Â î Ð!

« " È....

! Â î ð, Ç î ð
¾¾¾¾Â î ÷ ì ð Ò Â " ¾¾¾¾
ñ " Á Â î ð
« " È Â Ñ¾ Ñ Á Â î É Ð!

À¾î... À Î ...

À Ê Ò ì ì ð
À Â ý Á Î Í, ù;
À Ê Â î ð À Â ý Á Î ò¾Ò À¾î¾÷, ù!

° î Â î É ! Â Â ÷ !

± ð Ñ Â î Ò ð¾ Í, ù
¾¾¾¾ Ù, Ù ì ì Î Ý ð Î ð
! Â Â ÷- « Ù Â Â ð!

§, î Â ð !

- Í, ù, ÷ ù ! Â î Ò
¿ Â¾¾ §, î Â Ò ð
Â î½ Ê ì, Ê Ð
- Í, Ç!ý
« Ú Â Ð ¿ Â¾¾
¬ É ó¾Ò "¾!

Á ý É Ê ð !

±¾¾¾¾, " Ç ò¾î Â î Ç Á î,
Á ý É Ê ð Ð Â Î Í, ù
¬ É î ð, « Â ÷, Ç!ý
! Â Â ÷ " Ç
Á Ê Ò Ð Â¾¾¾¾¾, ù!

« . Í Â î À î Â¾¾÷, " Ê,
þ Â ñ¾î ð ¬ ñ Î.

¾ÁÆ÷ À ñ Àî Î

"ø, ¾î ýÈî Á ñ, ¾î ýÈî, ïÄö, ¾î ð¾ Ó ý ¾î ýÈÀ ã ð¾ ìÈ çð
¾Áú ìÈ".ÀÆ" Á Äîðó¾ çð Þ Èð¾ø À ñ Äî Î ±ðÐ " É °Èðð Äîðó¾¾î, ÞÖì,
SÄ ñ Îð.´ Øì ð, À ñ Ò, ÀÆì, ÀÆì, í, û SÄî ýÈ " Á À ñ ÄîðÉý
, Öð, ¾î Ö " ÁðÄî Î ±ÉÄð.

Äð Î Äîúì " :

Á " È ì ì ÄîÇì, ïð ÄîÇ í ìð Á, Ç ÷ ¾ð, ½ Á " Äð SÄ ½ ìðð, Ä ñ ½ ìý
Äð " ÁÄð, ùÄ " È ì, ïððð, ÁÉ " ¾ ÄÄ ì ìðð Äì " Ç Äî ñ ðÈ ÄÇ ÷ððð, Äðð¾
Äðð Äîðó¾È " Á µðÄððð, Þ íð " Äî ÄÉ, Öð¾ð Ì ÌðÄî ÍüÈð " ¾ð
¾ðÄðð, ¼ × " Ç Ä ½ í, ìðð, Äý, " Ä ÄÇ ÷ððð ðý Áðð " ¼ðð Þ ý Äð
Äðì, ìðð Äîúó¾É÷,
½ Ä Ù ì ì « Ú í " Á " ½ ÄÇ ÷ðð Þ ý ÒüÈ " ¾,

¾î ý Úúó ¾ð¾ ¾ððÇðÄî ÷
Þ È ì, ¾ Ä Ù ñ ¾ Äý
Ñ ñ ¾¾ý Ä, úó¾ý
Äî ñ Í ¾ø Ó, SÉ'

±ý Òð ì Úó¾î " , Äî¾Äîø « ÈÄÄðð.

Äðð¾¾ðð:

Äî S¾È Äðð Äîðó¾È Òì ì, ÄÄÈÄ " ½ ÄÇ ÷ðð " ¼ ÄÈ " ÈÄ¾í ì
ì, ì Ìðð, " ùÇ í ì ÇÄð ðý ðÜÄð ãð¾ Þ ý Ó, í, ìðÈ, Þ ÈÄ " Á ÜÈ, Äðð¾¾ðð, Äðð¾
Äî S¾î Üð ÞðÄð¾î ìý Þ ý ÈÄ " ÁÄî ì, ¼ " ÁÄð ÄîÇ í, üÜ.

'SÄðÄì ì " Äðð « È ì °ð Ó, ðð¾îðð
Sçì ì, ì ì " Äðð Äððð.'

±É Äðð¾ý Äý " Á " Ä Ä Ù Ü Ä ÷ ± Ì ð¾Äðð, ìÈ ÷.

¾ÁÆîý Äðð:

¾îø, ïðÄ ÷ Äðð ±ý Ä " ¾ð ÄðÄ¾ð ±ý Ù ì ì °ðÄîø
ìÈðÄì, ìÈ ÷, øÄ, ¾Ü, ñ, Þ " °, ì " ¾ ±ýÈ çý ìð ÄüÈÄð Äð¾¾
ÄÈ ì ìð. ±ÉSÄ SÄîü ì°Äð, ìð ìð " ñ " Á Äðð, ì, ì " ¼ Äðð Ä " ÄÄ
ÞÄì, Äì, Çð Þ " ½ ð¾ ÜÈðð¾È.

'Ä " Á " Ä Ä ñ Î Sçì ì, Ä, ñ, Äð ì ìð S¾îø ÄÄð.¾ý " Èð ¾î ì ì, ìýÈ
ÇüÈý SÄð ÄüÈð ì ìð ÄÄ " Èðð ì, ìý Ú Ä, úÄîÉð. SÄîððÈí, ìð ì¾ð
±ýÄð ÞÄîÉ ì°Äð " ìð, ì, ìÄSÉì SÄ ÷ ðð¾ Äð ì ìS°Äî¾¾ý Ä ÷ Äð
¾ðð " ÖÄÈ ð ñ ì ðððð ñ ½ ì ìð ±É ç ì ½ Ä¾ì, ìðð " Ä ÷ çð¾ý.
±ý Ä, ý SÄîððÈí, ìÈÄð " ñ ÄÄîÄý « Äý Äîð ñ ¼ « í, ð " ¾
° ì ¾ð¾ Ä SÄý ±ýÈðð ÓðÄ, ù " Öð¾.

ÄÈÄî Î:

Ä " Ä ÷ Ó ý Èý Ú " üÈSÄî Î " Ä ÷ çð¾ Äðð, " Ç ç ì " É × Ü ÷ Ä¾ý Äîðð Ì
« ùÄ¾í, Çð ç ì, ø' ´ ýÈ " Äðð ÄÆ ç¾ð¾ð Äðð Î¾ð ¾ÁÆ ÷ À ñ Äî ìð.

', ìð°, ìðS, ìð çðÄ " ¼ ç Ì, ø

0%¾ Ì °ÀÄüô !ÀÕõÀ °¼ Å;úó¾! Ä ý
¡Õ ã ý Ú Å ° ÄÄü . ø\$Ä; Î Ò½ Å'

$$\pm \gamma \hat{U} \tilde{A}_{j1/4} \cdots \tilde{A}_{j\hat{A}\delta^0 \uparrow 3/4} \cdots 1/2 \hat{A}_{j\emptyset} \cdots 1/2 \tilde{A}_{j\tilde{O}}.$$

! . i " 1/4Á1/4Õ:

$p\partial\tilde{A}i\frac{3}{4}A\div i\tilde{\partial}, -\partial\tilde{C}A\div i\tilde{\partial}, p\tilde{E}o\frac{3}{4}A\div i\tilde{\partial}, p\tilde{A}i\frac{3}{4}A\div i\tilde{\partial}\tilde{A}i\tilde{\partial}\tilde{A}i\tilde{E}y\tilde{E}i$
 $\tilde{A} \sim \tilde{A}\tilde{A}i\tilde{D} \mid i\tilde{\partial}\frac{3}{4}o\frac{3}{4}\tilde{A}\tilde{E}i\tilde{y}\tilde{A}\tilde{u}\tilde{C}i\tilde{n} \sim \tilde{A}, \frac{1}{4}\tilde{S}\tilde{A}\tilde{O}\tilde{A}\tilde{u}\tilde{C}\tilde{o}, \tilde{u} \pm \tilde{E}$
 $\tilde{S}\tilde{A}i\tilde{u}\tilde{E}o\tilde{A}\frac{1}{4}\tilde{E} \div \frac{3}{4}\tilde{A}\tilde{i}\tilde{u}\tilde{A}\tilde{y}\tilde{E} \div \tilde{u} \frac{3}{4}\tilde{o} \mid i \sim \frac{1}{4} \frac{3}{4}\tilde{E}\tilde{o}\frac{3}{4}i\tilde{o} \tilde{o}\tilde{A}\tilde{A} \div \tilde{A}\tilde{i}\tilde{y}\tilde{A}\tilde{E}\tilde{o} \mid \tilde{C}\tilde{i}\tilde{A}\tilde{o}$
 $\mid \tilde{A}\tilde{i}\tilde{y}\tilde{U}\tilde{o} \mid \tilde{A}\tilde{i}\tilde{O}\tilde{U}\tilde{o}\tilde{A} \sim \tilde{A}\tilde{A}i\tilde{D}\tilde{A}\tilde{E}i, \tilde{E} \div \frac{3}{4}\tilde{A}\tilde{i}\tilde{u}\tilde{A}\tilde{C} \div \tilde{o}\frac{3}{4}\tilde{A}\tilde{y}\tilde{E} \div \tilde{u}.$

ÓÊ× ¨ Æ:

$\frac{3}{4}A\grave{A}E\grave{I}y\grave{A}i\acute{u}$ " $\hat{A}$ $p\grave{u}A\grave{I}U$ « $\frac{3}{4}u\grave{I}$, $y\acute{U}$ $\frac{3}{4}E\grave{I}$ $p\grave{A}i$, $\frac{1}{2}o\grave{A}$ $i\grave{o}D$, $\hat{A}\hat{A}$
 $p\grave{A}i$, $\hat{A}i$, $u\pm i\grave{o}D$, $i\delta\frac{1}{4}i\grave{o}$ « " $\hat{A}\hat{A}$, $\frac{3}{4}i\acute{y}U$ | $\frac{3}{4}i\delta\hat{I}$ $\hat{A}i\acute{u}$ " $p\grave{o}\hat{A}\hat{E}o\frac{3}{4}i\acute{y}$ $\hat{A}\hat{E}$
 $\S\hat{A}\hat{n}\hat{I}$ | $\hat{A}y\hat{E}$, $i\hat{u}$ " $\hat{A}$ | , $\hat{n}\hat{I}$, $\S\hat{A}\hat{E}i\delta\frac{1}{4}\hat{A}\hat{O}\hat{o}$ $\hat{A}y\hat{A}\hat{u}\hat{O}$ « $\S\hat{A}\hat{u}\hat{I}$ $i\hat{o}$
 $\hat{A}\hat{n}\hat{A}i\delta\hat{E}\hat{u}$ | $\hat{A}i\hat{U}\hat{o}$ ° $\hat{E}\hat{o}$ | $\hat{A}\hat{u}\hat{E}\hat{D}$ $i\hat{o}$ $\frac{3}{4}\hat{A}\hat{E}\div$ $\hat{A}\hat{n}\hat{A}i\hat{I}$.

-À. Í Ó ¾ Ã Ä å Í Á,
þ ã ñ ¼ j ö ¬ ñ î.

“ Å “ ÄŜâĭ Ĥ Àĥĭ “ ĭĥ ĭ°øÅĐ ÓĐ,ĭø , ÉÁĭÉ “ À “ Âò à ĭ,ĥĭ
ĭ°øÅ¾ü ĭĥ ĭ°ÁĬ ĭö.”

- † ¡Ä! Ä÷ø¼ý

“ÀÄÓ ” È ŠÂ; ı̂ô í , û
‘ ÕÓ ” È ÓÊ ! Â Î Í , û.”

“ Õ ÁÉ¼ý ,øÅîˆ Âô ÒÈì ½ì ì ò ŠÀì Ð
« Å ý Åìúì Â ý ÓÊ× Å ˆ Å °ÁÁ í , ÇŠÂ
±¾÷Š, ì ù Ç ŠÅ ñ ÊÂÕì ì ò.”

-À Ç ð ¼

"| °ø Ö õ S À | Ð ã ° ò Ð | , , ñ S ¼ | °ø Ö Í , ú
¾ Ö Õ À À | ÷ ¾ | ø Þ Ö Ì , | Ð Á | ú Ì " . ."

“À ñ ĩ ì ì | °øÄ ¬ ° Àð¼|ø ¾| ý
 Á ° ÄÂ ý ¬ î | Å Ð | °øÄ ÓÊÔõ.”

¾Áúò ¾¼Œ;ǎÊý |ÀŒ ¨ Á, û

Ä Ä | Á Ô õ | ° | ø Ä ü ì | Å ò^{3/4} ð Î
 Å Ú | | ñ S^{1/4} Ø Ø^{3/4} | Å Í " , Ä ð
 Í^{3/4} Ø^{3/4} Ä Ì | | ü " È Í Ä ò ì |

Í, SÀi, Åiú " Å çòÐ
 ç!ý Û Â÷ ®ó¼ " Áó¼!ç!ý " É !ÄüÊÄü ¼Áú Á ñ §½ i!

¾ÄÄ " É ¾ý, ÖÄ " ÈÄø !, i ñ Î
 ç!ü §¾i Úõ !Äü !È Î ì õ !Äi(, ü)-
 ÄÊó¼ Á ñ ½!ý ÁÉõ ÄÊÿÐ..

ÓõÄ¾í, " Ç - üÇ¼ì, ð, ã × Ä, ò¾ " É , iòÐç!ý Û
 ÄiÄ¾ò " ¾¾i í ì õ Äiü ¼SÄ!
 À°!Äý Û ÄÖSÄi÷ ì Äi°ò " ¾Äi°Ç!òÐ
 Äí " ÁÄi ì ð ÄÄõ !ÄiÖSÇ!
 Ä " ÇóÐ, ¼ì ì õ ÄÄø, Ç!SÄ, ¼Äøõ ÁÉõ,
 ÄÖõ çõ ì ½õ.....

Ä¾õ !, i ñ ¼Äi " É Á ñ ÊÄÎ õ Ä, Ç!Öõ
 °Éõ !, i ñ ¼SÄí " °¾È!SÄi Î õ - ¼ÄÖõ
 'üÜ " ÁÄiö 'Ç!ÄüÜ ° í, Äiøõ
 'ôÄüÈ ¼Áú Á ñ SÉ!
 ç!ý " É ¾iúò¾ÄÄ÷ " ÇÖõ Äiúò¾ ç!ü ì õ
 ì ½õ SÄiø ±Ä÷ ì ì õ - ñ §¾i!

- ý " É ÄÊò§¾i÷ ÄÄiÖôÄý, p ù × Ä " ÄÊì,
 ç!Ä " ¼ò¾ Äü Ü Äý -Ä " ¼ò¾ ì Èü
 ±ØôÖõ ì Äø ÓÆí, ð Î õ ±ð Î ¾ì ì õ,
 Ä " È°iüÈð Î õ ¼Áú Äi°ò " ¾.

õÄý ÄiÄ¾ ÄÊó¼ Á ñ ½!ø
 « ÊSÄ " É Ä " ¼òÐ, ç!ý Õ, ú Äi¼
 °i!ôÄÄ÷ " ÇÖõ °ÈôÄ ì ì õ ±ý Û Â÷ Á ñ §½!
 - ý " É Ä ½ í ì, SÈiõ, ÁÉ¾iø « øÄ
 ±í, Ü
 Äi ½ õ¾iø!

-¾Áüî !°øÄý,
 ¾. - Éó¼ Äf ö §°, Äý,

ÄÖö, iÄö þÐÄi, ð î ö

þÇ í, i " Ä ¬¾Ä ý
 Í ð |¾i | ì ð s; ÄÄ¾Ö ö
 |¾i ð í î | °ø Ö ö |¾ý ÈSÄ- ± í s, | °ý ÈiSÄi?
 ¬¾ø ÄÇ ÷ ì ì ö, - Ä ÷ |, i î ì ì ö
 ¿ø Äj Ö |, i ñ ¼, i üSÈ ± í s, | °ý ÈiSÄi?
 Ä " Ä |, i î ì ì ö - Ä, Ètø
 Í Äi °i |, i üSÈ
 - " É ÄüÄ " É ó "¾ÄiSÄ Äi í, sÄ ñ î sÄi?
 |ÄÖ, |ÄÖ ö |¾i Ètø ÑðÄö
 - ý " É « "¾ðÐÄð¾s¾i |ÄðÈ ì ì üsÇ?

ÄÉ ö |¾i î ö ¿ ñ Ä " ½ sÄiø
 ± ý Ðý Äö Äi í, | |, i ü Ü ö
 ¬üÈ í " ÄsÄ...
 , iø ¿ " É ì ì ö ¿¾sÄ ± í s, | °ý ÈiSÄi?
 ¿ | °ý È þ¾ÄÄi ö
 |¾i Èü °i " Ä, ü Äó¾É sÄ
 s¾i ö ± ý, ñ, Ü ì ì ¿
 Ñ ñ |Äi ÖÇi ÉiSÄi?

þÉ ö, ì ½ ö, sÄ¾Äý Èt
 ± í, " Ç ²óÐö |Äi Ü " Äð¾iSÄ
 - ý ¾iÈ " É s°i¾i |, Äi
 Äó¾É þì, ð¾¼í, ü
 |¾i Ètø ÄÇ ÷ î °i ± ý Ü ö |ÄÄ ÷ î | °iø Ät
 ü " Èi |, iø ¾Äö þÖó¾É ÄÄ í, ü « ý Ü
 |ÄiÈÄÄø Óý sÈüÈö
 Ä¾Ätø " Ä «¾" É Öö
 ÄiüÈÉ, iüÈ " ÄÄiö
 |ÄiÜ ì, Ätø " Ä äÄt «¾" É
 ¬¾ÄÄ, Ä í, ü ¬¾Èt ý
 « Ät × ± ý Ü ö ì " ½, " Ç |¾i î ð¾É.

ÄÇ ÷ î °i ± ý ÄÐ
 ÄtÄü " È « ÄtðÄ¾ø " Ä
 þÇ ö |ÄiÈÄiÇ sÉ
 þÉÄiÄÐ °ó¾" É | °ö

ÄÇ í, Çt ý, ñ ½i tø
 üÇ ÷ î °i s¾" ÄÄtø " Ä
 þÖðÄ "¾Ä î ðÐ ÄÈðÄ¾ü ì
 ¬ " ° ²sÉi?
 ÄÇ í ì " Ç ÄÇ ÷ ì ì ö
 ¿ø Ä |ÄiÈÄiÇÉiö - Ö¾ÈtøSÄiö!
 - Ä ÷ Ät " É, i ñ sÄiö!

¬. °iÄ °ó¾Äi, 2¬ð ¬ ñ î-þ

² ý « ðÁ_i ÁÈóĐ ŠÀ_i É_iö!!

à ì ð ÓØ^¾_iö " ÄÄÄ_tø " Ä
 « ^¾ðÊð^¾_i ý ±ØôØ ÌË_i÷, û
 þÄó^¾_ÄÄ_iö ÄøĐÄ_i Ì
 « " Ä Ì " ÊÄ_iö Ì Ç_tòĐ
 « Ä[°]ÄÄ_iöð ^¾Öðð " Ä_tø
 ±ý " Éô Ä_i÷òĐ[°] Ì_iÌ ÌÈĐ
 ,øÄ_iöð ŠÄ_iÉ þðÄ_t,
 ÑÉ_tì Ì Ì Ì í, Äö,
 öð ä ð^¼_t,
 ±øÄ_iŠÄ Ä[°]ôÄð Í Ä_tð^¼Đ
 þó^¾ Ì_i Š^¼_ø Ä_iúì " , Ä_tø
 Ä_i÷òĐ Ä_i÷òĐ
 ,øÄ_t ^¾ó^¾_iö
 Ì, Ì Ì[°]ð - ý ÄÊÄ_tø
 ±ý " É[°]Ä_iöðĐ
 « ý ð ^¾ÄÄð Í ð
² ý « ðÁ_i ÁÈóĐ
 ŠÄ_iÉ_iö « ðÁ_i?

« ðÁ_i!!!

« Äû þÖ_i Ì ð Ä " Ä
 ±Äý Ü^¼₄
 ±ý " É ' ý Úð
 Ì[°]öÄ ÓÊÄ_iĐ
 « Ä " Ç ÄÈ Ì ð ŠÄ_iĐ
 « ó^¾₄ ^¼×Ç_iø Ü^¼₄
 ±ý " É , ðÄ_iüÈ
 ÓÊÄ_iĐ,
 « Äû ^¾_i ý ' « ðÁ_i'

f Ì. Í ó^¾Ä_tÄ_iÄ_iÄ^½ ý
 ä ý È_ið Ì Ì Í

आपकी यादोंमें.....

खामोशीकादामनथामेहुए
 मदहोशीकाआलमबनाए हुए
 आपकी यादोंमें
 हरेकलम्होंकोसीपकीतरहसंजोतेजारहेहैं ।।
 आपकीसोचसेपरे
 आपकेविचारोंसे अनि7
 आपकीअदाओं के कायल
 आपकी यादोंमें
 हरेकलम्होंकोसीपकीतरहसंजोतेजारहेहैं ।।
 आपकाअपने लक्ष्य के प्रति एकाग्रता
 वोपूनर्निमाण की विचारधारा
 वोसामाजिकदष्टिकोण
 शायद यही देख
 हमआपकी यादोंमें
 हरेकलम्होंकोसीपकीतरहसंजोतेजारहेहैं ।।
 आपकी वो धूमिलसीछवि
 वोआखोंमेंबसीनयनरूपीकाया
 सचमुचइसलिए तो
 हमआपकी यादोंमें
 हरेकलम्होंकोसीपकीतरहसंजोतेजारहेहैं ।।

श्वेताकुमारी

फिरEी बढ़नाथा.....

सफरकठिनथा, परनामुमकिननहीं
 सोचनिरालीथी, परनिश्चलनहीं
 मंजिलउज्जवलथी,परआसाननहीं
 रास्ते जर्ख भरेथे,परकंटीलेनहीं
 फिरEी बढ़नाथाउसेपानेको ।।
 उसेपानेकारास्ता
 सागरमेंभव्वरसाथा
 रेतमेंतपीहुई धूमिलसाथा
 समुद्रमेंउठीविशाललहरोंसाथा
 फिरEी बढ़नाथाउसेपानेको ।।
 उसेपाने की चाहत
 धरतीसेनिकलीज्वारभाटा के समानथी
 सूरजसेनिकलीकिरणों के समानथी
 फिरEी बढ़नाथाउसेपानेको ।।
 उसेपाकरजो खुशीमिली
 इसतरहमानोंपूरेरोमरोममें
 रक्तकाप्रवाहतेज होगयाहो
 औरजुबोंचीख चीख कर कह रहीहो
 लोआखिरमैनैतुम्हेंपाहीलिया ।।

श्वेताकुमारी

हाय ये महंगाई

हायहाय हाय ये महंगाई
 कमरतोड़ती ये महंगाई
 पाँचसालों के बाद
 दुसरीसरकारमहंगाई बढ़ानेआई
 हाय हाय हाय ये महंगाई
 हाय हाय ये ज़ालीममहंगाई
 यहाँइंसान बन गयाकसाई
 कोईभूख सेरोए
 कोई यूहीसरपटकके मरजाए
 हाय हाय हाय ये महंगाई
 कमरतोड़ती ये महंगाई
 हाय हायकिसानबेचारा
 दरदरकाहेतो वो मारा
 कभीसाहूकारतोकभीबैंकोंमेंजाए
 बैंकोंमेंबीज कादाम देख
 उसकीआँखोंसेऑसूआजाए
 चारोंतरफहैभूखमारी
 भूख की बिमारी ये बढ़ाई
 आजक्याहुआसाहूकारदलालोंमें
 किसबातकीहै ये लड़ाई
 हाय हाय हाय ये महंगाई
 ज़ालीम बन गया ये महंगाई
 सौसौकरोड़ की आबादीमें
 मारडालीये महंगाई
 रोटीकपड़ातोदूर
 कैसेकरे ये पढ़ाई
 हाय हायसरकारकाबाड़ा
 महंगाईहोगईहदसे ज़्यादा
 एकसरकारहटेभीतो
 दूजालाए सौतनमहंगाई
 हाय हाय हाय ये महंगाई
 कमरतोड़ती ये महंगाई ।।
 मुकेशकुमार

सबसेसुंदरनारी: धरतीमाँहमारी

सबसेसुंदरनारी धरतीमाँहमारी
 माँ की तरहबुलाती, कलीयोंसेभीन्यारी
 नाजानेकितनेहीकष्टझेलते
 सबसेसुंदरनारी धरतीमाँहमारी
 वक्षोंसेइसकीगोद खिले
 हरेभरे खेतलगतेइसकोप्यारीप्यारी
 प्यासलगेभीतोकभीप्यासीहीरहजाती
 सबसेअनोखीनारी धरतीमाँहमारी
 भू, भूमि, धराइसकीप्याय हैसारी
 पाप, पुण्य, स्वर्ग, नरक, हैइनकीक्यारी
 अलगदुनियावालेकोभीभाए माँप्यारी
 सबसेसुंदरनारी धरतीमाँहमारी
 सबसेअदभुतनारी धरतीमाँहमारी ।।
 मुकेशकुमार

असफलता, सफलताप्राप्तकरनेका एक महत्वपूर्णहिस्साहै

संकल्पहीमनुष्य का बल है ॥

विवेक, बहादुरीकाउत्तमअंशहै ॥

एकाग्रतासेहीविजय मिलतीहै ॥

गुस्साआपकोछोटाबनाताहै, क्षमाआपकोविस्तारदेतीहै

मेहनत, हिम्मतऔरलगनसेकल्पनासाकारहोतीहै ॥

कर्मकरनेमेंहीअधिकारहै, फलमेंनहीं ॥

उम्मीदपरदुनियाकायमटिकीहै ॥

वेकभीअकेलेनहींहोते, जिनकेपासआदर्शहै ॥

TELUGU QUOTES

అభ్యయం కూసు శివదయ

కోట శివదయ కూట కైరళ

ఏమి లేన యెడలలే ఆమ దమ చేతు మహా వ్యక్తులు

అశోదహానిక పలుగు యొక్కతవ

అలసయలవ్వు తు శివపు

అందలే జాతు అందక పోతే కాళ్ళ

ఆమ చేను మేనోళ్ళ దూడ గోట మేసితయ ?

అయ్యుచే వరకు అమ ప సయ్యగుతుంద

ఏ ఎడకు ఆ గొడుగు

చేదహోత ఉన్ నుత పోయినట్లు

శివదయతీన పాడు శివత పశుమ

చెడపకురా చెడమ

నేరు ముచదయితే ఉరు ముచదమతురద

కోక మ క్కకు దొండ పుడు

చెట్లకు రాళ్ళదట్టలు

గుమ్మడి కాయల దొంగ అంటే భుజాలు తదుమకొననడట

గుడ్డ వచ్చి శివము పెక్కిరదినట్లు

ఎవరు శత్రుకున్ గోతలే వారె పడతారు

దురాశ దఃఖమ నకు చెట

శివక తు లేన పోడిక చేమడ శివక

చేదహోత ఉన్ నుత పోయినట్లు

ఆకిల దోచ యెరుగదు , శివడు సుఖమెరుగదు

MALAYALAM QUOTES

മിന്നുന്നതെല്ലാം പൊന്നല്ല.

(All that glitters is not gold.)

പലതുള്ളി പെരുവെള്ളം.

(Little drops of water makes the mighty ocean.)

വിത്തുഗുണം പത്തുഗുണം.

(As you sow, so you reap.)

മൂത്തവർ ചൊല്ലും മുതുമെല്ലിക്ക, ആദ്യം കയ്ക്കും പിന്നെ മധുരിക്കും.

(An elder's words is like gooseberries; at first sour then sweet.)

ആനക്കാര്യം പറയുമ്പോൾ ആണോ ചേനക്കാര്യം?

(When you are discussing about elephant don't talk about yam matters.)

മുറ്റത്തെ മുല്ലക്കു മണമില്ല.

(The jasmine shrub in your front yard has no fragrance.)

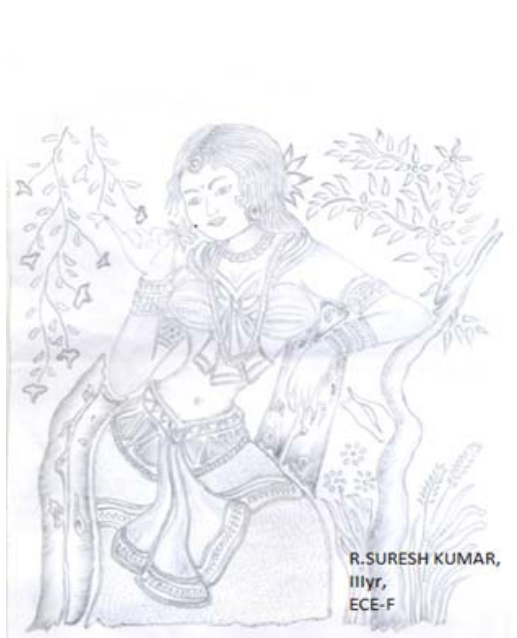
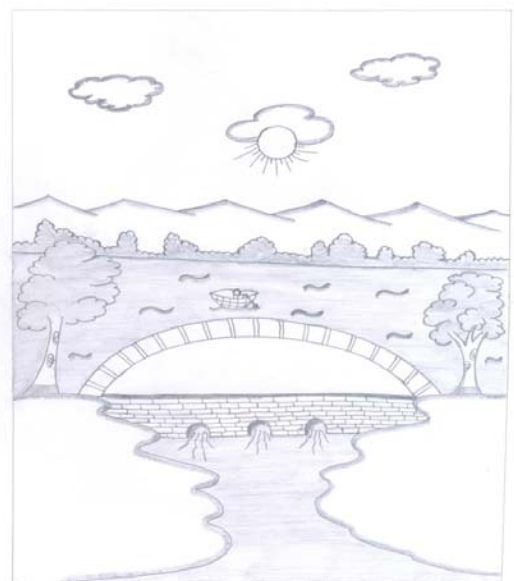
UPCOMING ENTRANCE EXAMS DATES

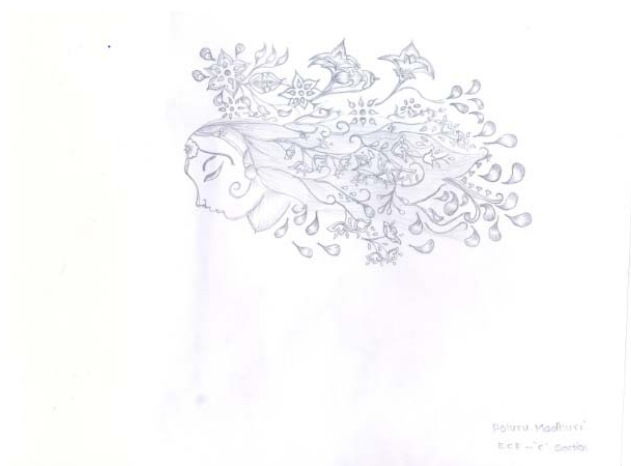
EXAM	DATE
CAT	27.10.10 – 24.11.10
XAT	2.1.11
FMS FOR MBA AND PH.D	5.12.10
FMS FOR MBA(PART TIME) AND MBA(HCA)	23.1.11
AIPGDET 2011	09 January 2011
AIPGMET 2011	09 January 2011
SNAP Test 2010	19 December 2010

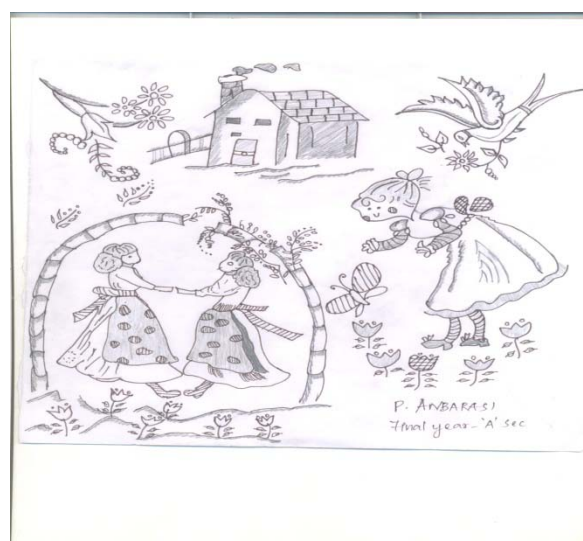
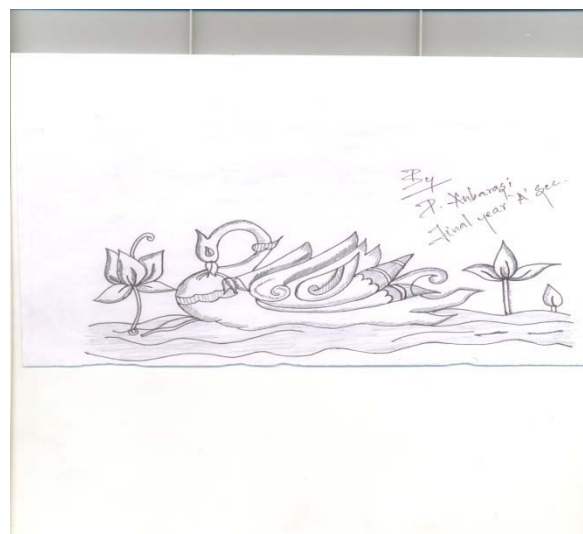
JMET 2011	12 December 2010
CETPPPC 2010	04 September 2010
IIM CAT 2010	27 October 2010
IRMA 2010	14 November 2010
IIFT 2010 MBA Entrance Exam	28 November 2010
IIBF 2010	14 November 2010
INAT 2010	22 December 2010
IBSAT 2010	26 December 2010

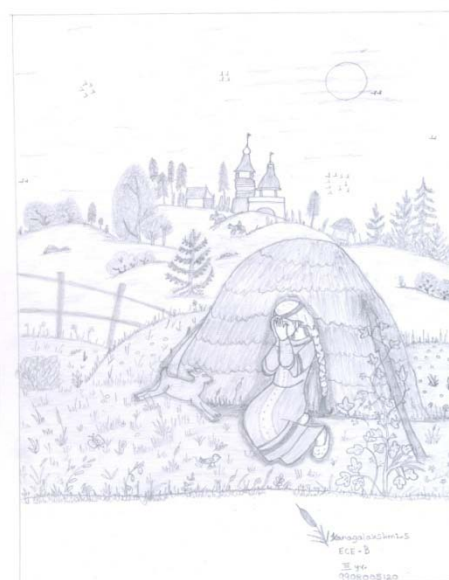
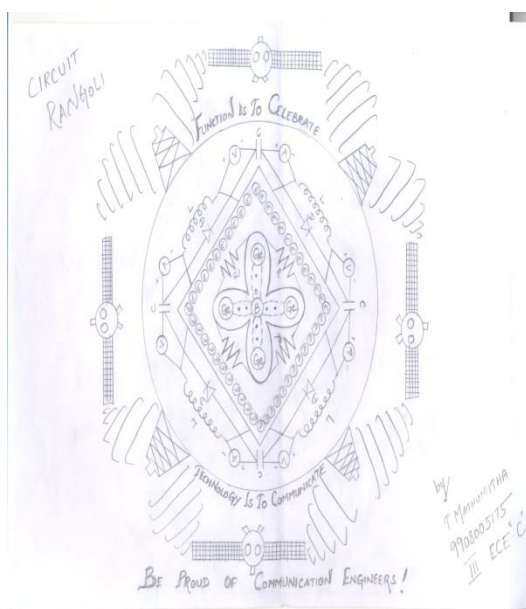
CA CPT	19 December 2010
CSIR UGC NET	19 December 2010
NCERT NTSE 2010	20 November 2010
Directorate of Government Examinations 2010	September / October 2010
NSTSE 2011	30 January 2011
AIMA Management Aptitude Test MAT	05 September 2010 & 11 September 2010
IIT JEE(TENTATIVE)	2 nd week of April 2011
AIEEE(TENTATIVE)	Last week of April 2011

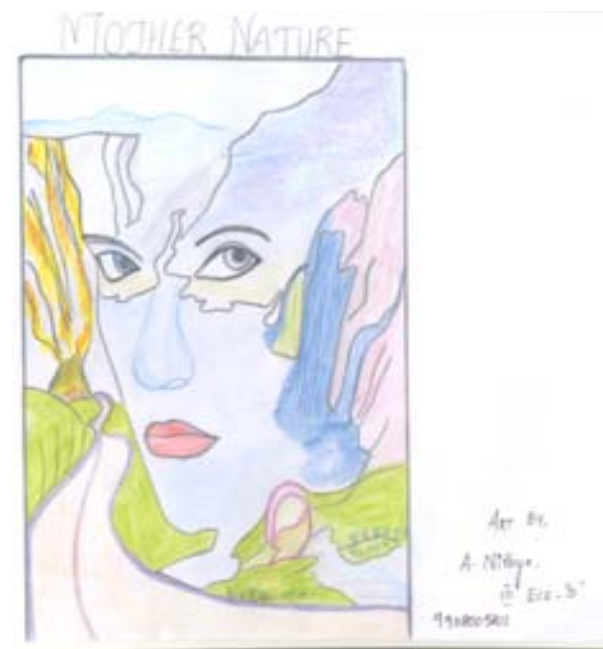
NMAT ONLINE	01st October, 2010 to 31st December, 2010
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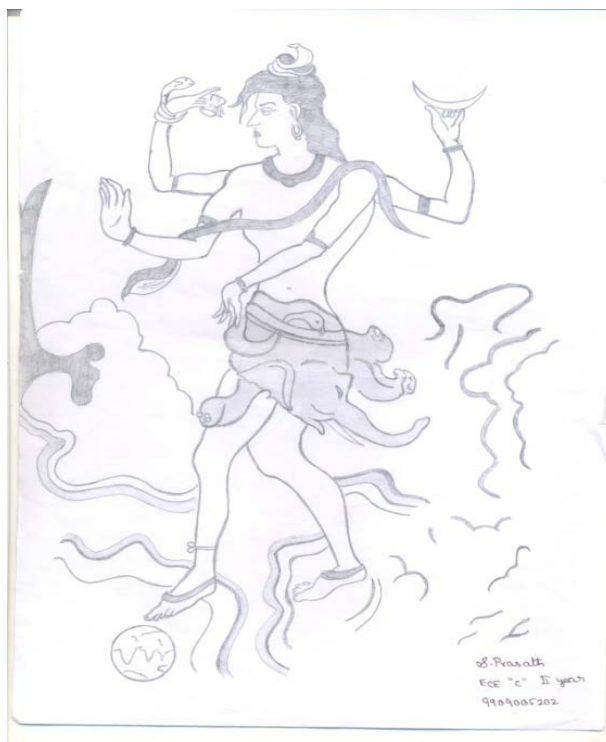












CAMPUS PLACEMENT

The task of filling up vacant positions in an organization is extremely important and must be done right in order to get the best people to work for you and take the Company forward with their skills and dedication.

Companies have been using a variety of methods to employ and select personnel to work for them and Campus Recruitment is one of the newest methods used by most Companies today.

What is Campus Recruitment...???

Campus Recruitment or Campus Placement has become a popular and commonly used method for selecting personnel in an organization. In Campus Recruitment, Our Client / Lemon Fresh Solutions visits some of the most important Technical and Professional Institutes in an attempt to hire young intelligent and smart students at source.

The most significant fact about Campus Recruitment is that instead of the job seekers approaching employers, the employers come to the most suitable prospective candidate with a job offer. It is common practice for Institutes today to hire a Placement Officer who coordinates with small, medium and large sized Companies and helps in streamlining the entire Campus Recruitment procedure.

Benefits of Campus Recruitment: -

- 1.Campus Recruitment proves to be beneficial for all the parties concerned, Institute, Lemon Fresh Solutions & the Employer.
- 2.Companies get the opportunity to choose from and select the best talent in a short span of time.
- 3.Companies end up saving a lot of time and efforts that go in advertising vacancies, screening and eventually selecting applicants for employment.

4.College students who are just passing out get the opportunity to present themselves to some of the best Companies within their industry of interest. Landing a job offer while still in college and joining just after graduating is definitely what all students dream of....!!!

Great interviews arise from careful groundwork. You can face your next interview if you:

1. Enter into a state of relaxed concentration. This is the state from which great basketball players or Olympic skaters operate. You'll need to quit the negative self chatter in your head through meditation or visualization prior to sitting down in the meeting. You'll focus on the present moment and will be less apt to experience lapses in concentration, nervousness, self-doubt and self-condemnation.
2. Act spontaneous, but be well prepared. Be your authentic self, professional yet real. Engage in true conversation with your interviewer, resting on the preparation you did prior to coming to the meeting. Conduct several trial runs with another person simulating the interview before it actually occurs. It's the same as anticipating the questions you'll be asked on a final exam.
3. Set goals for the interview. It is your job to leave the meeting feeling secure that the interviewer knows as much as he or she possibly can about your skills, abilities, experience and achievements. If you sense there are misconceptions, clear them

- up before leaving. If the interviewer doesn't get around to asking you important questions, pose them yourself (diplomatically) and answer them. Don't leave the meeting without getting your own questions answered so that you have a clear idea of what you would be getting yourself into. If possible, try to get further interviews, especially with other key players.
4. Know the question behind the question. Ultimately, every question boils down to, "Why should we hire you?" Be sure you answer that completely. If there is a question about your meeting deadlines, consider whether the interviewer is probing delicately about your personal life, careful not to ask you whether your family responsibilities will interfere with your work. Find away to address fears if you sense they are present.
 5. Follow up with an effective "thank you" letter. Don't write this letter lightly. It is another opportunity to market yourself. Find some areas discussed in the meeting and expand upon them in your letter. Writing a letter after a meeting is a very minimum. Standing out among the other candidates will occur if you thoughtfully consider this follow up letter as an additional interview in which you get to do all the talking. Propose useful ideas that demonstrate your added value to the team.
 6. Consider the interviewer's agenda. Much is on the shoulders of the interviewer. He or she has the responsibility of hiring the right candidate. Your ability to do the job will need to be justified. "Are there additional pluses here?" "Will this person fit the culture of this organization?" These as well as other questions will be heavily on the interviewer's mind. Find ways to demonstrate your qualities above and beyond just doing the job.
 7. Expect to answer the question, "Tell me about yourself." This is a pet question of prepared and even unprepared interviewers. Everything you include should answer the question, "Why should we hire you?" Carefully prepare your answer to include examples of achievements from your work life that closely match the elements of the job before you. Obviously, you'll want to know as much about the job description as you can before you respond to the question.
 8. Watch those nonverbal clues. Experts estimate that words express only 30% to 35% of what people actually communicate; facial expressions and body movements and actions convey the rest. Make and keep eye contact. Walk and sit with a confident air. Lean toward an interviewer to show interest and enthusiasm. Speak with a well-modulated voice that supports appropriate excitement for the opportunity before you.
 9. Be smart about money questions. Don't fall into the trap of telling the interviewer your financial expectations. You may be asking

for too little or too much money and in each case ruin your chances of being offered the job. Instead, ask what salary range the job falls in. Attempt to postpone a money discussion until you have a better understanding of the scope of responsibilities of the job.

10. Don't hang out your dirty laundry. Be careful not to bare your soul and tell tales that are inappropriate or beyond the scope of the interview. State

your previous experience in the most positive terms. Even if you disagreed with a former employer, express your enthusiasm for earlier situations as much as you can. Whenever you speak negatively about another person or situation in which you were directly involved, you run the risk (early in the relationship) of appearing like a troubled person who may have difficulty working with others.

Possible GD Topics

Balance between professionalism and family.

1. Effect of cinema on Youth.
2. Education in India compared to foreign nations.
3. Environment-Who's Responsibility.
4. About Hockey being the primary game in India.
5. Present state of Indian Cricket team.
6. Love marriage/Arranged marriage.
7. Advantages of Co-education.
8. How to deal with international terrorism.
9. Should we pursue our policy of dialogue with Pakistan?
10. Beauty contests degrade womanhood.
11. The rise of regional blocs threatens independent nations like India.
12. Six billion and one bronze!
13. Is dependence on computers a good thing?
14. Why can't we be world players in industry as we are in software?
15. Multinational corporations: Are they devils in disguise?

Social Topics:

16. Bride burning and dowry may look bad, but are an integral part of India.
17. The education system needs serious reforms.
18. Showing Violence and Crimes should not be allowed in films and on television.
- Let us legalize gambling.
19. Is management an art or science?
20. The Rush for MBA is really a rush for big money.
21. The objective of Management is to maximize profits.
22. The Internet is an exercise in hype.

23. Family owned business vs professionally run businesses.

Creative Topics:

- 24. The Wheel is Turning Round and Round.
- 25. If I was the Finance Minister/Prime Minister.
- 26. There is no right way to do a wrong thing.
- 27. Do Beauty and Brains Go Together?
- 28. When I woke up in the morning I saw...
- 29. A ship docked in harbor cannot face the storms.
- 30. Up the Down Staircase.
- 31. Marriage is a social trap.

ECONOMIC TOPICS:

- 32. Developing countries need trade, not aid.
- 33. Why do we lag behind China?
- 34. Capitalism is a very flawed system but the others are so much worse.
- 35. Flexibility of labour laws is the key to attracting more Foreign Direct Investment.

POLITICAL TOPICS:

- 36. Reserving seats for women in Panchayat has not only been a farce but has distracted from developing a more genuine voice of women.
- 37. Voters, not, political parties are responsible for the criminalization of politics.
- 38. The voters are required to be well informed and educated about their candidates so that they can elect the right aspirant by their own assessment.

APTITUDE QUESTIONS

Q1. Two series are 16,21,26.... and 17,21,25.....

What is the sum of first hundred common numbers

- (a) 101100
- (b) 110100
- (c) 101110
- (d) 110101

Q2. A, B and C are three speakers. They have to speak randomly along with another 5 speakers in a function.

A has to speak before B and B has to speak before C. What is the probability?

Q3 If $dy = (\sec x + y \tan x) dx$, Then the curve is

- (a) $x = y \cos x$
- (b) $x = y \sin x$
- (c) $x = y \tan x$
- (d) $x = y \sec x$

Q3. For the word SURITI, if you arrange the letters in dictionary order then what is its rank?

- (a) 234
- (b) 235
- (c) 236
- (d) 237

Q4. Two series are 16,21,26.... and 17,21,25.....

What is the sum of first hundred common numbers

- (a) 101100
- (b) 110100
- (c) 101110
- (d) 110101

Q5. There are two sections in a question paper each contain five questions. A student has to answer 6 questions.

Maximum no. of questions that can be answered from any section is 4.

How many ways he can attempt the paper?

- (a) 50
- (b) 100
- (c) 120
- (d) 200

ANSWERS

Q1. Ans. (a)

Q2. Ans. 1/6

Q3. Ans. (c)

Q4. Ans. (a)

Q5. Ans. (d)

FREDERICK BAKEWELL



English physicist, Frederick Bakewell developed in 1847 a facsimile system with tin-foil covered revolving drums for transmitting and receiving recorded pictures. Frederick Bakewell, an English physicist, was one of the first to demonstrate facsimile transmission.

Working on a different principle, Frederick Bakewell designed a system that had many of the features of modern facsimile machines. In 1847 he patented a chemical telegraph. Actually, Bakewell improved Bain's proposal by replacing the pendulums with synchronized rotating cylinders. The first public demonstration of the Bakewell's facsimile system took place in 1851 at the World's Fair in London.

Bakewell's system involved writing or drawing on a piece of metal foil with a special insulating ink. The foil was then wrapped around a cylinder (C). This cylinder formed a part of a machine, which rotated it slowly on its axis, as in a lathe.

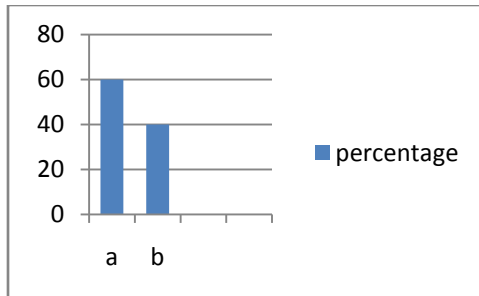
The cylinder rotated at a uniform rate by means of a clock mechanism (M). A metal stylus driven by a screw thread (T) traveled along the surface of the cylinder as it turned, tracing out a path over the complete piece of foil. Each time the stylus crossed a line of the insulating ink, the electrical current running through the foil to the stylus was interrupted. At the receiver, a similar pendulum-driven stylus marked chemically treated paper with an electric current as the receiving cylinder rotated.

The main problems with Bakewell's machine were how to keep the two cylinders moving at the same rate (in synchronization) and to make sure that both transmitting and receiving styli started at the same point on the cylinder at the same time (in phase). Despite problems with synchronization, Bakewell's machine was capable of transmitting handwriting and simple line drawings along telegraph wires. It therefore had the potential of allowing, for example, signatures to be verified by different branches of a bank. But Bakewell's system never became commercial.

Later, in 1861, Bakewell's system was improved by an Italian priest, Abbe Caselli. Caselli was able to use it to send handwritten messages as well as photographs.

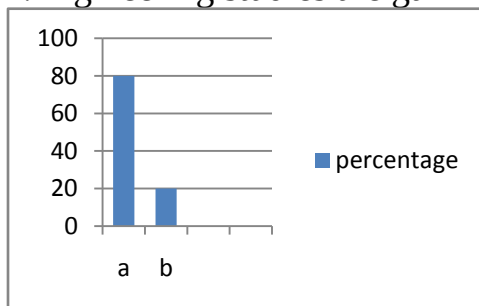
POLL QUESTIONS

1.Are we really into this race of engineers to reach our destiny?



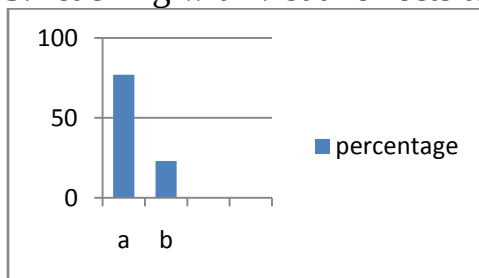
a)yes b)no

2.Engineering studies are gaining more attraction due to



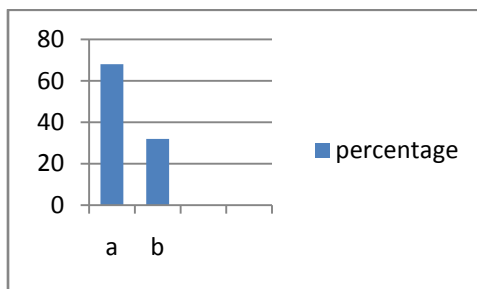
a)quantity b)quality

3.Teaching with visual effects turns out more effective than the black board



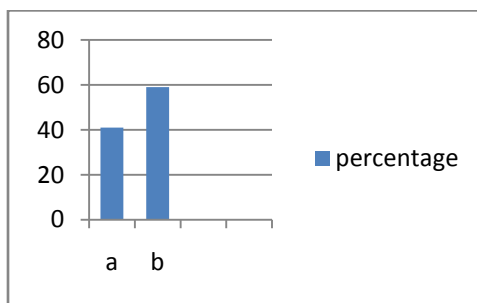
(a)yes (b)no

4. Do most of the engineering graduates wish to be placed in the core companies?



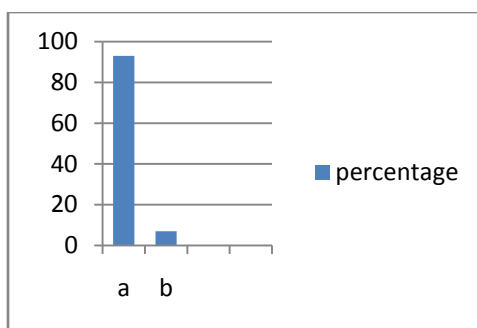
(a)yes (b)no

5. Students are compelled to opt engineering or they do it on their own interest.



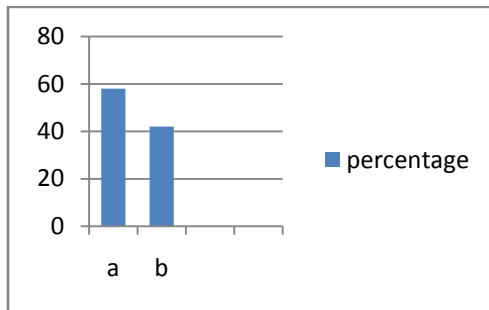
(a) They are compelled (b)out of their own interest

6. Is education a tool for making money ?



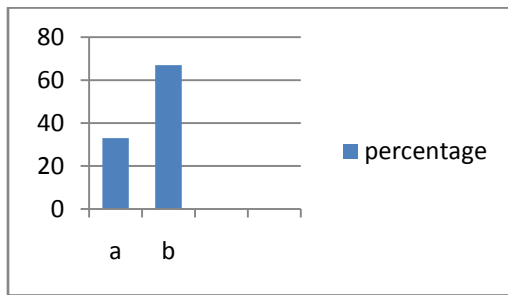
(a)yes (b)no

7.Engineering students are good at personality with



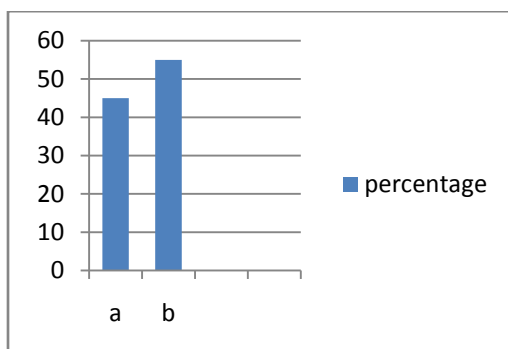
(a)formal dresses (b)casuals

8.Do engineering graduates use laptops for academic purpose really



(a)yes (b)no

9.How engineering students are differing from others



(a)way of thinking (b)behavior

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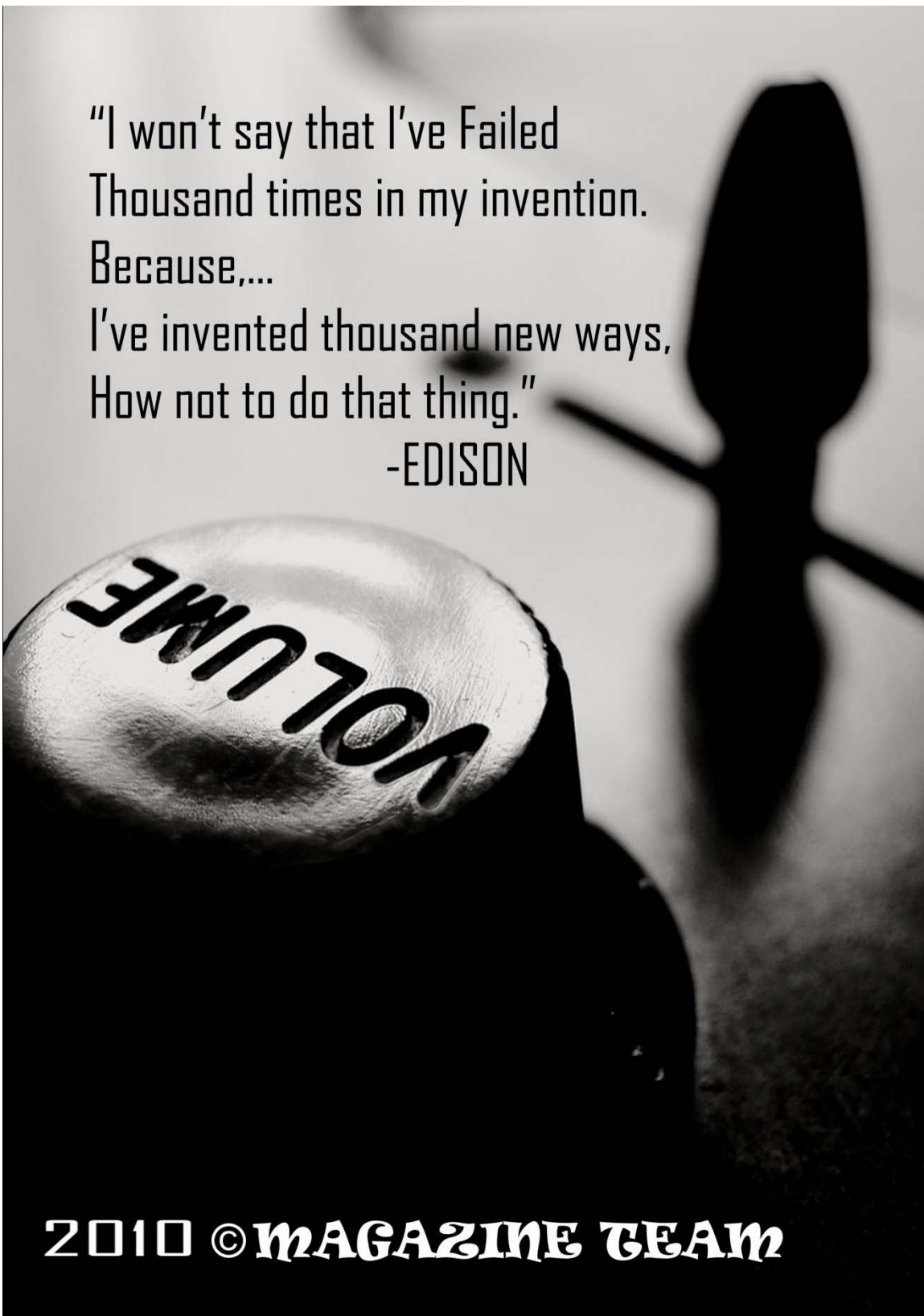
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"I won't say that I've Failed
Thousand times in my invention.
Because,...
I've invented thousand new ways,
How not to do that thing."
-EDISON

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